



PETROLEUM RESERVOIR ENGINEERING

PGG 431

(Day & Time Periods: Monday 1-3pm)

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PGG 313 Course Description

- General composition of Petroleum.
- Fundamental properties of fluid permeated rocks.
- Properties of Porous media containing multiple fluid saturations.
- Fundamentals of the behaviour of hydrocarbon fluids.
- Determination and application of reservoir fluid properties.
- Properties of Water.
- Data Evaluation for reservoir calculations.
- The material balance

PGG 431 Course Description

- Study of gas-condensate and undersaturated reservoirs including recovery methods.
- Oil reservoirs under simultaneous dissolved gas drive, Gas cap drive and water drive.
- Water influx.

PGG 431 Course Description (Reworked)

- The Material Balance revisited - Application of Material Balance Analysis to different hydrocarbon reservoir systems – dry gas, gas-condensate, undersaturated oil and saturated oil reservoirs under different drive mechanisms.
- Water Influx
- Secondary recovery methods – immiscible displacement by gas and water.
- An overview of Enhanced Oil Recovery.
- Factors affecting Recovery Efficiency.

Required Background

- PGG 321 – Fluid Flow Through Porous Medium
- PGG 223 – Petroleum Geology
- PGG 322 – Formation Evaluation
- PGG 313 – Basic Reservoir Engineering
- PGG 311– Petroleum Engineering Laboratory I
- PGG 325 – Petroleum Engineering Laboratory II
- Knowledge from other courses might come in handy

References

- **Ezekwe, N., *Petroleum Reservoir Engineering Practice*, 2011 Pearson Education, Inc.**
- Lyons, W., *Working Guide to Reservoir Engineering*, 2010 Gulf Publishing
- Donnez, P., *Essentials of Reservoir Engineering*, 2007 Technip
- Ahmed, T. and McKinney, R., *Advanced Reservoir Engineering* 2005 Elsevier Inc
- Dake, L.P., *The Practice of Reservoir Engineering*, 2001 Elsevier
- Craft, B.C. and Hawkins M.F., *Applied Petroleum Reservoir Engineering*, 1991 Prentice-Hall Inc
- Dake, L.P., *Fundamentals of Reservoir Engineering*, 1998 Elsevier Science

Learning Outcomes

- Understand fundamental concepts for Reservoir Engineering Analysis
- Understand basic hydrocarbon recovery processes
- Apply knowledge to different hydrocarbon reservoir systems for the purpose of evaluating past performance and forecasting future production.

Basis for Grades

Components

Attendance	10%
Mid Semester Test / Assignment	15%
Group Presentations	15%
Final Examination	<u>60%</u>
	<u>100%</u>

Ground Rules

- Keep your phones on vibration or silent mode
- You are free to note typographical errors on the slides. **Notify me after the class.** Please take note of the slide number.
- If you have a question, raise your hand. Note that all questions can't be entertained in class. Any questions I can't answer will be treated in the next class.
- The number of attendees at each class would be taken. This would be used to confirm day's attendance list.

Motivating Quote

True teaching, then, is not that which gives knowledge, but that which stimulates pupils to gain it. One might say that he teaches best who teaches least; or that he teaches best whose pupils learn most without being taught directly.

- John Milton Gregory

The General Material Balance Equation – Lecture 1

(Chapter 7 & 9: Main Text)

N.B.: Commonly used abbreviations

MBE – Material Balance Equation

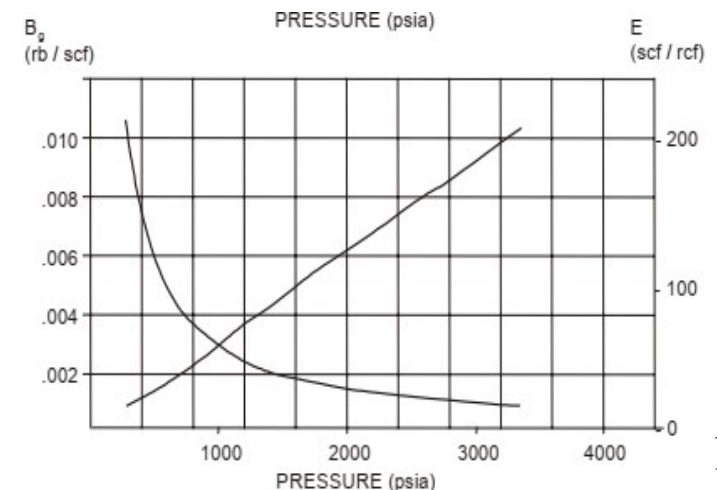
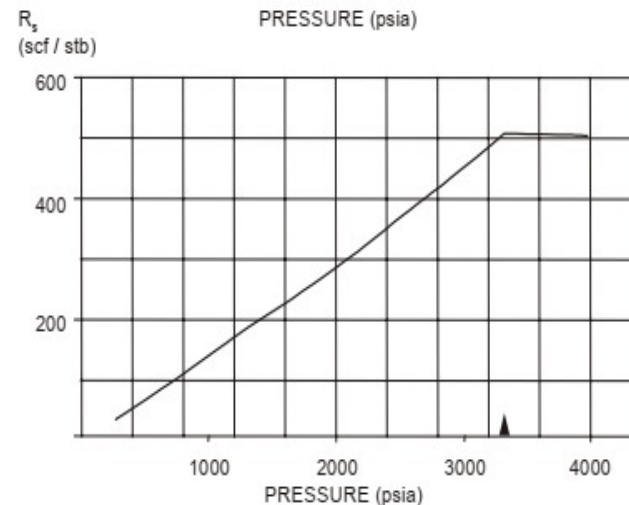
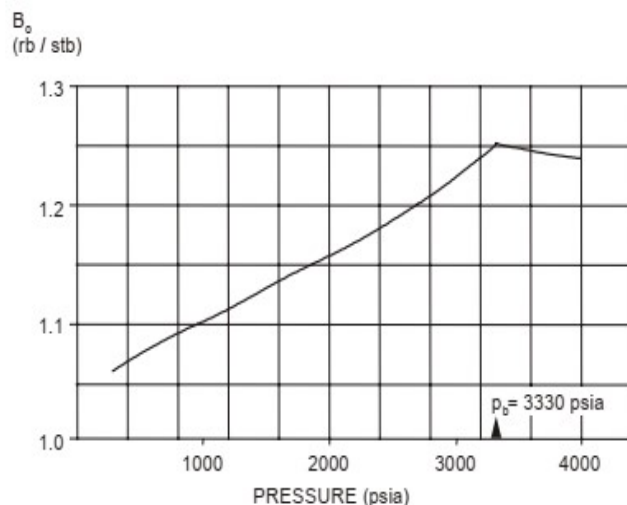
GMBE – General Material Balance Equation

Definition of Relevant Terms (Revision)

- What is an initially undersaturated reservoir?
- What is a volumetric reservoir?
- How is the total (two-phase) formation volume factor B_t defined?

Definition of Relevant Terms (Revision)

- Initially-undersaturated reservoirs are those whose initial but not necessarily final pressure is greater than the saturation (dew or bubble point) pressure.
- Volumetric reservoirs are those whose hydrocarbon pore volume does not change.
- The Two-Phase Oil Formation Volume Factor is an oil PVT property defined as: $B_t = B_o + (R_{si} - R_s)B_g$



Introduction

- The description of the equation that is developed as a material balance equation is in a strict sense not accurate. Actually, it is a **volumetric balance based on reservoir barrels** instead of a material balance based on mass.
- The GMBE is derived from the basis that the reservoir pore volume as existing at its initial conditions is constant. Thus, the sum of the volume changes of gas, oil, water, and the formation rock caused by net fluid production is equal to zero.
- The GMBE is based on an oil reservoir with a primary gas cap at initial conditions and reservoir pressure designated as p_i . At a later time, t , reservoir pressure is assumed to have been reduced from p_i to p due to production of oil, water, and gas. During the production period, it is assumed that there was water influx into the reservoir from an aquifer. Also assumed is that water and/or gas was injected into the reservoir.

The MBE Concept

- The concept the MBE was presented by Schilthuis in 1936 and is simply based on the principle of the volumetric balance.
- It states that the cumulative withdrawal of reservoir fluids is equal to the combined effects of fluid expansion, pore volume compaction, and water influx.
- In its simplest form, the equation can be written on a volumetric basis as:

$$\text{Initial volume} = \text{volume remaining} + \text{volume removed}$$

The MBE Concept

- The MBE is designed to treat the reservoir as a single tank or region that is characterized by homogeneous rock properties and described by an average pressure, i.e., no pressure variation throughout the reservoir, at any particular time or stage of production.
- Therefore, the MBE is commonly referred to as a tank model or zero-dimensional (0-D) model.
- These assumptions are of course unrealistic since reservoirs are generally considered heterogeneous with considerable variation in pressures throughout the reservoir. However, it has been shown that the tank-type model accurately predicts the behaviour of the reservoir in most cases if accurate average pressures and production data are available.

Basic Assumptions in the MBE

- Constant Temperature:
 - Pressure–volume changes in the reservoir are assumed to occur without any temperature changes. If any temperature changes occur, they are usually sufficiently small to be ignored without significant error.
- Uniform Reservoir Characteristics:
 - The reservoir has uniform porosity, permeability, and thickness characteristics. In addition, the shifting in the gas–oil contact or OWC is uniform throughout the reservoir
- Fluid recovery:
 - The fluid recovery is considered independent of the rate, number of wells, or location of the wells. The time element is not explicitly expressed in the material balance when applied to predict future reservoir performance.

Basic Assumptions in the MBE

- Pressure Equilibrium:
 - All parts of the reservoir have the same pressure and fluid properties are therefore constant throughout. Minor variations in the vicinity of the wellbores may usually be ignored. Substantial pressure variation across the reservoir may cause excessive calculation error.
- Constant Reservoir Volume:
 - Reservoir volume is assumed to be constant except for those conditions of rock and water expansion or water influx that are specifically considered in the equation. The formation is considered to be sufficiently competent that no significant volume change will occur through movement or reworking of the formation due to overburden pressure as the internal reservoir pressure is reduced.

Basic Assumptions in the MBE

- At most, there are two HC phases: oil and gas
- At most, there are two HC components: stock-tank oil and separator (surface) gas
 - These define the popular two-component formulation
- The reservoir oil-phase consist of stock-tank oil and separator-gas
 - This accounts for the effects of dissolved- or solution- gas.
- The reservoir gas-phase consist of only separator-gas
 - This ignores the possibility of volatilized-oil and restricts application to black-oils and dry-gases and precludes application to volatile-oils, gas-condensates, or wet-gases.
- There are no compositional gradients within the system
 - The system is treated as a tank without gradients

Necessary Requirements for Material Balance Application

- Adequate collection of reliable data. All production data should be recorded with respect to the same time period. If possible, gas cap and solution gas production records should be maintained separately.
- PVT samples or data sets represent the actual fluid compositions and that reliable and representative laboratory procedures have been used.
- Average pressure decline trend can be defined
- Interstitial water saturation
- Rock and Water compressibility

Necessary Requirements for Material Balance Application

- Average pressure decline trend can be defined. This is critical when correctly weighting or averaging the individual well pressures is required.
 - An example is when the producing formations are comprised of two or more zones of different permeabilities. In this case, the pressures are generally higher in the zone of low permeability and because the measured pressures are nearer to those in high-permeability zones, the measured static pressures tend to be lower and the reservoir behaves as if it contained less oil.
 - This is also true for fields that are not fully developed, because the average pressure can be that of the developed portion only, whereas the pressure is higher in the undeveloped portions.

Necessary Requirements for Material Balance Application

- Establishing an average pressure decline trend can be possible even if there are large pressure differentials across the field under normal conditions. Averaging *individual well pressure* declines can possibly be used to determine a uniform trend in the entire reservoir.
- Since the MBE is usually applied at regular intervals of 3–6 months, i.e., $\Delta t = 3\text{--}6$ months, throughout the lifetime of the field, the average field pressure can be expressed in terms of the incremental net change in underground fluid withdrawal, $\Delta(F)$, as:

$$\bar{p}_r = \frac{\sum_j \bar{p}_j \Delta(F)_j / \Delta \bar{p}_j}{\sum_j \Delta(F)_j / \Delta \bar{p}_j}$$

- where the total underground fluid withdrawal at time t and $t + \Delta t$ are given by:

$$F_t = \int_0^t [Q_o B_o + Q_w B_w + (Q_g - Q_o R_s - Q_w R_{sw}) B_g] dt$$

$$F_{t+\Delta t} = \int_0^{t+\Delta t} [Q_o B_o + Q_w B_w + (Q_g - Q_o R_s - Q_w R_{sw}) B_g] dt$$

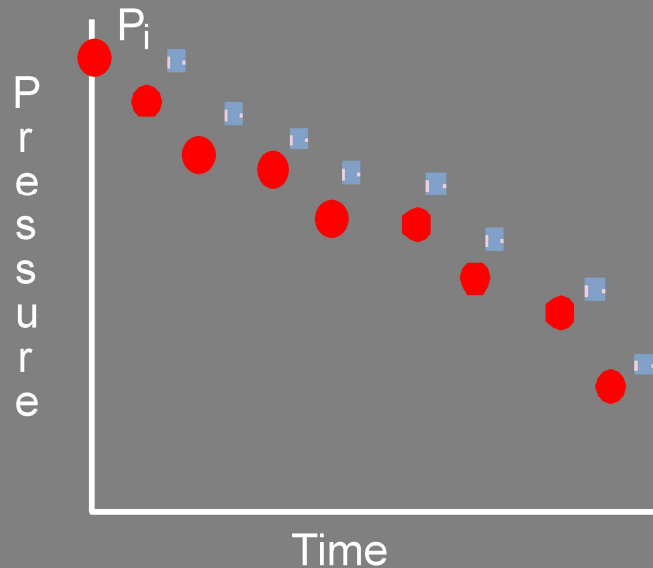
Average Pressure Decline Trend

Ideally

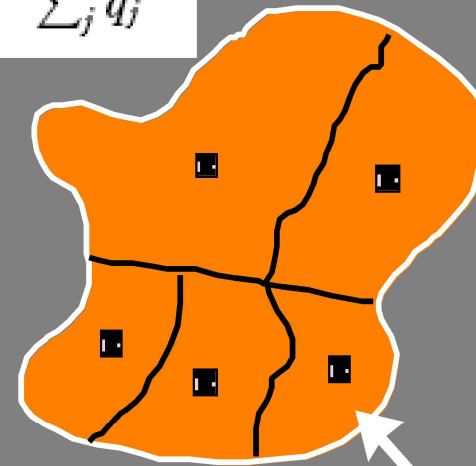
$$\bar{p}_r = \frac{\sum_j (\bar{p}V)_j}{\sum_j V_j}$$

In Practice

$$\bar{p}_r = \frac{\sum_j (\bar{p}q)_j}{\sum_j q_j}$$



Non-equilibrium
pressure decline



Drainage
boundaries

p_j, q_j, V_j

$$p = \frac{\sum_j (p_j q_j / p'_j)}{\sum_j (q_j / p'_j)} \quad \bar{p}_r = \frac{\sum_j [(\bar{p}q)_j / (\partial \bar{p} / \partial t)_j]}{\sum_j [q_j / (\partial \bar{p} / \partial t)_j]}$$

(p' = rate of pressure change)

GENERAL MATERIAL BALANCE EQUATION

Developing the MBE

- Before deriving the material balance, it is convenient to denote certain terms by symbols for brevity. The symbols used conform where possible to the standard nomenclature adopted by the Society of Petroleum Engineers.

• p_i Initial reservoir pressure, psi • p Volumetric average reservoir pressure • p Change in reservoir pressure = $p_i - p$, psi • p_b Bubble point pressure, psi • N Initial (original) oil-in-place, STB • N_p Cumulative oil produced, STB • G_p Cumulative gas produced, scf • W_p Cumulative water produced • R_p Cumulative gas–oil ratio, scf/STB • GOR Instantaneous gas–oil ratio, scf/STB • R_{si} Initial gas solubility, scf/STB • R_s Gas solubility, scf/STB • S_{wi} Initial water saturation	• B_{oi} Initial oil formation volume factor, bbl/STB • B_o Oil formation volume factor, bbl/STB • B_{gi} Initial gas formation volume factor, bbl/scf • B_g Gas formation volume factor, bbl/scf • W_{ing} Cumulative water injected, STB • G_{inj} Cumulative gas injected, scf • W_e Cumulative water influx, bbl • m Ratio of initial gas cap gas reservoir volume to initial reservoir oil volume, bbl/bbl • G Initial gas cap gas, scf • PV Pore volume, bbl • c_w Water compressibility, psi^{-1} • c_f Formation (rock) compressibility, psi^{-1}
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Material Balance before Production

Original pore volume = gas-cap volume + oil volume + connate water volume

$$PV_i = G B_{gci} + N B_{oi} + W B_{wi} \quad (RB)$$

Hydrocarbon pore volume = gas-cap volume + oil volume

$$HCPV_i = G B_{gci} + N B_{oi} \quad (RB)$$

Where

N = Oil (STB)

W = Water (STB)

G = Gas (SCF)

B_{gci} = Initial Gas- Cap FVF (RB/SCF)

B_{oi} = Initial Oil FVF (RB/STB)

B_{wi} = Initial Water FVF (RB/STB)

Define the ratio m as:

$$m = \frac{\text{initial volume of gas cap in bbl}}{\text{volume of oil initially in place in bbl}} = \frac{GB_{gi}}{NB_{oi}}$$

Eq. M1

Original saturation

$$S_{gi} + S_{oi} + S_{wi} = 1$$

$$G B_{gi} = PV S_{gi}$$

$$N B_{oi} = PV S_{oi}$$

$$W B_{wi} = PV S_{wi}$$

$$\text{Initial oil volume} + \text{initial gas cap volume} = (PV)(1 - S_{wi})$$

$$NB_{oi} + mNB_{oi} = (PV)(1 - S_{wi})$$

Solving for PV gives:

$$PV = \frac{NB_{oi}(1 + m)}{1 - S_{wi}} \quad \text{Eq. M2}$$

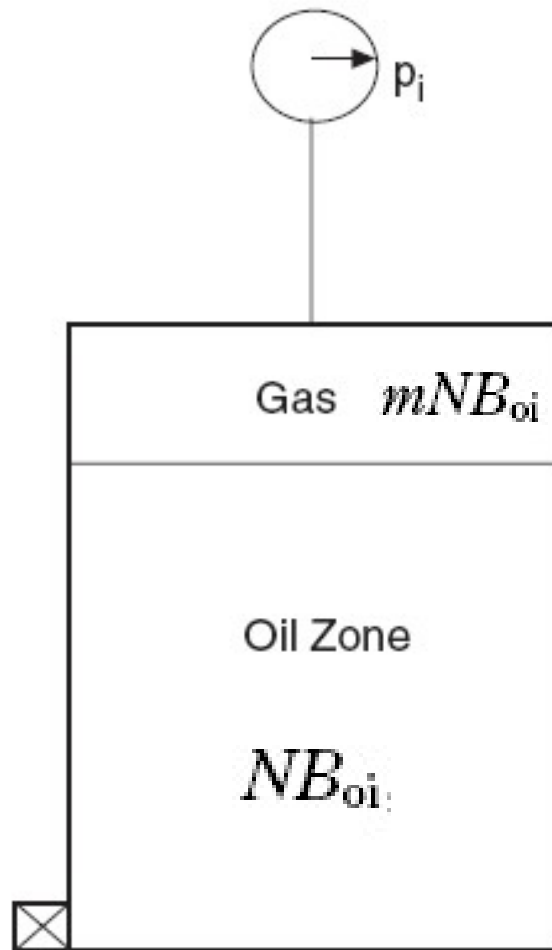
Developing the MBE

- Treating the reservoir PV as an idealized container as illustrated in Figure M1, volumetric balance expressions can be derived to account for all volumetric changes which occur during the natural productive life of the reservoir.
- The MBE can be written in a generalized form as follows:
PV occupied by the oil initially in place at p_i
+ PV occupied by the gas in the gas cap at p_i
= PV occupied by the remaining oil at p
+ PV occupied by the gas in the gas cap at p
+ PV occupied by the evolved solution gas at p
+ PV occupied by the net water influx at p
+ change in PV due to connate water expansion and
+ pore volume reduction due to rock expansion
+ PV occupied by the injected gas at p
+ PV occupied by the injected water at p

Eq. M3

Developing the MBE

Initial Conditions



Later Condition

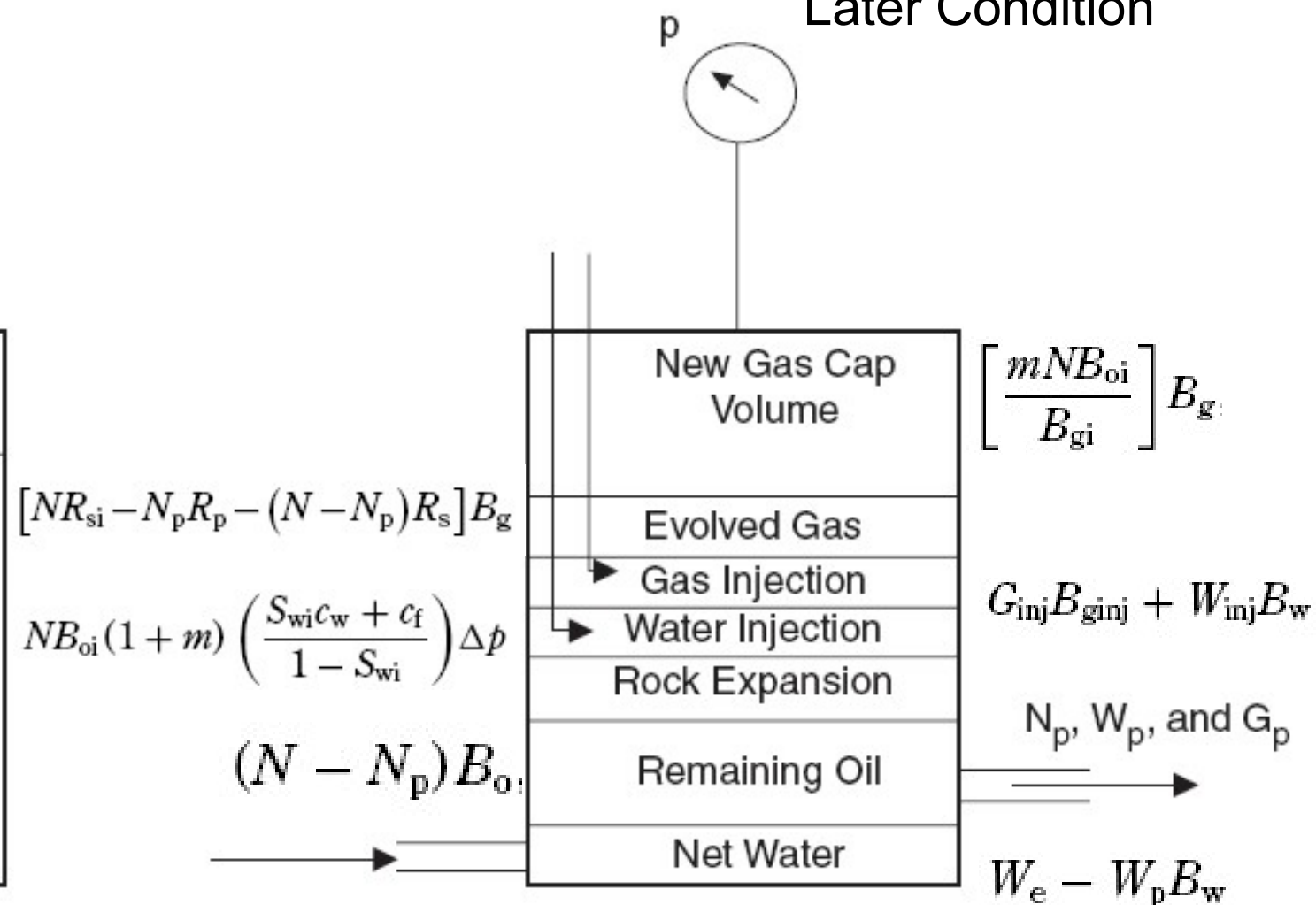


Figure M1: **Tank Model Concept** Source: Ahmed and Tarek (2005)

Development of Terms for MBE after production (at reservoir pressure, p)

1) HCPV occupied by the remaining oil = $(N - N_p)B_o$, bbl

Eq. M4

2) HCPV occupied by the gas cap: $\left[\frac{mNB_{oi}}{B_{gi}} \right] B_g$

Eq. M5

3) HCPV occupied by the evolved solution gas less gas produced: $[NR_{si} - N_p R_p - (N - N_p)R_s]B_g$

Eq. M6

4) PV occupied by the net water influx: $W_e - W_p B_w$

Eq. M7

5) PV occup. by the injected gas & water: $G_{inj}B_g + W_{inj}B_w$

Eq. M8

Development of Terms for GMBE

6) Connate water and rock expansion:

$$c = \frac{-1}{V} \frac{\partial V}{\partial p}$$

Reduction in Hydrocarbon Pore Volume (HCPV) =
Connate water and rock expansion

Eq. M9

$$-d(\text{HCPV}) = (c_w V_w + c_f V_p)(p_i - p) = (c_w V_p S_{wi} + c_f V_p)(p_i - p)$$

$$-d(\text{HCPV}) = (c_w S_{wi} + c_f)(V_p)(p_i - p) = (1 + m)NB_{oi} \left[\frac{c_w S_{wi} + c_f}{1 - S_{wi}} \right] \Delta p$$

Substituting Equations M1 and M2 with M4 through M9 in Equation M3 and rearranging gives the MBE represented volumetrically in reservoir barrels as:

$$NB_{oi} + NmB_{oi} - N[B_o + (R_{si} - R_s)B_g] + N_p[B_o + (R_{si} - R_s)B_g] + N_p(R_p - R_{si})B_g - \frac{NmB_{oi}B_g}{B_{gi}} = G_{inj}B_g + W_{inj}B_w + W_e - W_pB_w + (1 + m)NB_{oi} \left[\frac{c_w S_{wi} + c_f}{1 - S_{wi}} \right] \Delta p$$

Eq. M10

Forms of the General MBE

$$N = \left(N_p B_o + (G_p - N_p R_s) B_g - (W_e - W_p B_w) - G_{inj} B_{ginj} - W_{inj} B_w \right) / \left((B_o - B_{oi}) + (R_{si} - R_s) B_g + m B_{oi} \left[(B_g / B_{gi}) - 1 \right] + B_{oi} (1 + m) \left[(S_{wi} c_w + c_f) / (1 - S_{wi}) \right] \Delta p \right)$$

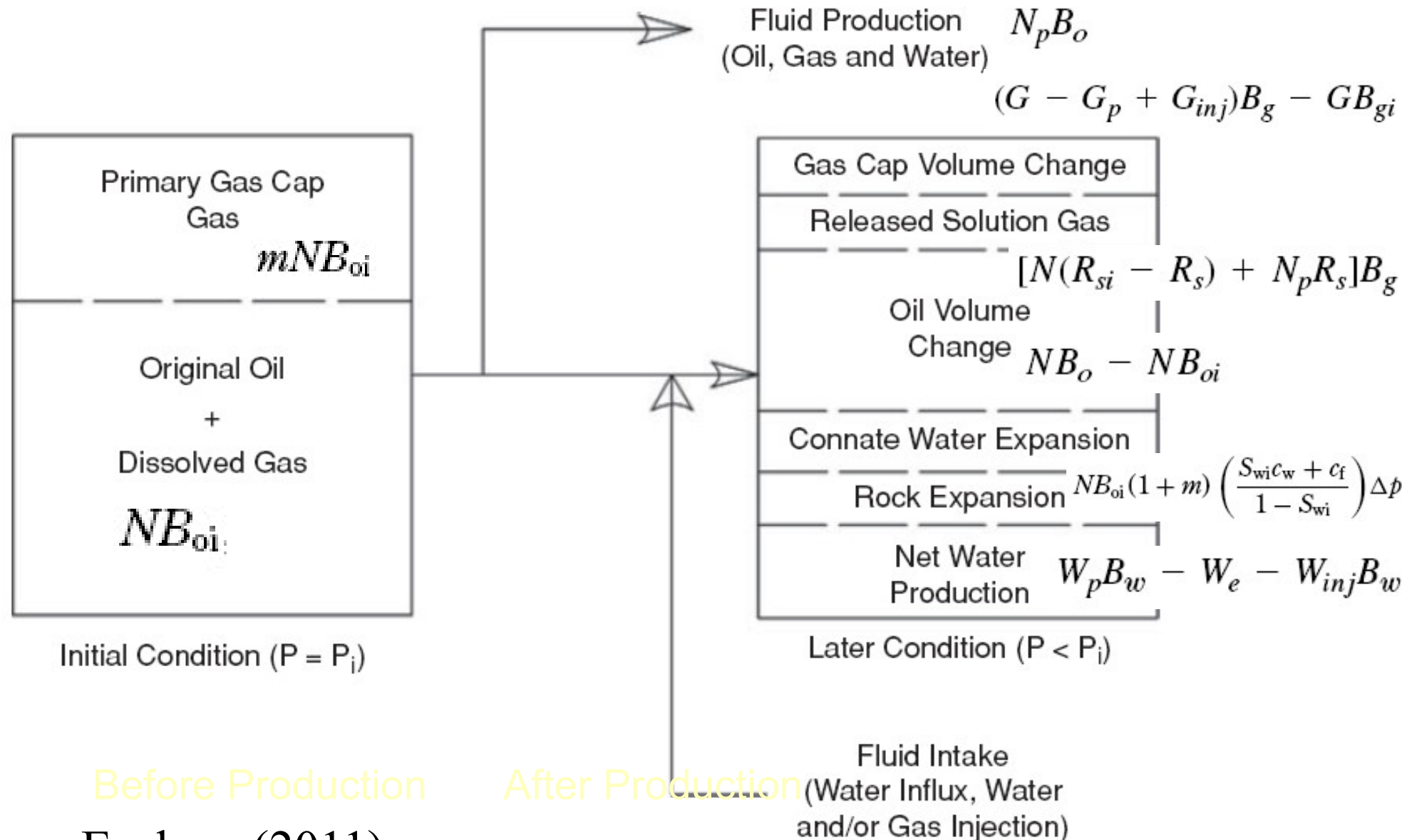
Eq. M11

- A more convenient form of the MBE can be arrived at (Eq. M13), by introducing the concept of the total (two-phase) formation volume factor B_t into Eq. M10. This oil PVT property is defined as: $B_t = B_o + (R_{si} - R_s) B_g$. Eq. M12 Thus, the MBE is also -

$$N_p [B_t + (R_p - R_{si}) B_g] + W_p B_w - G_{inj} B_g - W_{inj} B_w = N \left[(B_t - B_{ti}) + (1 + m) B_{ti} \left(\frac{c_w S_{wi} + c_f}{1 - S_{wi}} \right) \Delta p + \frac{m B_{ti}}{B_{gi}} (B_g - B_{gi}) \right] + W_e$$

Eq. M13

Tank Model Concept



Reservoir Drive Indices

- In a combination drive reservoir where all the driving mechanisms are simultaneously present, it is of a practical interest to determine the relative magnitude of each of the driving mechanisms and its contribution to the production. This objective can be achieved by rearranging the MBE in the following generalized form:

$$\begin{aligned} & \frac{N(B_t - B_{ti})}{A} + \frac{NmB_{ti}(B_g - B_{gi})/B_{gi}}{A} + \frac{W_e - W_p B_w}{A} \\ & + \frac{NB_{oi}(1+m) \left[\frac{c_w S_{wi} + c_f}{1 - S_{wi}} \right] (p_i - p)}{A} \\ & + \frac{W_{inj} B_{winj}}{A} + \frac{G_{inj} B_{ginj}}{A} = 1 \end{aligned}$$

$$DDI + SDI + WDI + EDI + WII + GII = 1.0$$

- with the parameter A as defined by:

$$A = N_p [B_t + (R_p - R_{si})B_g]$$

GMBE for dry and wet gas reservoirs

- The GMBE was developed for a saturated oil reservoir with a primary gas cap. However, it can be modified to apply to gas reservoirs (dry gas and wet gas).
- The GMBE can only be applied to gas condensate reservoirs, if it is assumed that the gas initially exists as a single phase (that means no condensation has occurred in the reservoir).
- Since at initial conditions, it is assumed there is no oil in the gas reservoir, $N = 0$ and $N_p = 0$. Also, it is assumed that there is no water or gas injection. Thus, for gas reservoirs, the GMBE (either Eq. (M11) or (M13) reduces to:

$$G_p B_g + W_p B_w = G B_{gi} \left(\frac{c_w S_{wi} + c_f}{1 - S_{wi}} \right) \Delta p + G(B_g - B_{gi}) + W_e$$

Eq. M14

Applications of Material Balance

- Valuable, analytical tool for evaluation of reservoir drive mechanisms
- As a precursor to reservoir simulation which is a more rigorous tool for reservoir evaluation. In practice, material balance methods can be of assistance in determining the type of reservoir model to construct, and assessing the quality of reservoir performance data for conducting reservoir modelling (SPE 90362, SPE 71819 , SPE 132781)
- Can provide an estimate of initial HC in place independent of geological interpretation. Thus, to verify volumetric estimates
- Can be used to predict or estimate reservoir pressure

Applications of Material Balance

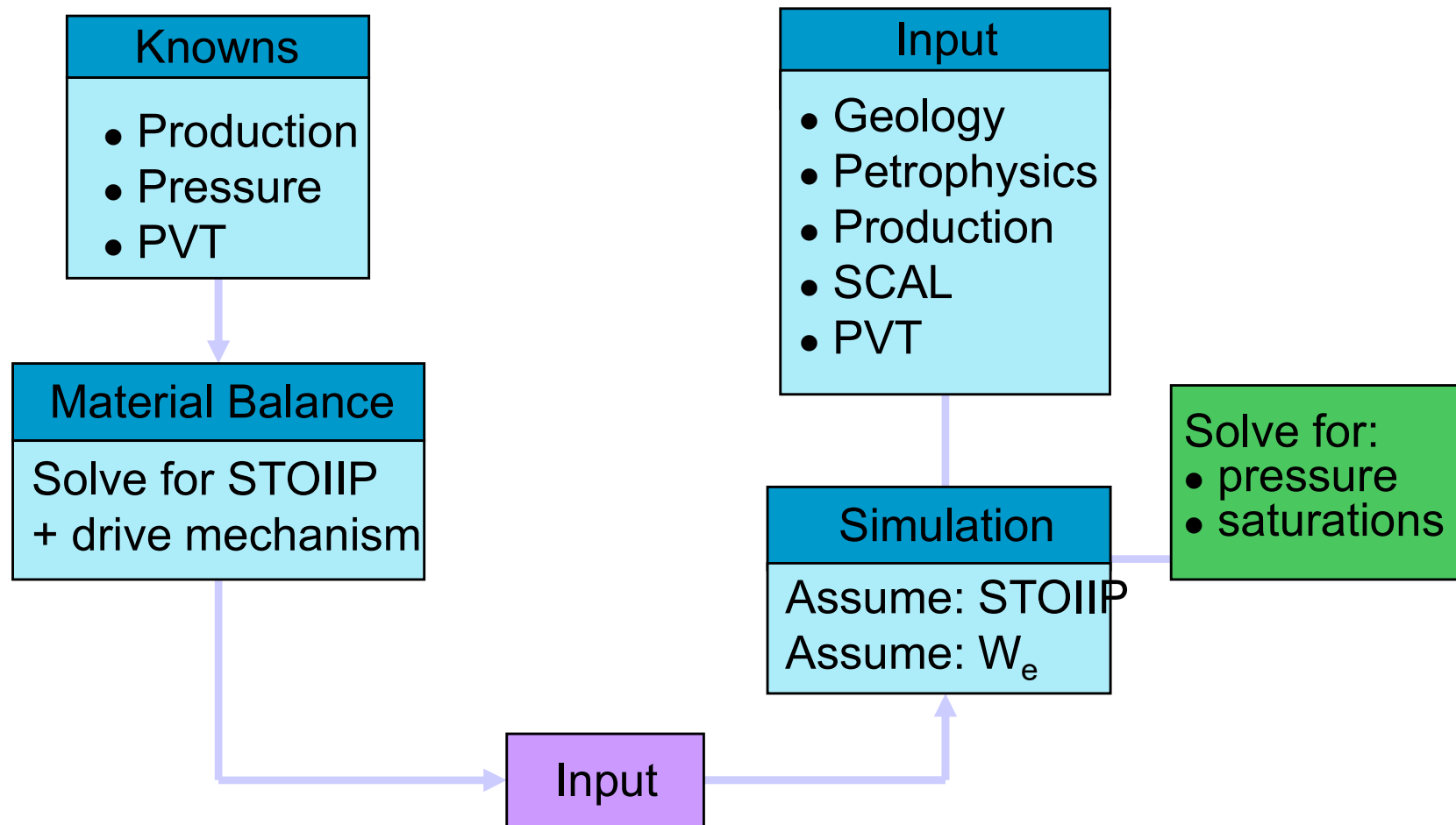
- Reservoir Characterization Tool
- Identify undrained hydrocarbons (Determination of the location of remaining oil) – SPE 108648
- Estimation of Present GOC & OWC
- Can be used as a forecasting tool *in certain situations*; complementary equations are usually required.
- Can be used to help evaluate operating strategies such as new wells, accelerated rate, compression;
- In some cases, it can be used to screen for enhanced recovery

History Matching the Pressure-Production data of individual wells

- Once the reservoir pressure declines below the bubble point pressure, it is essential to perform the following tasks:
 - Generate, the pseudo-relative permeability ratio k_{rg}/k_{ro} for the entire reservoir or for individual wells drainage area.
 - Assess the solution gas driving efficiency.
 - Examine the field gas–oil ratio as compared to the laboratory solution gas solubility R_s to define the bubble point pressure and critical gas saturation.

Material Balance / Numerical Simulation

History Matching Field Performance



THE MATERIAL BALANCE AS AN EQUATION OF A STRAIGHT LINE

The Straight-line MBE

- An insight into the general MBE, i.e., Equation M.11, may be gained by considering the physical significance of the following groups of terms from which it is comprised;
 - $N_p[B_o + (R_p - R_s)B_g]$, reservoir volume of cumulative oil & gas produced;
 - $[W_e - W_p B_w]$, net water influx that is retained in the reservoir;
 - $[G_{inj} B_{ginj} + W_{inj} B_w]$, cumulative fluid injection in the reservoir;
 - $[m B_{oi}(B_g/B_{gi} - 1)]$, net expansion of the gas cap that occurs with the production of N_p stock-tank barrels of oil (as expressed in bbl/STB of original oil-in place).
- There are essentially three unknowns in Equation M.11;
 - (1) the original oil-in-place N ,
 - (2) the cumulative water influx W_e , and
 - (3) the original size of the gas cap as compared to the oil zone size m .

The Straight-line MBE

- In developing a methodology for determining the above three unknowns, Havlena and Odeh (1963, 1964) expressed Equation M11 in the following form:

$$- F = N[E_o + mE_g + E_{f,w}] + (W_e + W_{inj}B_w + G_{inj}B_{ginj}) \quad \boxed{\text{Eq. M15}}$$

- in which the terms F , E_o , E_g , and $E_{f,w}$ are defined as follows
 - F : Underground Withdrawal, $F = N_p [B_o + (R_p - R_s)B_g] + W_p B_w$
 - E_o : expansion of oil and its originally dissolved gas,

$$E_o = (B_o - B_{oi}) + (R_{si} - R_s)B_g$$
 - E_g : expansion of the gas cap gas, $E_g = B_{oi} [(B_g / B_{gi}) - 1]$
 - $E_{f,w}$: expansion of the initial water and the reduction in the PV

$$E_{f,w} = (1 + m)B_{oi} \left[\frac{c_w S_{wi} + c_f}{1 - S_{wi}} \right] \Delta p$$

The Straight-line MBE

- Havlena and Odeh examined several cases of varying reservoir types with Equation M.15 and pointed out that the relationship can be rearranged in the form of a straight line.
- For example, in the case of a reservoir which has no initial gas cap (i.e., $m = 0$) or water influx (i.e., $W_e = 0$), and negligible formation and water compressibilities (i.e., c_f and $c_w = 0$), Equation M.15 reduces to: $F = NE_o$
- This expression suggests that a plot of the parameter F as a function of the oil expansion parameter E_o would yield a straight line with slope N and intercept equal to 0.
- The straight-line method requires the plotting of a variable group versus another variable group, with the variable group selection depending on the mechanism of production under which the reservoir is producing.

The Straight-line MBE

- The most important aspect of this method of solution is that it attaches significance to the sequence of the plotted points, the direction in which they plot, and to the shape of the resulting plot.
- The significance of the straight-line approach is that the sequence of plotting is important and if the plotted data deviates from this straight line there is some reason for it.
- This significant observation will provide the engineer with valuable information that can be used in determining the following unknowns: initial oil-in-place N ; size of the gas cap m ; water influx W_e ; driving mechanism; average reservoir pressure.

The Straight-line MBE

- Ahmed and McKinney (2005) presented six cases where the straight-line form of the MBE can be applied.
 - Case 1: Determination of N in volumetric undersaturated reservoirs
 - Case 2: Determination of N in volumetric saturated reservoirs
 - Case 3: Determination of N and m in gas cap drive reservoirs
 - Case 4: Determination of N and W_e in water drive reservoirs
 - Case 5: Determination of N , m , and W_e in combination drive reservoirs
 - Case 6: Determination of average reservoir pressure p_{avg}

The Straight line MBE – Case 1

Volumetric Undersaturated Oil Reservoirs

- For a volumetric and undersaturated reservoir, the conditions associated with driving mechanism are:
 - $W_e = 0$ since the reservoir is volumetric
 - $m = 0$ since the reservoir is undersaturated
 - $R_s = R_{si} = R_p$ since all produced gas is dissolved in the oil
- Equation M13 reduces to:
$$F = N(E_o + E_{f,w})$$
 M.15a
- Or:
$$N = \frac{F}{E_o + E_{f,w}}$$
 M.16
- Where:

$$F = N_p B_o + W_p B_w$$

$$E_o = B_o - B_{oi}$$

$$E_{f,w} = B_{oi} \left[\frac{c_w S_w + c_f}{1 - S_{wi}} \right] \Delta p$$

The Straight-line MBE – Case 1

Volumetric Undersaturated Oil Reservoirs

- When a new field is discovered, one of the first tasks of the reservoir engineer is to determine if the reservoir can be classified as a volumetric reservoir, i.e., $W_e = 0$.
- The classical approach of addressing this problem is to assemble all the necessary data (i.e., production, pressure, and PVT) that is required to evaluate the right-hand side of fluid and rock expansion term.
- The term $F/(E_o + E_{f,w})$ for each pressure and time observation is plotted versus cumulative production N_p or time. Dake (1994) suggested that such a plot can assume different shapes as shown in Fig. C1a. N.B.: *The downward trend in points on curve B as time progresses denotes the diminishing degree of energizing by the aquifer.*

The Straight-line MBE – Case 1

Volumetric Undersaturated Oil Reservoirs

- Figures

Fig. C1a

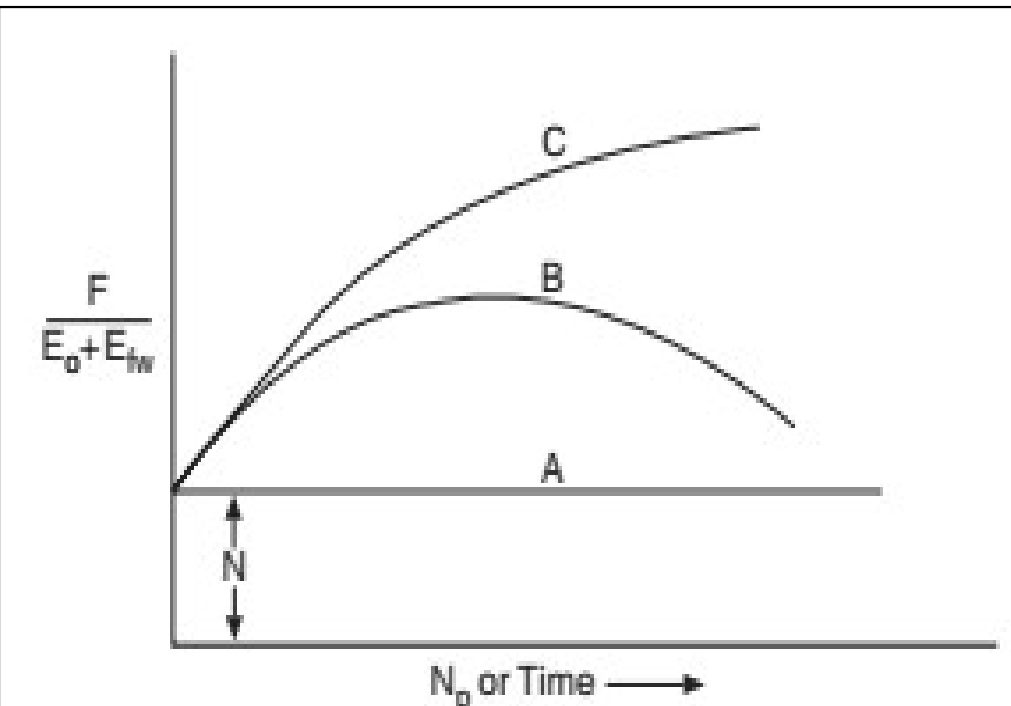
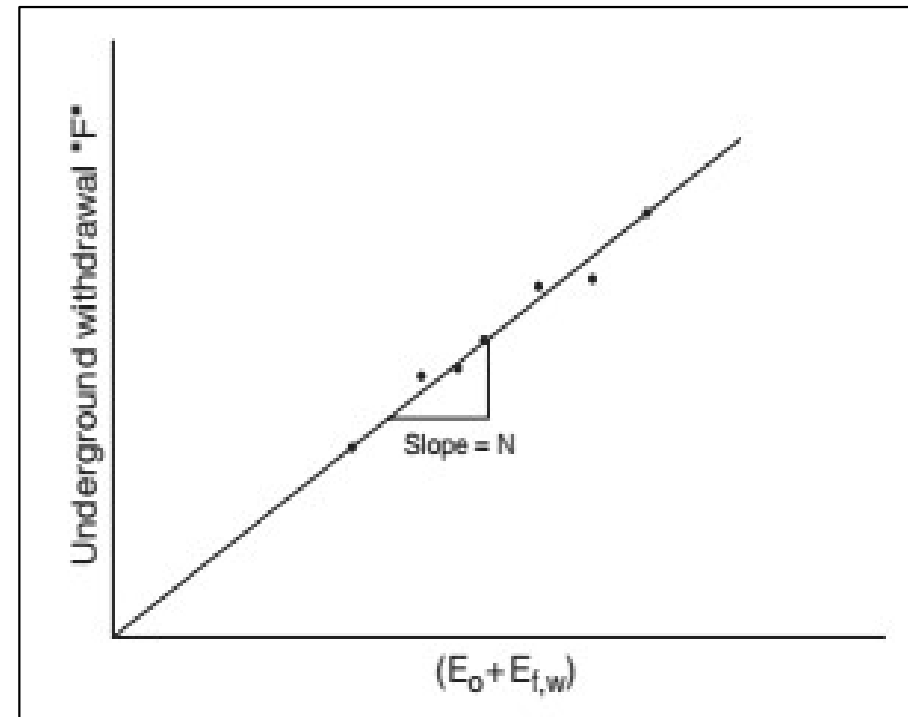


Fig. C1b



The Straight-line MBE – Case 1

Volumetric Undersaturated Oil Reservoirs

- Dake (1994) pointed out that in water drive reservoirs, the shape of the curve, i.e., $F/(E_o + E_{f,w})$ versus time, is highly rate dependent. For instance, if the reservoir is producing at a higher rate than the water influx rate, the calculated values of $F/(E_o + E_{f,w})$ will dip downward, revealing a lack of energizing by the aquifer, whereas if the rate is decreased the reverse happens and the points are elevated.
- A linear plot of the underground withdrawal F vs. $(E_o + E_{f,w})$ indicates that the field is producing under volumetric performance, i.e., no water influx, and strictly by pressure depletion and fluid expansion. On the other hand, a non-linear plot indicates that the reservoir should be characterized as a water drive reservoir

Example Application

Undersaturated Reservoir

- Problem: A small reservoir is being produced by depletion above the bubble point pressure with no water drive. Use material balance to calculate the oil produced from initial conditions to bubble point conditions considering (a) only the expansion of hydrocarbons and (b) expansion of hydrocarbons and formation water.
- Given:
 - $N = 8 \text{ MMstb}$;
 - $P_i = 3,000 \text{ psia}$;
 - $P_b = 2,500 \text{ psia}$;
 - $c_w = 3 \times 10^{-6} \text{ psi}^{-1}$;
 - $S_{wc} = 0.30 \text{ PV}$
 - $B_{oi} = 1.3 \text{ rb/stb}$
 - $B_o = 1.45 \text{ rb/stb}$
 - $c_f = 5 \times 10^{-6} \text{ psi}^{-1}$

Example Application

Undersaturated Reservoir

$$0 \quad 0 \quad 0 \quad 0 \quad 0$$

$$GB_{gci} + NB_{oi} = (G + G_i - G_{pc})B_{gc} + (N - N_p)B_o + (W_i - W_p)B_w + [NR_{si} - (N - N_p)R_s - G_{ps}]B_{gs} + (c_f + c_w S_{wi}) \left(\frac{NB_{oi}}{(1 - S_{wi})} \right) \Delta p + W_e$$

$$(a) \quad NB_{oi} = (N - N_p)B_o \quad \text{or} \quad N_p = \frac{N(B_o - B_{oi})}{B_o}$$

$$N_p = \frac{8E6(1.45 - 1.3)}{1.45} = 827586 \text{ stb}$$

$$(b) \quad NB_{oi} = (N - N_p)B_o + (c_w S_{wc}) \left[\frac{NB_{oi}}{(1 - S_{wc})} \right] \Delta p$$

$$\text{or } N_p = \frac{N(B_o - B_{oi})}{B_o} + \frac{(c_w S_{wc})(NB_{oi})\Delta p}{B_o(1 - S_{wc})}$$

$$N_p = \frac{8E6(1.45 - 1.30)}{1.45} + \frac{(3E - 6(0.3))(8E6(1.3))(500)}{1.45(1 - 0.3)}$$

$$= 827586 + 4610 = 832196 \text{ stb}$$

Volumetric Depletion Fields

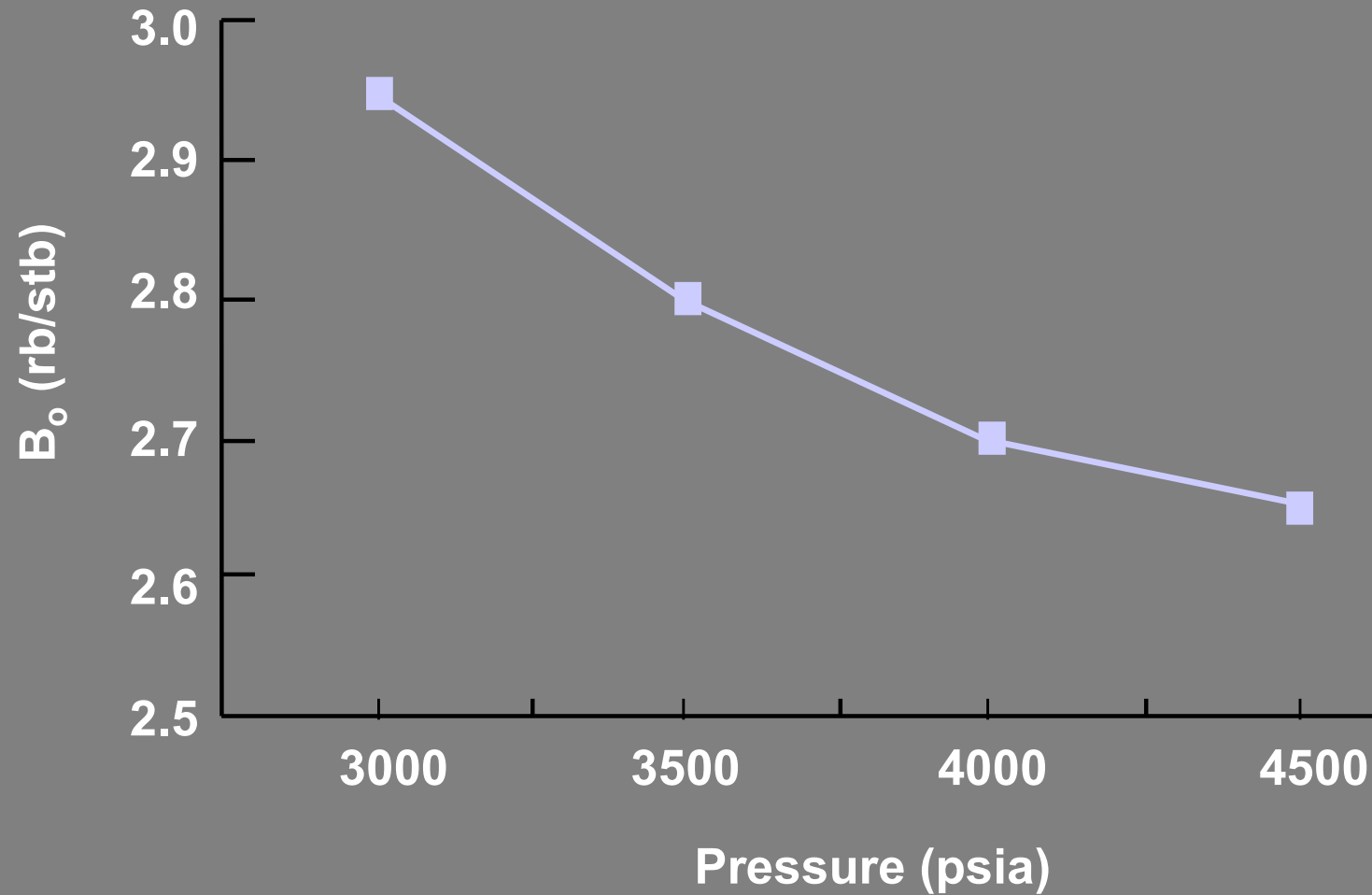
Volatile Oil Reservoir Example

Problem: A small, but complex reservoir is being produced by depletion above the bubble point pressure. It contains volatile oil with a severe degree of heterogeneity, although average permeability is less than 10 mD. No account need be taken of any natural water influx since this is a lens type accumulation completely devoid of any aquifer support. Use material balance to determine the effective STOIIP contributing to the pressure / production history.

Given:

STOIIP	=	95 Mstb (volumetric estimate)
P_i	=	4250 psia
P_b	=	2900 psia
B_{oi}	=	2.654 rb/stb
B_w	=	1.020 rb/stb
c_w	=	$3 \times 10^{-6} \text{psi}^{-1}$
c_f	=	$5 \times 10^{-6} \text{psi}^{-1}$
S_{wc}	=	0.30 PV

FVF for Volatile Reservoir



Volatile Oil Reservoir

Pressure & Cumulative Production Statistics

Well No.	Datum Pressure (psia)	Δp (psi)	Cumulative Oil (stb)	Produced Water (stb)
1	3511	739	239830	2730
2			15140	150
3	3546	704	242330	2780
4	3522	728	140900	1740
5			17170	1630
6	3507	743	204310	2100
7	3520	730	277730	2500
Totals (stb)			1137410	13630

$$\bar{p} = 3521 \text{ psia} \quad \Delta p = 729 \text{ psia}$$

Governing Equations

$$\begin{aligned}
 \cancel{G} \cancel{B}_{gci}^0 + N B_{oi} = & (\cancel{G} + \cancel{G}_i - \cancel{G}_p) \cancel{B}_g^0 + (N - N_p) B_o + (\cancel{W}_i - \cancel{W}_p) \cancel{B}_w^0 + \cancel{G}_s \cancel{B}_{gs}^0 \\
 & + \cancel{W}_e^0 + (c_f + c_w S_{wi}) \left(\frac{N B_{oi}}{1 - S_{wi}} \right) \Delta p
 \end{aligned}$$

$$\text{Now } c_o = \frac{B_o - B_{oi}}{B_{oi} \Delta p} \qquad c_{eff} = \frac{c_o S_{oi} + c_w S_{wi} + c_f}{1 - S_{wi}}$$

$$N_p B_o + W_p B_w = N B_{oi} c_{eff} \Delta p$$

$$N B_{oi} = \frac{N_p B_o + W_p B_w}{c_{eff} \Delta p}$$

Volatile Oil Reservoir

Problem Solution

$$c_o = \frac{B_o - B_{oi}}{B_{oi} \Delta p} = \frac{2.790 - 2.654}{2.654 (729)} = 70.29 \times 10^6 \text{ psi}^{-1}$$

$$c_{\text{eff}} = \left(\frac{[70.29 (0.7) + 3 (0.3) + 5]}{0.7} \right) \times 10^6 = 78.72 \times 10^6 \text{ psi}^{-1}$$

$$[NB_{oi}]_{\text{MB}} = \frac{N_p B_o + W_p B_w}{c_{\text{eff}} \Delta p} = \frac{1137410 (2.79) + 13630 (1.02)}{78.72 \times 10^6 (729)} = 55.54 \text{ Mrb}$$

$$[NB_{oi}]_{\text{VOL}} = 95 (2.654) = 252.13 \text{ Mrb}$$

$$\frac{[NB_{oi}]_{\text{MB}}}{[NB_{oi}]_{\text{VOL}}} = \frac{55.54}{252.13} = 0.22$$

The Straight-line MBE – Case 2

Volumetric Saturated Oil Reservoirs

- An oil reservoir that originally exists at its bubble point pressure is referred to as a “saturated oil reservoir.” The main driving mechanism in this type of reservoir results from the liberation and expansion of the solution gas as the pressure drops below the bubble point pressure.
- The only unknown in a volumetric saturated oil reservoir is the initial oil-in-place N . Normally, the water and rock expansion term $E_{f,w}$ is negligible in comparison to the expansion of solution gas; however, it is recommended to include the term in the calculations.
- Equation M.15a is used:

$$F = N(E_o + E_{f,w})$$

M.15a

The Straight-line MBE – Case 2

Volumetric Saturated Oil Reservoirs

- However, the parameters F and E_o that constitute the above expression are given in an expanded form to reflect the reservoir condition as the pressure drops below the bubble point. The underground withdrawal F and the expansion term $(E_o + E_{f,w})$ are defined by:

$$F \text{ in terms of } B_o \quad F = N_p[B_o + (R_p - R_s)B_g] + W_p B_w$$

$$\text{or equivalently in terms of } B_t \quad F = N_p[B_t + (R_p - R_{si})B_g] + W_p B_w$$

$$E_o \text{ in terms of } B_o \quad E_o = (B_o - B_{oi}) + (R_{si} - R_s)B_g$$

$$\text{or equivalently in terms of } B_t \quad E_o = B_t - B_{ti}$$

and:

$$E_{f,w} = B_{oi} \left[\frac{c_w S_w + c_f}{1 - S_{wi}} \right] \Delta p$$

The Straight-line MBE – Case 2

Volumetric Saturated Oil Reservoirs

- Equation M.15 indicates that a plot of the underground withdrawal F , evaluated by using the actual reservoir production data, as a function of the fluid expansion term $(E_o + E_{f,w})$ should result in a straight line going through the origin with a slope of N .
- The above interpretation technique is useful in that, if a simple linear relationship such as Equation M.15 is expected for a reservoir and yet the actual plot turns out to be non-linear, then this deviation can itself be diagnostic in determining the actual drive mechanisms in the reservoir.
- For instance, Equation M.15 may turn out to be nonlinear because there is an unsuspected water influx into the reservoir, helping to maintain the pressure.

The Straight line MBE – Case 3

Gas Cap Drive Reservoirs

- For a reservoir in which the expansion of the gas cap gas is the predominant driving mechanism, the effect of water and pore compressibilities as a contributing driving mechanism can be considered negligible as compared to that of the high compressibility of the gas.
- However, whenever a gas cap is present or its size is to be determined, an exceptional degree of accuracy of pressure data is required. The specific problem with reservoir pressure is that the underlying oil zone in a gas cap drive reservoir exists initially near its bubble point pressure.
- Therefore, the flowing pressures are obviously below the bubble point pressure, which exacerbates the difficulty in conventional pressure buildup interpretation to determine average reservoir pressure.

The Straight line MBE – Case 3

Gas Cap Drive Reservoirs

- Assuming that there is no natural water influx or it is negligible (i.e., $W_e = 0$), the Havlena and Odeh material balance can be expressed as: $F = N[E_o + mE_g]$ [M.17]
- in which the variables F , E_o , and E_g are given by:

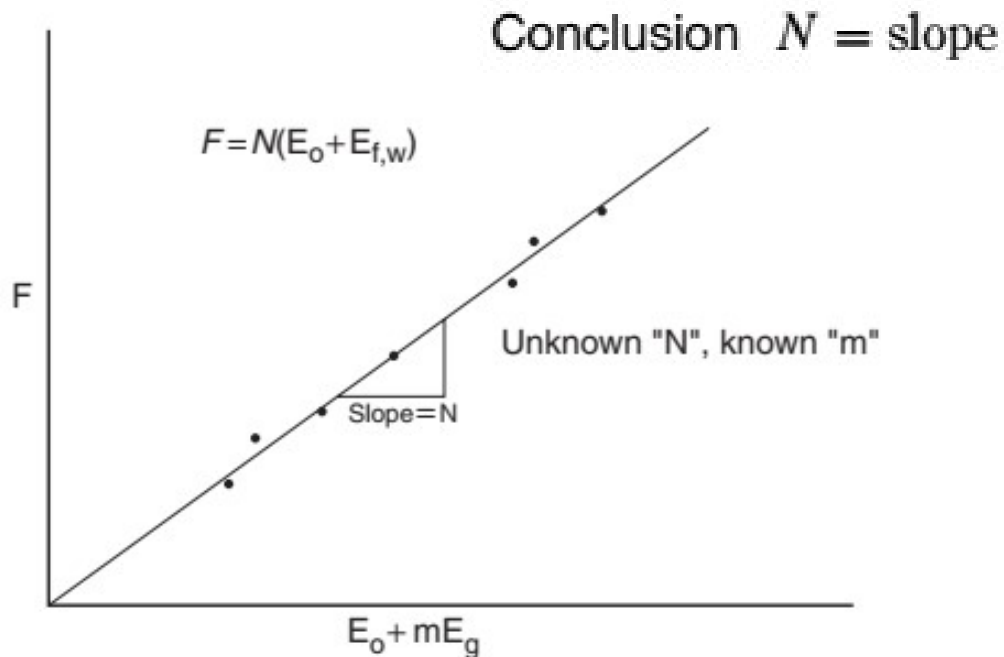
$F = N_p[B_o + (R_p - R_s)B_g] + W_pB_w$ $= N_p[B_t + (R_p - R_{si})B_g] + W_pB_w$	$E_o = (B_o - B_{oi}) + (R_{si} - R_s)B_g$ $= B_t - B_{ti}$	$E_g = B_{oi}[(B_g/B_{gi}) - 1]$
--	---	----------------------------------

- The methodology in which Eq. M.17 can be used depends on the number of unknowns in the equation. There are three possible unknowns in Eq. M.17. These are:
 - (1) N is unknown, m is known;
 - (2) m is unknown, N is known;
 - (3) N and m are unknown.
- The practical use of Eq M.17 in determining the three possible unknowns is presented below.

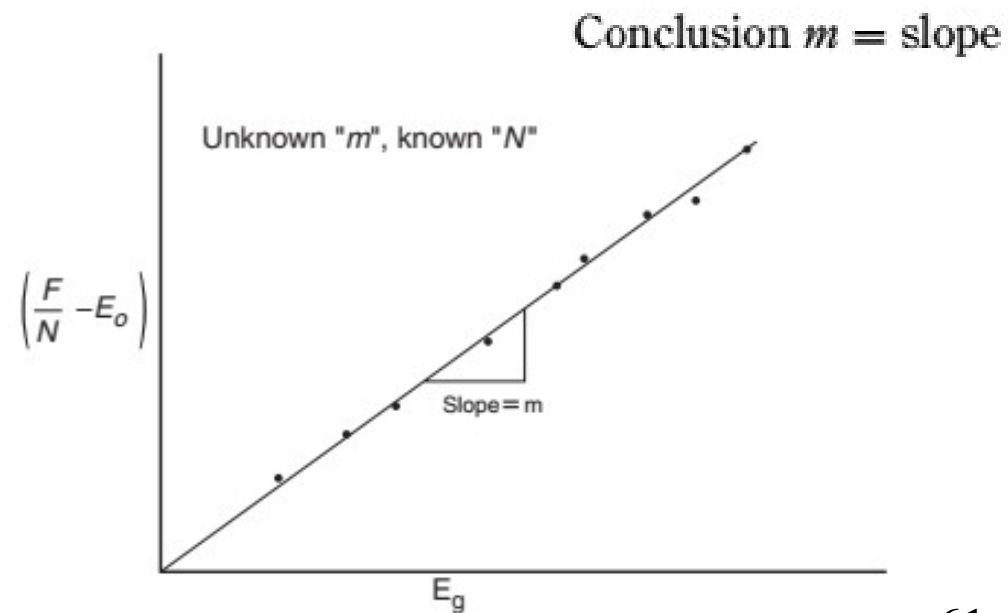
The Straight line MBE – Case 3

Gas Cap Drive Reservoirs

- **Unknown N , known m** Eq. M.17 indicates that a plot of F vs $(E_o + mE_g)$ on a Cartesian scale would produce a straight line through the origin with a slope of N , as shown in below. In making the plot, the underground withdrawal F can be calculated at various times as a function of the production terms M_p and R_p .



- **Unknown m , known N** Eq. M.17 can be rearranged as an equation of straight line, to give: $\left(\frac{F}{N} - E_o\right) = mE_g$
- This relationship shows that a plot of the term $(F/N - E_o)$ vs. E_g would produce a straight line through the origin with a slope of m



The Straight-line MBE – Case 3

Gas Cap Drive Reservoirs

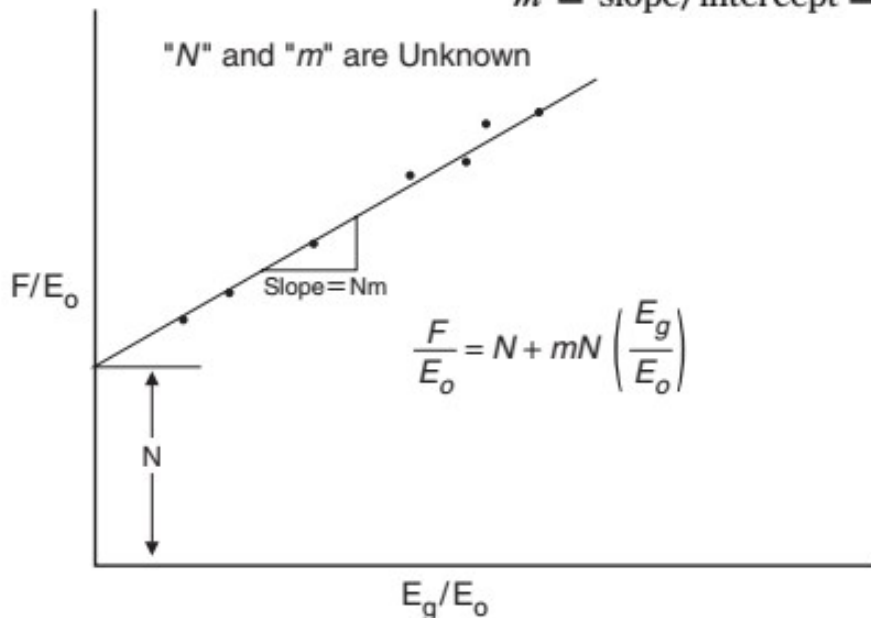
- ***N and m are unknown*** If there is uncertainty in both the values of N and m , Eq. M.17 can be re-expressed as:

$$\frac{F}{E_o} = N + mN \left(\frac{E_g}{E_o} \right)$$
- A plot of F/E_o versus E_g/E_o should then be linear with intercept N and slope mN .

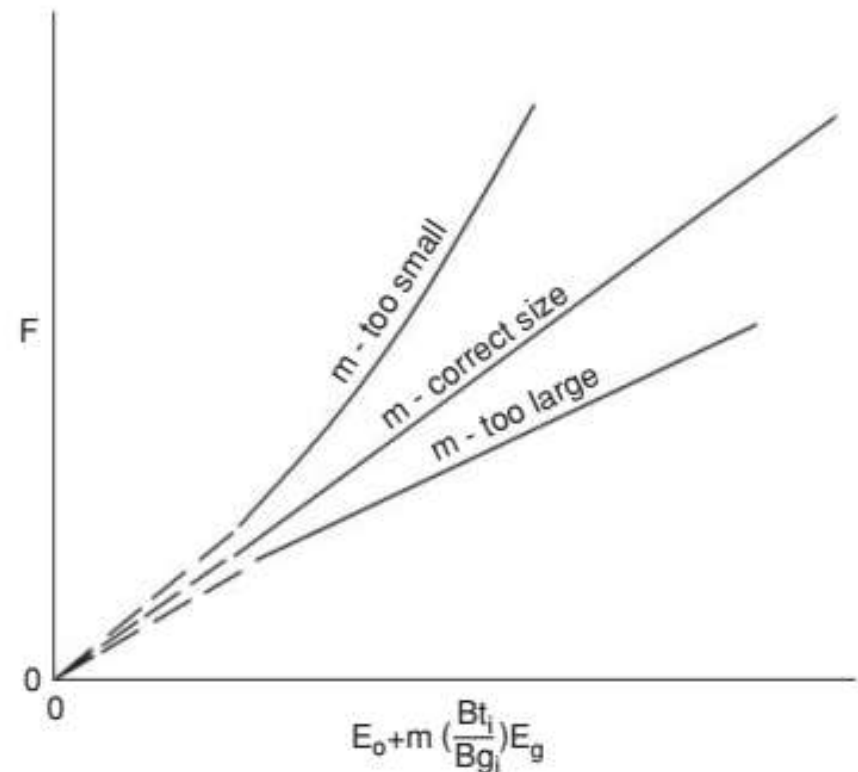
Conclusions N = intercept

mN = slope

m = slope/intercept = slope/ N



- ***N and m are unknown*** If m is known, a plot of F vs. $[E_o + m(B_{ti}/B_{gi})E_g]$ will give a straight line passing through the origin with the slope equal to N . If m is too small, the line will curve upward. If m is too large, the line will curve downward.



The Straight line MBE – Case 4

Water Drive Reservoirs

- In a water drive reservoir, identifying the type of the aquifer and characterizing its properties are perhaps the most challenging tasks involved in conducting a reservoir engineering study. Yet, without an accurate description of the aquifer, future reservoir performance and management cannot be properly evaluated.
- The term $E_{f,w}$ can frequently be neglected in water drive reservoirs. This is not only for the usual reason that the water and pore compressibilities are small, but also because a water influx helps to maintain the reservoir pressure and, therefore, the Δp appearing in the $E_{f,w}$ term is reduced, or:

$$F = NE_o + W_e \quad [M.18]$$

- The above equation hold where no initial gas cap exists.

The Straight line MBE – Case 4

Water Drive Reservoirs

- In attempting to use Eq. M.18 to match the production and pressure history of a reservoir, the greatest uncertainty is always the determination of the water influx W_e .
- In fact, in order to calculate the water influx the engineer is confronted with what is inherently the greatest uncertainty in the whole subject of reservoir engineering. The reason is that the calculation of W_e requires a mathematical model which itself relies on knowledge of aquifer properties. These, however, are seldom measured since wells are not deliberately drilled into the aquifer to obtain such information.
- For a water drive reservoir with no gas cap, Eq. M18 can be rearranged and expressed as:

$$\frac{F}{E_o} = N + \frac{W_e}{E_o} \quad \boxed{\text{M.19}}$$

The Straight-line MBE – Case 4

Water Drive Reservoirs (VEH Unsteady State Model)

- The van Everdingen and Hurst unsteady-state model is given by:

$$W_e = B \Sigma \Delta p W_{eD} \quad (\text{M.20}) \quad \text{with } B = 1.119 \phi c_t r_e^2 h f$$
- Van Everdingen and Hurst presented the dimensionless water influx W_{eD} as a function of the dimensionless time t_D and dimensionless radius r_D that are given by:

$$t_D = 6.328 \times 10^{-3} \frac{kt}{\phi \mu_w c_t r_e^2}$$

$$c_t = c_w + c_f$$

$$r_D = \frac{r_a}{r_e}$$

where:

t = time, days

k = permeability of the *aquifer*, md

ϕ = porosity of the *aquifer*

μ_w = viscosity of water in the *aquifer*, cp

r_a = radius of the *aquifer*, ft

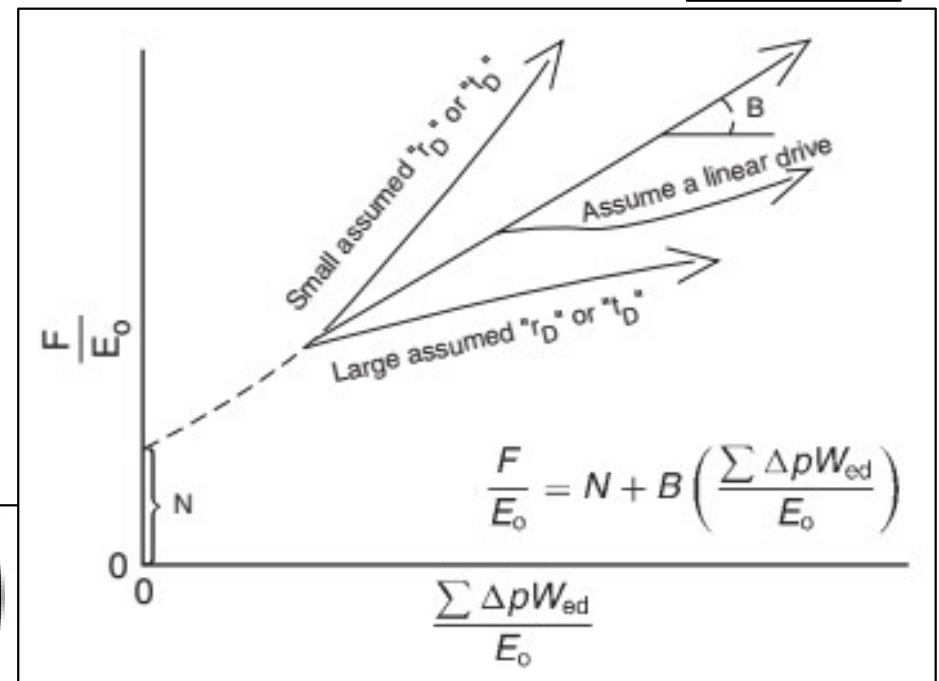
r_e = radius of the reservoir, ft

c_w = compressibility of the water, psi^{-1}

- Combining the above equations with gives

$$\frac{F}{E_o} = N + \frac{W_e}{E_o}$$

$$\frac{F}{E_o} = N + B \left(\frac{\Sigma \Delta p W_{eD}}{E_o} \right)$$



The Straight-line MBE – Case 4

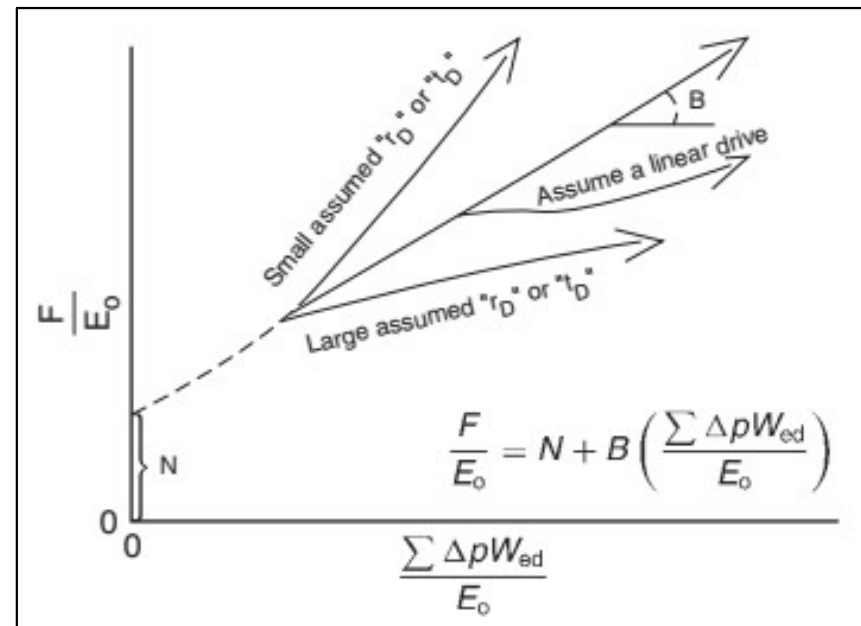
Water Drive Reservoirs (VEH Unsteady State Model)

- The proper methodology of solving the above linear relationship is summarized in the following steps.
 - Step 1. From the field past production and pressure history, calculate the underground withdrawal F and oil expansion E_o .
 - Step 2. Assume an aquifer configuration, i.e., linear or radial.
 - Step 3. Assume the aquifer radius r_a and calculate the dimensionless radius r_D .
 - Step 4. Plot F/E_o vs. $(\Sigma \Delta p W_{eD})/E_o$ on a Cartesian scale. If the assumed aquifer parameters are correct, the plot will be a straight line with N being the intercept and the water influx constant B being the slope. It should be noted that four other different plots might result.

The Straight-line MBE – Case 4

Water Drive Reservoirs (VEH Unsteady State Model)

- The four other different plots might result are:
 - 1) Complete random scatter of the individual points, which indicates that the calculation and/or the basic data are in error.
 - 2) A systematically upward-curved line, which suggests that the assumed aquifer radius (or dimensionless radius) is too small.
 - 3) A systematically downward-curved the indicating that the selected aquifer radius (or dimensionless radius) is too large.
 - 4) An S-shaped curve indicates that a better fit could be obtained if a linear water influx is assumed.



The Straight-line MBE – Case 4

Water Drive Reservoirs (VEH Unsteady State Model)

- It should be noted that in many large fields, an infinite linear water drive satisfactorily describes the production–pressure behavior. For a unit pressure drop, the cumulative water influx in an infinite linear case is simply proportional to $\sqrt{t}$ and does not require the estimation of the dimensionless time t_D .
- Thus, the Van Everdingen and Hurst dimensionless water influx W_{eD} in Equation M20 is replaced by the square root of time, to give:

$$W_w = B \sum \left[\Delta p_n \sqrt{(t - t_n)} \right]$$

- Therefore, the linear form of the MBE can be expressed as:

$$\frac{F}{E_o} = N + B \left(\frac{\sum \Delta p_n \sqrt{t - t_n}}{E_o} \right)$$

The Straight line MBE – Case 5

Combination Drive Reservoirs

- This relatively complicated case involves the determination of the following three unknowns:
 - (1) initial oil-in-place N ;
 - (2) size of the gas cap m ;
 - (3) Water influx W_e .

- Havlena and Odeh differentiated Eq. M14 with respect to pressure and rearranged the resulting equation to eliminate m , to give:

$$\frac{FE_g' - F'E_g}{E_oE_g' - E_o'E_g} = N + \frac{W_eE_g' - W_e'E_g}{E_oE_g' - E_o'E_g} \quad \boxed{\text{M21}}$$

- in which the reversed primes denote derivatives with respect to pressure. That is:

$$E_g' = \frac{\partial E_g}{\partial p} = \left(\frac{B_{oi}}{B_{gi}} \right) \frac{\partial B_g}{\partial p} \approx \left(\frac{B_{oi}}{B_{gi}} \right) \frac{\Delta B_g}{\Delta p}$$

$$E_o' = \frac{\partial E_o}{\partial p} = \frac{\partial B_t}{\partial p} \approx \frac{\Delta B_t}{\Delta p}$$

$$F' = \frac{\partial F}{\partial p} \approx \frac{\Delta F}{\Delta p}$$

$$W_e' = \frac{\partial W_e}{\partial p} \approx \frac{\Delta W_e}{\Delta p}$$

The Straight line MBE – Case 5

Combination Drive Reservoirs

- A plot of the left-hand side of Eq. M21 versus the second term on the right for a selected aquifer model should, if the choice is correct, provide a straight line with unit slope whose intercept on the ordinate gives the initial oil-in-place N . After having correctly determined N and W_e , Eq. M18 can be solved directly for m , to give:

$$m = \frac{F - NE_o - W_e}{NE_g}$$

- Note that all the above derivatives can be evaluated numerically using one of the finite difference techniques; e.g., forward, backward, or central difference formula.

The Straight line MBE – Case 6

Average reservoir pressure

- To gain any understanding of the behavior of a reservoir with free gas, e.g., solution gas drive or gas cap drive, it is essential that every effort be made to determine reservoir pressures with accuracy.
- In the absence of reliable pressure data, the MBE can be used to estimate average reservoir pressure if accurate values of m and N are available from volumetric calculations.
- It is determined iteratively using the following steps using the Case 3 Scenario (Gas Cap Drive Reservoirs):

$$F = N [E_o + mE_g]$$

M17

The Straight line MBE – Case 6

Average reservoir pressure

- Step 1. Select the time at which the average reservoir pressure is to be determined and obtain the corresponding production data, i.e., N_p , G_p , and R_p .
- Step 2. Assume several average reservoir pressure values and determine the left-hand side F of Equation M16 at each assumed pressure. That is:
$$F = N_p[B_o + (R_p - R_s)B_g] + W_pB_w$$
- Step 3. Using the same assumed average reservoir pressure values of step 2, calculate the right-hand side of Eq. M16.
- Step 4. Plot the left- and right-hand sides of the MBE, as calculated in steps 2 and 3, on Cartesian paper as a function of assumed average pressure. The point of intersection gives the average reservoir pressure that corresponds to the selected time of step 1.
- Step 5. Repeat steps 1 through 4 to estimate reservoir pressure at each selected depletion time.

Assignment

- 1) Solve Questions 1, 3, 5 - 7 on Pages 324 & 325 Advanced Reservoir Engineering (Ahmed & McKinney, 2005)

End of Lecture 1

Gas Reservoirs

– Lecture 2

(Chapter 8: Main Text)

Introduction

- Gas reservoirs are classified as dry, wet or condensate depending on path of depletion (in terms of pressure & temp.) relative to the phase envelope of the reservoir fluids.
- For dry gas reservoirs, no liquids are formed either in the reservoir or production facilities because the path of depletion is completely outside the phase envelope (Figure G.1).
- For wet gas reservoirs, no liquids are condensed within the reservoir, but liquids may condense in the production facilities because temperature and pressure conditions in the production facilities may fall within the two-phase region (Figure G.1).
- Retrograde gas condensate reservoirs may initially exist as single-phase fluid systems but condense liquid phases in the reservoir as pressure is reduced by production (Figure G.2).

Introduction

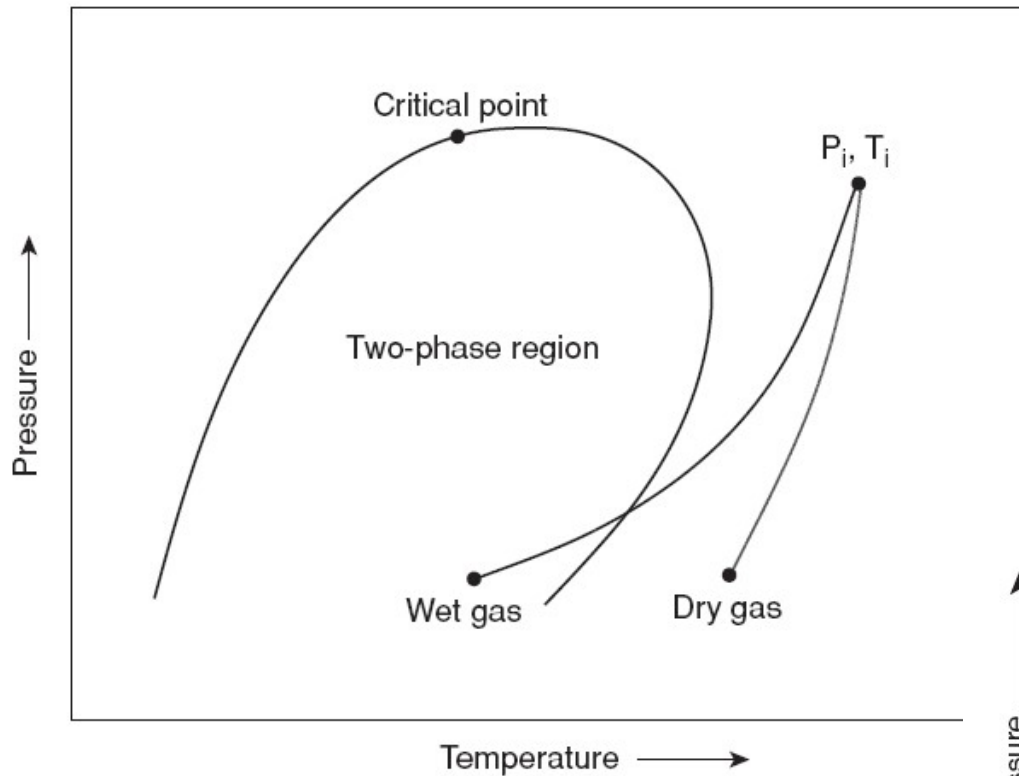


Figure G.1 Phase diagram showing dry gas and wet gas behaviour based on the phase envelope

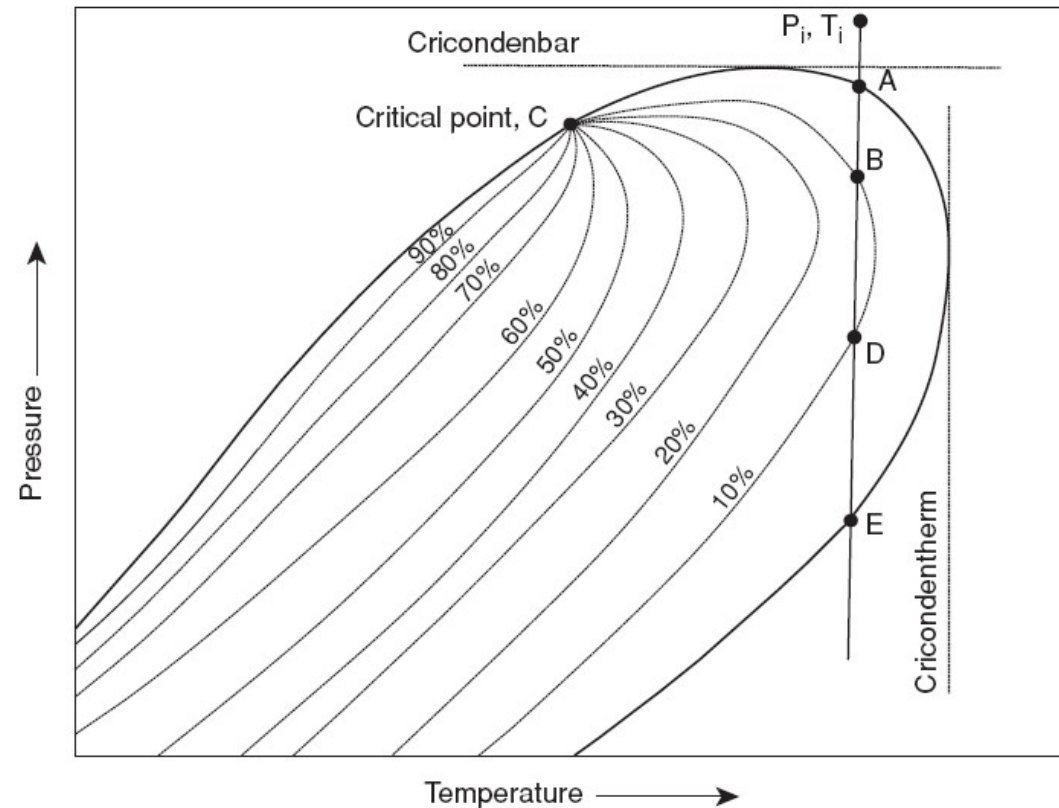


Figure G.2 Phase diagram showing retrograde behaviour

Introduction

- Gas reservoirs are further classified as **volumetric** or **non-volumetric** reservoirs.
 - Volumetric gas reservoirs are defined as completely isolated, closed systems with approximately constant hydrocarbon pore volumes. Volumetric gas reservoirs are presumed not to gain significant pressure support or fluid influx from outside sources, such as water influx from aquifers or neighbouring shale (non-reservoir) layers.
 - Non-volumetric gas reservoirs exhibit evidence of pressure support or influx of fluids (mostly water) from outside sources, such as aquifers or neighbouring shale intervals. A special group of gas reservoirs are classified as overpressured gas reservoirs. These reservoirs are sometimes called geopressured or abnormally pressured gas reservoirs. Geopressured gas reservoirs have pressure gradients that are sometimes approximately twice as high as those of normally pressured gas reservoirs.

Volumetric Calculations for Dry Gas Reservoirs

- Original Gas in Place in scf: $G = \frac{7758 \times A \times h \times \phi \times (1 - S_{wi})}{B_{gi}}$

where $B_{gi} = \frac{0.005021 z_i T_R}{p_i}$ (RB/scf) or $B_g = 0.02827 \frac{ZT}{p}$ ft³/scf

- At any intermediate pressure, the cumulative volume of gas produced from the reservoir is the difference between the OGIP and the volume of gas remaining in the reservoir. This is represented as: $G_p = G - G_r$ (8.3)
- In Eq. (8.3), G_p is the cumulative volume of produced gas, and G_r is the volume of gas remaining in the reservoir at the intermediate pressure. From Eq. (8.1), $G_r = \frac{7758 \times A \times h \times \phi \times (1 - S_{wi})}{B_g}$
- At an abandonment pressure, p_a , Eq. (8.3) becomes: $G_p = G - G_a$
- Thus, $G_p = 7758 \times A \times h \times \phi \times (1 - S_{wi}) \left[\frac{1}{B_{gi}} - \frac{1}{B_{ga}} \right]$

Volumetric Calculations for Dry Gas Reservoirs

- The ultimate recovery factor, for a volumetric gas reservoir is defined as:

$$R = \frac{G - G_a}{G} = \frac{G_p}{G} \quad (8.7)$$

- Substituting Eqs. (8.1) and (8.6) into Eq. (8.7), the ultimate gas recovery factor, is:

$$R = 1 - \frac{B_{gi}}{B_{ga}} = 1 - \frac{p_a z_i}{p_i z_a}$$

- Note that the basic assumption in developing the equations for a volumetric dry gas reservoir is that initial water saturation does not change over the productive life of the reservoir. Gas recovery factors for volumetric dry gas reservoirs are usually high, approaching 90% in many cases. Even at low reservoir pressures, recovery from volumetric gas reservoirs can be improved by utilizing compressors to reduce well head pressures. However, it is important to be aware that there are few true volumetric gas reservoirs.

Volumetric Calculations for Wet Gas and Retrograde Gas Condensate Reservoirs

- The original gas in place for a wet gas reservoir can be calculated with the same equations as for a dry gas reservoir.
- For a retrograde condensate reservoir, the pressure of a retrograde gas condensate reservoir is below its dew point pressure, volumetric calculation of in-place volumes and reserves will be in error since condensation must have taken place in the reservoir. Liquid condensation changes the volume and composition of gas remaining in the reservoir. This affects the uniformity of the composition of gas in the reservoir, which is the basic assumption in volumetric calculations.
- The best approach for estimating reserves for condensate reservoirs in this condition is to conduct thorough constant volume depletion (CVD) experiments with fluid samples obtained from the reservoir, and by performing compositional simulation of the depletion process with the CVD data.

Volumetric Calculations for Wet Gas and Retrograde Gas Condensate Reservoirs

- The original wet gas or retrograde condensate gas in place, G_c , can be calculated volumetrically as:
$$G_c = \frac{7758 \times A \times h \times \phi \times (1 - S_{wi})}{B_{gi}}$$
- The main difference between the application of the volumetric equation to a dry gas reservoir versus a wet gas or a retrograde gas condensate reservoir is calculation of the initial gas FVF, B_{gi} . For wet or retrograde condensate gases, calculation of wellstream compositions should include all the liquids condensed in the production facilities.
- The best approach is to sample the gas and liquid streams at the separators and stock tank. The total composition of the reservoir gas (wet or condensate) is then determined by the recombination method.

Volumetric Calculations for Wet Gas and Retrograde Gas Condensate Reservoirs

- The cumulative wet or retrograde gas condensate produced, is given by:

$$G_{cp} = G_p + \frac{133,316(\gamma_o N_p)}{M_o}$$

- Where G_p is cumulative dry gas produced in scf; γ_o is the specific gravity of stock tank oil (condensate); M_o is cumulative stock tank oil (condensate) produced in STB; and is the molecular weight of stock tank oil (condensate), lbm/lb-mole.

GAS MATERIAL BALANCE (IN TERMS OF P/Z)

Gas Material Balance

- If enough production–pressure history is available for a gas reservoir in terms of:
 - cumulative gas production G_p as a function of pressure,
 - gas properties as a function of pressure at reservoir temperature, and
 - the initial reservoir pressure, p_i ,
- then the gas reserves can be calculated without knowing the areal extend of the reservoir or the drainage area of the well A , thickness h , porosity ϕ , or water saturation S_w .
- The material balance equation can be expressed in numerous forms depending on the type of the application and the driving mechanism.

Dry Gas Material Balance

- By forming a mass or mole balance on the gas, the general material balance for a gas reservoir represented by an idealized gas container is given as:

$$\frac{p_{sc} G_p}{RT_{sc}} = \frac{p_i V}{ZRT} - \frac{p [V - (W_e - B_w W_p)]}{ZRT}$$

$$n_i = n_r + n_p \quad (G1)$$

(G2)

- Where

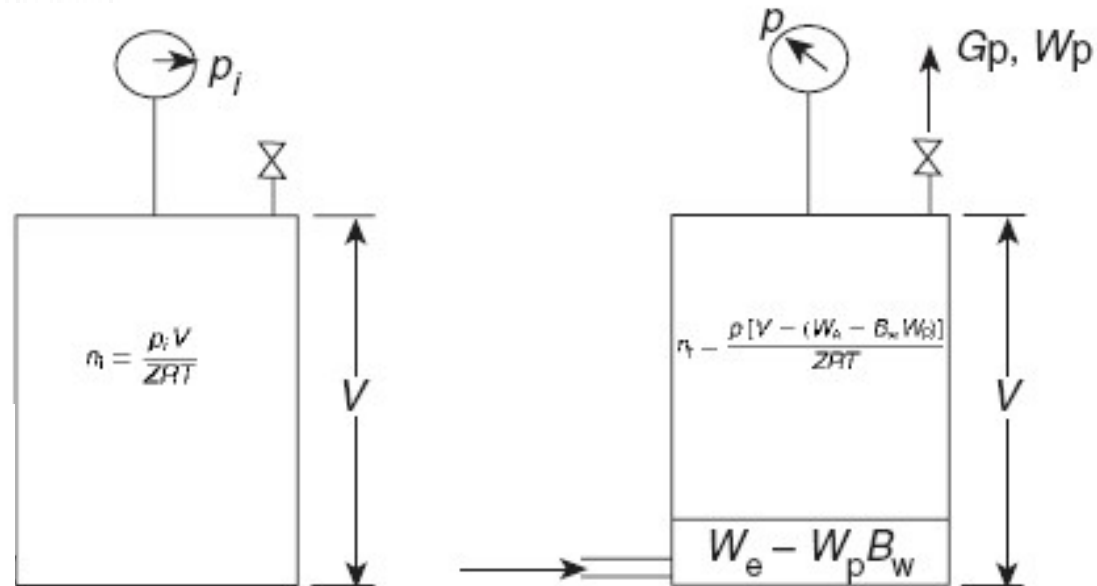
p_i = initial reservoir pressure
 G_p = cumulative gas production, scf
 p = current reservoir pressure
 V = original gas volume, ft³
 Z_i = gas deviation factor at p_i

Z = gas deviation factor at p

T = temperature, °R

W_e = cumulative water influx, ft³

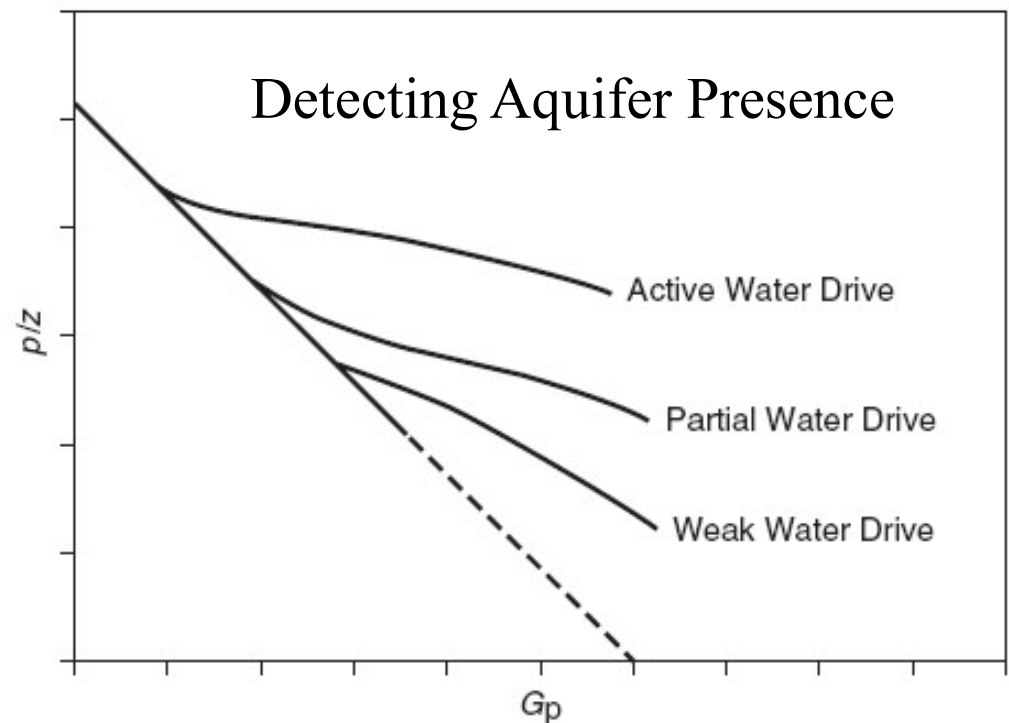
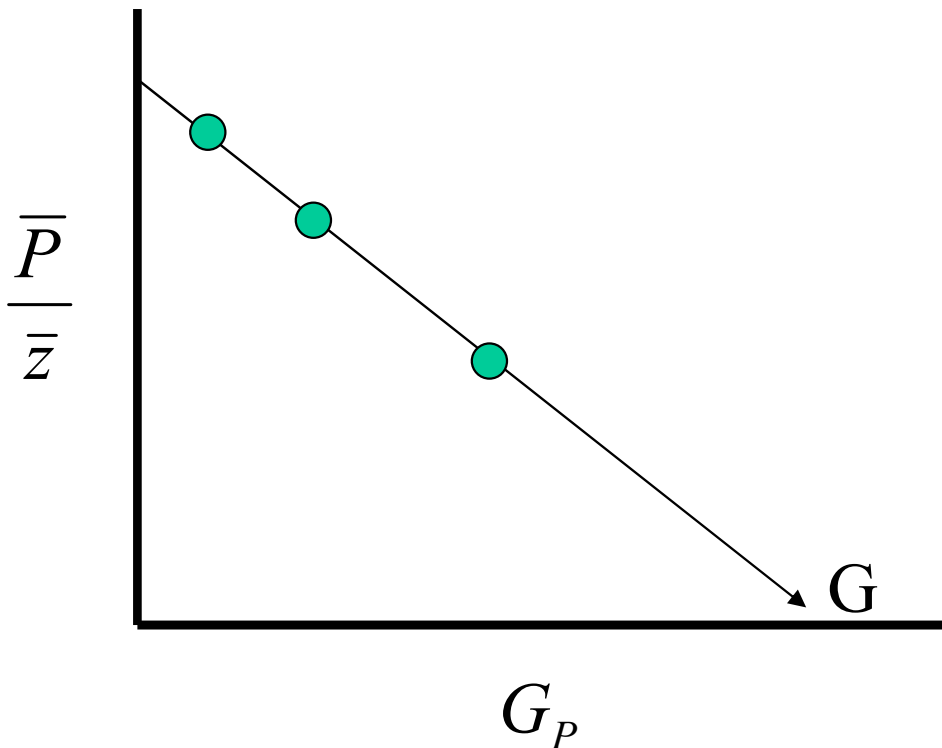
W_p = cumulative water production, stock-tank ft³



- In general, dry gas reservoirs can be classified either as:
 (1) volumetric gas reservoirs; or (2) water drive gas reservoirs.

Volumetric Gas Reservoirs

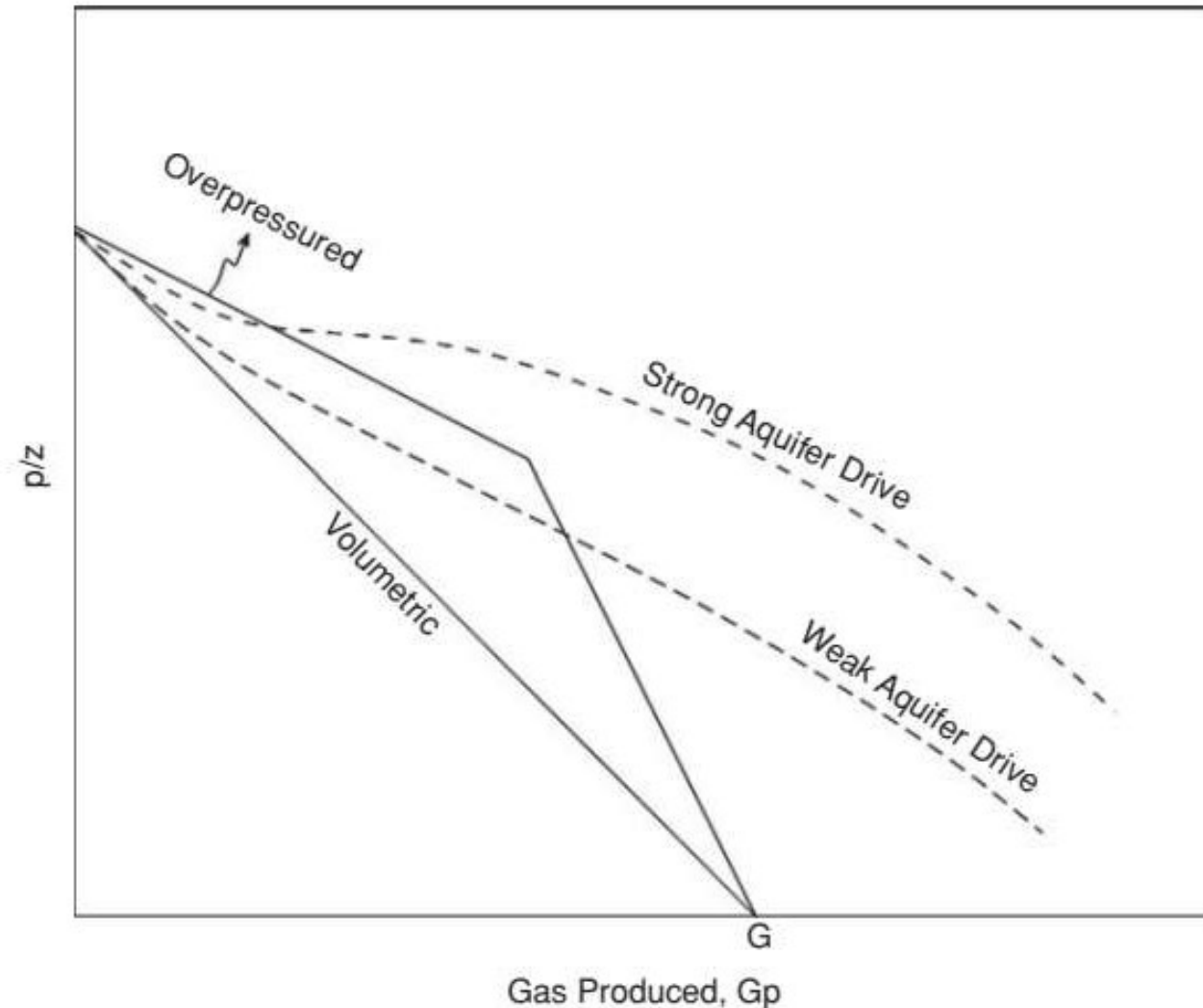
- For a volumetric reservoir and assuming no water production, Equation (1) becomes:
$$\frac{\bar{P}}{\bar{Z}} = \frac{P_i}{Z_i} \left[1 - \frac{G_P}{G_i} \right] \quad (G3)$$
- Note that the reservoir gas volume, V can be expressed as $V = B_{gi}G$ or $G = E_{gi}V$



Gas Material Balance

- A plot of Equation (G3) yields a straight line for volumetric gas reservoirs. When the intercept on the y-axis is equal to from which initial pressure can be estimated if not available.
- When the intercept on the x-axis is equal to the original gas in place, The plot is widely misapplied to all types of gas reservoirs. It is important to remember that it applies in a strict sense only to volumetric gas reservoirs.
- Volumetric reservoirs by definition have no fluid influxes either from aquifers or any other external sources. However, the plot can be used in a diagnostic sense to qualitatively demonstrate the existence of potential energy support for the reservoir from external sources.

Volumetric Gas Reservoirs



- Sketch of plots for various gas reservoir drive mechanisms.

- Some of the characteristic shapes of the plot for different reservoir drive mechanisms are shown here for volumetric, geopressured, weak, and strong water-drive gas reservoirs.
- It is sometimes permissible to apply the plot to a retrograde gas condensate reservoir at pressures below the dew point if the amount of liquid condensation in the reservoir is presumed to be less than 10 percent of the hydrocarbon pore volume.

Volumetric Gas Reservoirs

- This popular equation (G3) indicates that extrapolation of the straight line to the abscissa, i.e., at $p/Z = 0$, will give the value of the gas initially in place as $G = G_p$.
- The graphical representation of Equation (G3) can be used to detect the presence of water influx, as shown in Figure (3). When the plot of p/Z vs. G_p deviates from the linear relationship, it indicates the presence of water encroachment.
- Field average p/Z From the individual well performance in terms of p/Z vs. G_p , the recovery performance of the entire field can be estimated from the following relationship:

$$\left(\frac{p}{Z}\right)_{\text{Field}} = \frac{p_i}{Z_i} - \frac{\sum_{j=1}^n (G_p)_j}{\sum_{j=1}^n \left[G_p / \frac{p_i}{Z_i} - \frac{p}{Z} \right]_j}$$

This equation is applicable as long as all wells are producing with defined static boundaries, i.e. under pseudosteady-state conditions.

Volumetric Gas Reservoirs – Example 1

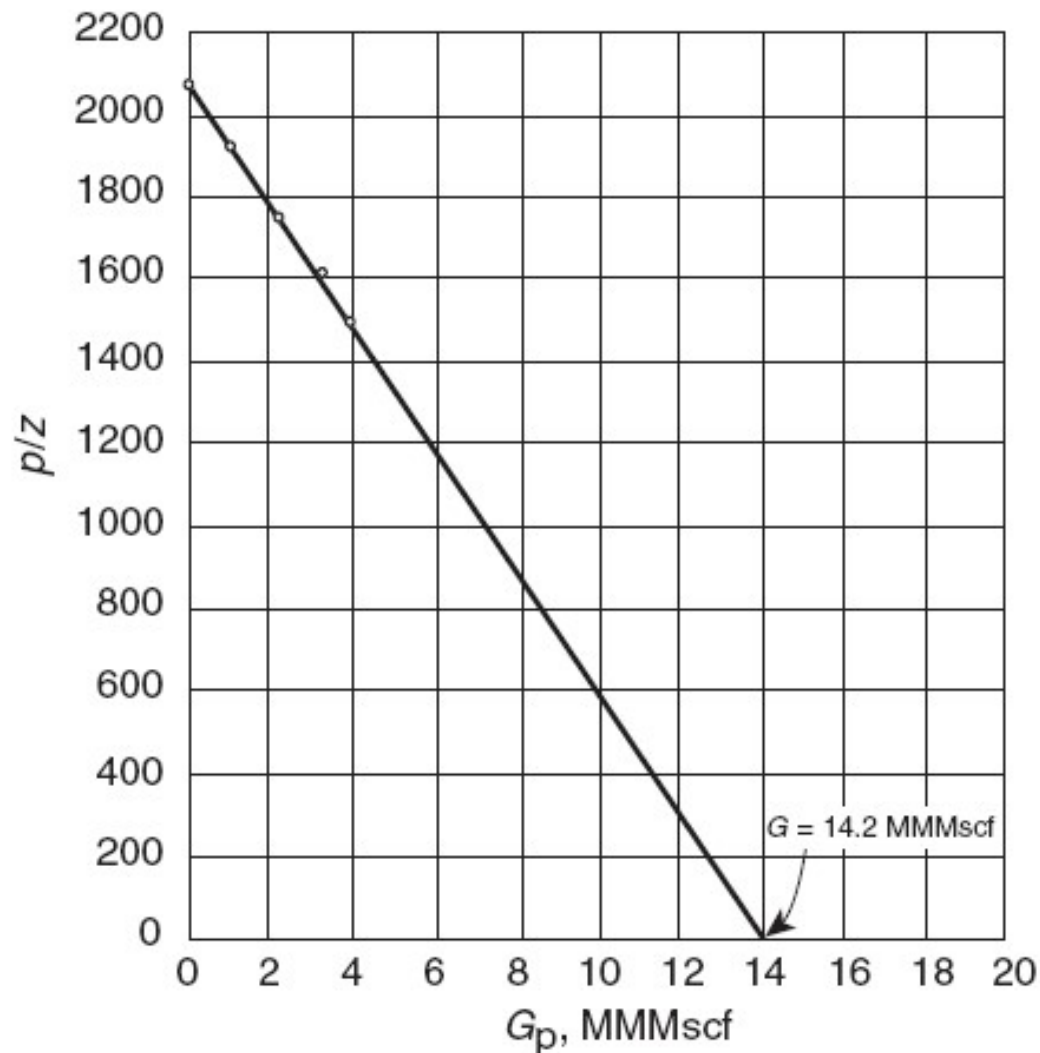
- A volumetric gas reservoir has the following production history.

<i>Time, t (years)</i>	<i>Reservoir pressure, p (psia)</i>	<i>Z</i>	<i>Cumulative production, G_p (MMMscf)</i>
0.0	1798	0.869	0.00
0.5	1680	0.870	0.96
1.0	1540	0.880	2.12
1.5	1428	0.890	3.21
2.0	1335	0.900	3.92

- The following data is also available: $\phi = 13\%$, $S_{wi} = 0.52$, $h = 54 \text{ ft}$, $A = 1060 \text{ acres}$, $T = 164^\circ\text{F}$
- Calculate the gas initially in place volumetrically and from the MBE.

Volumetric Gas Reservoirs – Example 1

- To Calculate the gas initially in place from the MBE, plot p/Z versus G_p as shown in below and determine G as: 14.2 MMMscf



Gas Recovery Factor

- The gas recovery factor (RF) at any depletion pressure is defined as the cumulative gas produced G_p at this pressure divided by the gas initially in place G :

$$RF = \frac{G_p}{G}$$

- Introducing the gas RF into Equation (2) gives:

$$\frac{p}{Z} = \frac{p_i}{Z_i} \left[1 - \frac{G_p}{G} \right]$$

- or $\frac{p}{Z} = \frac{p_i}{Z_i} [1 - RF]$

- Solving for the RF at any depletion pressure gives:

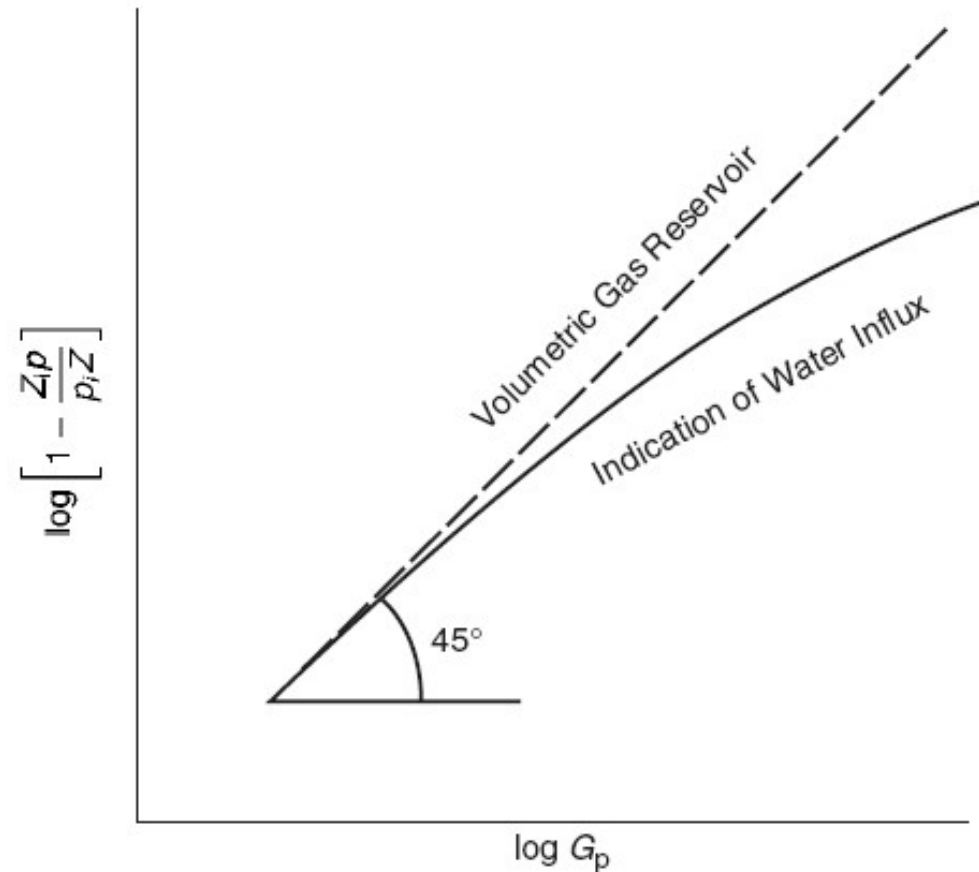
$$RF = 1 - \left[\frac{p}{Z} \frac{Z_i}{p_i} \right]$$

Water Drive Gas Reservoirs

- The plot of p/Z versus cumulative gas production G_p is a widely accepted method for solving gas material balance under depletion drive conditions. The extrapolation of the plot to atmospheric pressure provides a reliable estimate of the original gas-in-place.
- If a water drive is present the plot often appears to be linear, but the extrapolation will give an erroneously high value for gas-in-place. If the gas reservoir has a water drive, then there will be two unknowns in the MBE, even though production data, pressure, temperature, and gas gravity are known. These two unknowns are initial gas-in-place and cumulative water influx.
- In order to use the MBE to calculate initial gas-in-place, some independent method of estimating We , the cumulative water influx, must be developed.

Gas Material Balance in graphical form

- There are several methods of expressing the MBE in a convenient graphical form that can be used to describe the recovery performance of a volumetric or water drive gas reservoir including:
 - Energy Plot
 - MBE as a straight line;
 - Cole plot; modified Cole plot;
 - Roach plot; modified Roach plot;
 - Fetkovich et al. plot;
 - Paston et al. plot;
 - Hammerlindl method.



Gas Material Balance in graphical form

– The Energy Plot

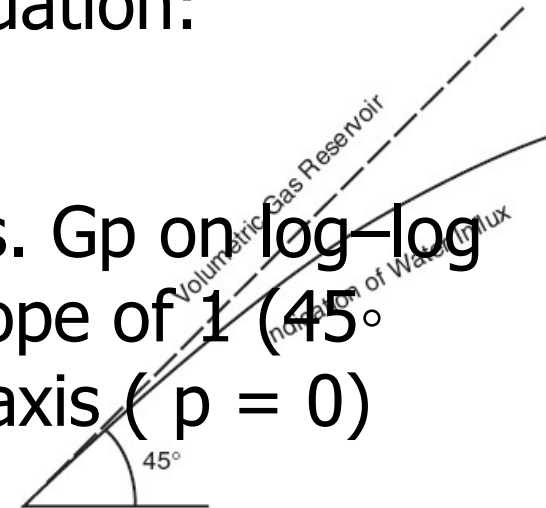
- This plot has been found to be useful in detecting water influx early in the life of a reservoir.

- Re-arranging $\frac{p}{Z} = \frac{p_i}{Z_i} \left[1 - \frac{G_p}{G} \right]$ to give $1 - \left[\frac{p}{Z} \frac{Z_i}{p_i} \right] = \frac{G_p}{G}$

- Taking the logarithm of both sides of this equation:

$$\log \left[1 - \frac{Z_i p}{p_i Z} \right] = \log G_p - \log G$$

- it is obvious that a plot of $\left[1 - \frac{Z_i p}{p_i Z} \right]$ vs. G_p on log-log coordinates will yield a straight line with a slope of 1 (45° angle). An extrapolation to 1 on the vertical axis ($p = 0$) yields a value for initial gas-in-place, G .



- If W_e is not zero, the slope of the plot will be less than one, and will also decrease with time, since W_e increases with time. An increasing slope can only occur due to either gas leaking from the reservoir or bad data, as this would imply that the gas occupied PV was increasing with time.

GAS MATERIAL BALANCE (IN TERMS OF GAS FORMATION VOLUME FACTOR)

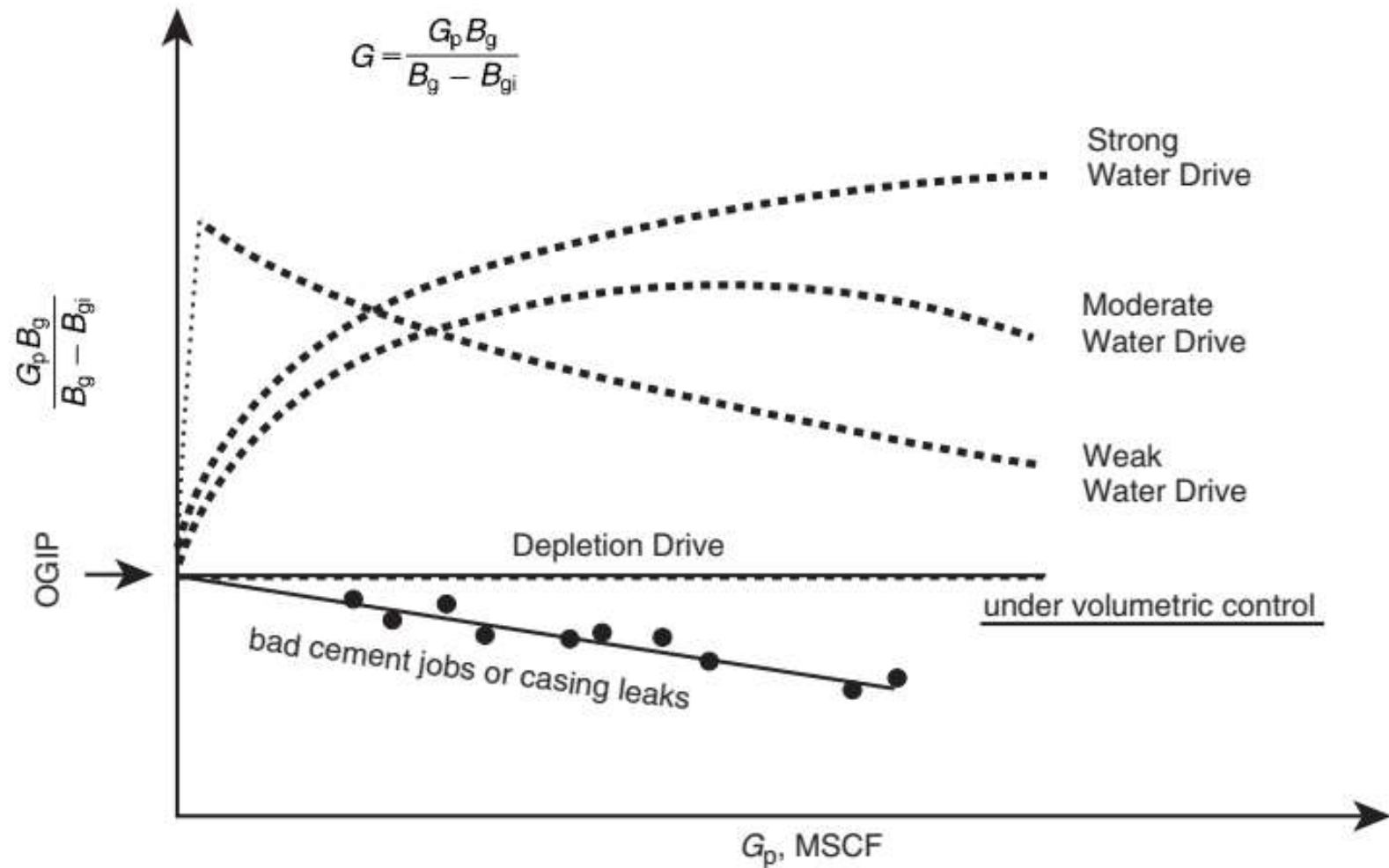
Gas Material Balance in graphical form

– Cole Plot

- It should be pointed out that the successive application of Eq. M13 can yield increasing or decreasing values of the gas initially in place G . Two different situations therefore exist:
 - 1) When the calculated value of the gas initially in place G appears to increase with time, the reservoir might be under drive. The invasion of water reduces the pressure drop for a given amount of production, making the reservoir appear larger as time progresses. The reservoir should in this case be classified as a water drive gas reservoir. Another possibility, if no known aquifer exists in the region, is that gas from a different reservoir or zone might migrate through fractures or leaky faults.
 - 2) If the calculated value of G decreases with time, the pressure drops more rapidly than would be the case in a volumetric reservoir. This implies loss of gas to other zones, leaky cementing job or casing leaks, among other possibilities.

$$G = \frac{G_p B_g}{B_g - B_{gi}}$$

Gas Material Balance in graphical form – Cole Plot



- It should be pointed out that the successive application of

Gas Material Balance in graphical form

– Effect of water influx on GIIP

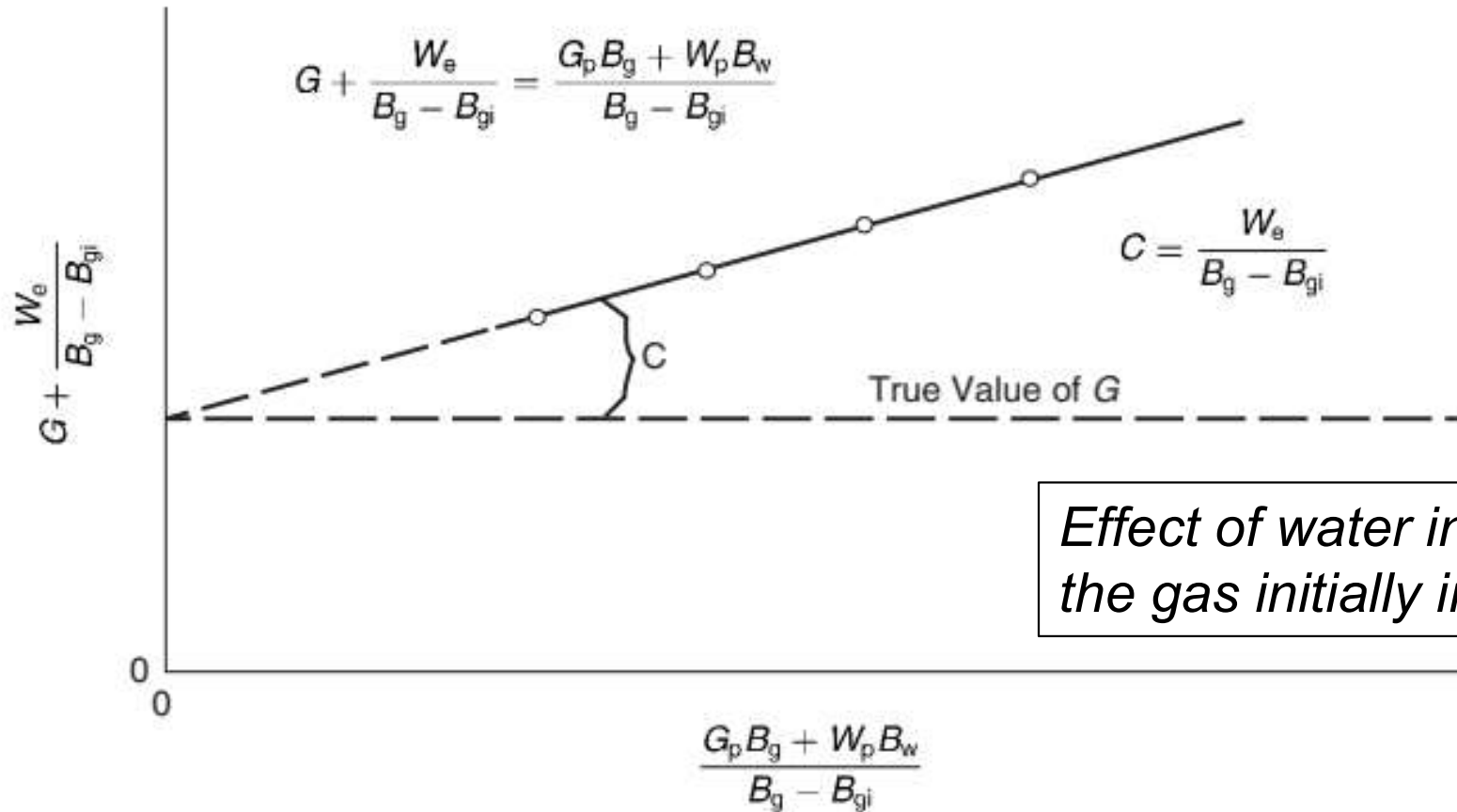
- If the gas reservoir has a water drive, then there will be two unknowns in the MBE, even though production data, pressure, temperature, and gas gravity are known. These two unknowns are initial gas-in-place and cumulative water influx. In order to use the MBE to calculate initial gas-in-place, some independent method of estimating W_e , the cumulative water influx, must be developed.
- Equation M13 can be modified to include the cumulative water influx and water production, to give:

$$G = \frac{G_p B_g - (W_e - W_p B_w)}{B_g - B_{gi}}$$

$$G + \frac{W_e}{B_g - B_{gi}} = \frac{G_p B_g + W_p B_w}{B_g - B_{gi}}$$

Gas Material Balance in graphical form

– Effect of water influx on GIIP



Effect of water influx on calculating the gas initially in place.

- This graphical technique can be used to estimate the value of W_e , because at any time the difference between the horizontal line (i.e., true value of G) and the straight line $G + [W_e/(B_g - B_{gi})]$ will give the value of $W_e/(B_g - B_{gi})$

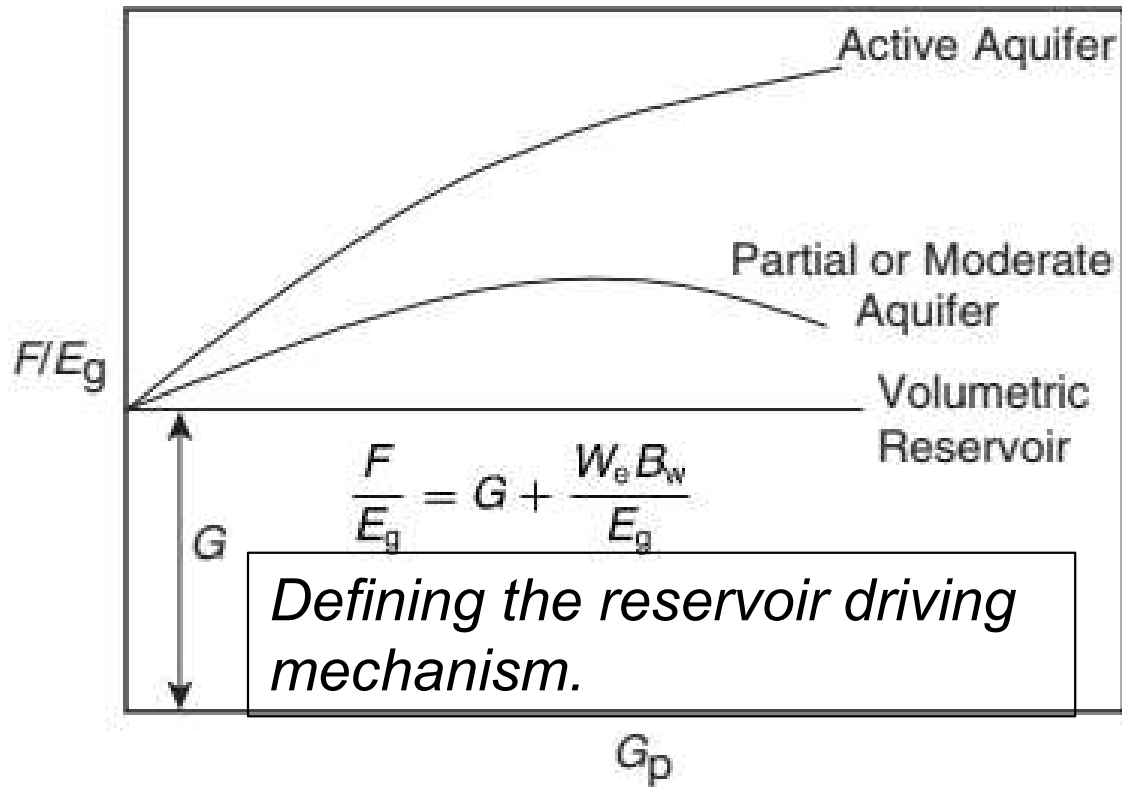
Gas Material Balance in graphical form

– Effect of water influx on GIIP

- Because gas often is bypassed and trapped by the encroaching water, recovery factors for gas reservoirs with water drive can be significantly lower than for volumetric reservoirs produced by simple gas expansion.
- In addition, the presence of reservoir heterogeneities, such as low-permeability stringers or layering, may reduce gas recovery further. While, ultimate recoveries of 80% to 90% are common in volumetric gas reservoirs, while typical recovery factors in water drive gas reservoirs can range from 50% to 70%.

Gas Material Balance in graphical form

– MBE as a Straight line



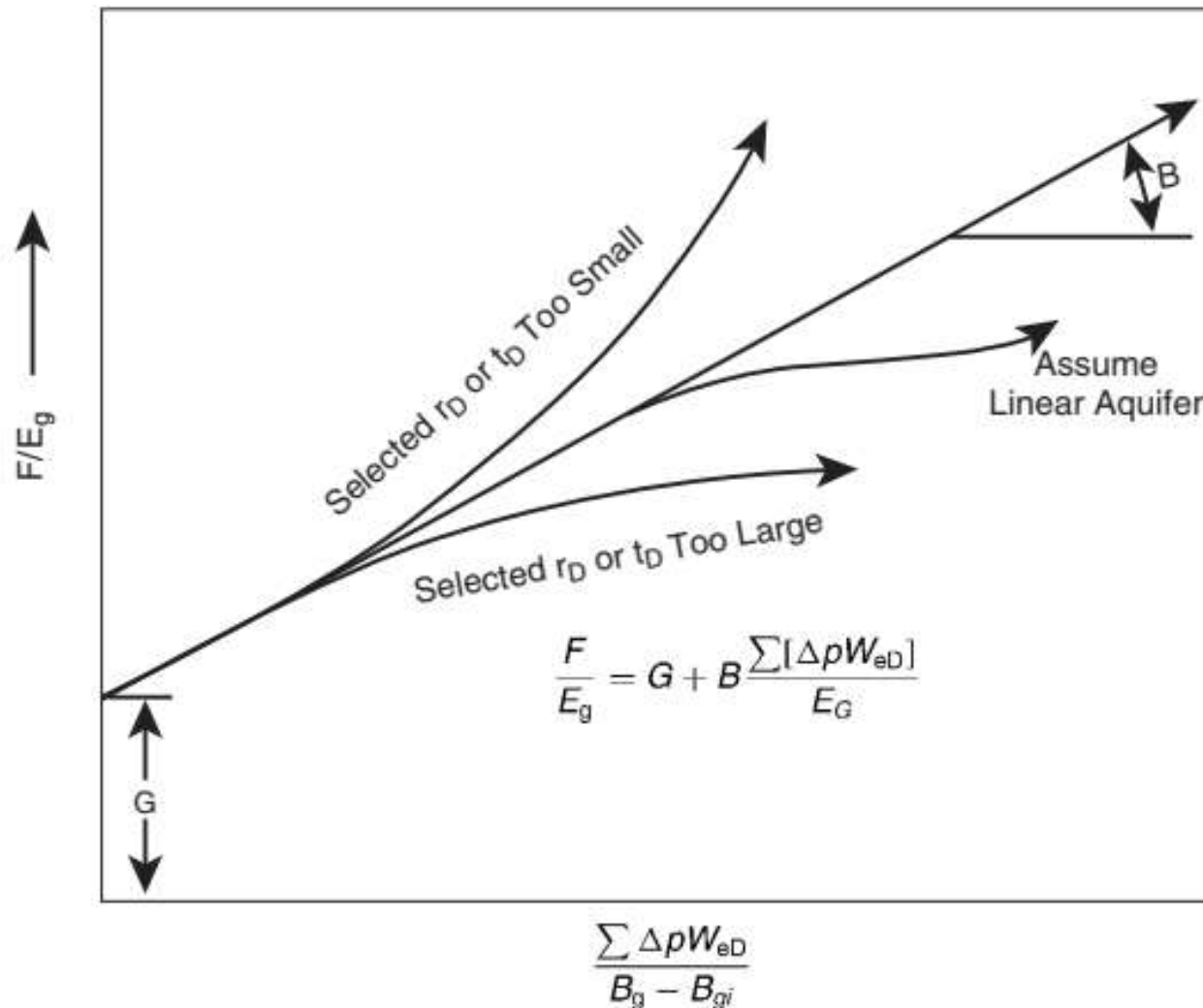
Non-linear plots will result if the aquifer is improperly characterized. A systematic upward or downward curvature suggests that the summation term is too small or too large, respectively, while an S-shaped curve indicates that a linear (instead of a radial) aquifer should be assumed.

- The points should plot sequentially from left to right. A reversal of this plotting sequence indicates that an unaccounted aquifer boundary has been reached and that a smaller aquifer should be assumed in computing the water influx term

Gas Material Balance in graphical form

– MBE as a Straight line

- The resulting straight line intersects the y axis at the initial gas-in-place G and has a slope equal to the water influx constant B ; as illustrated in Figure 3.23.



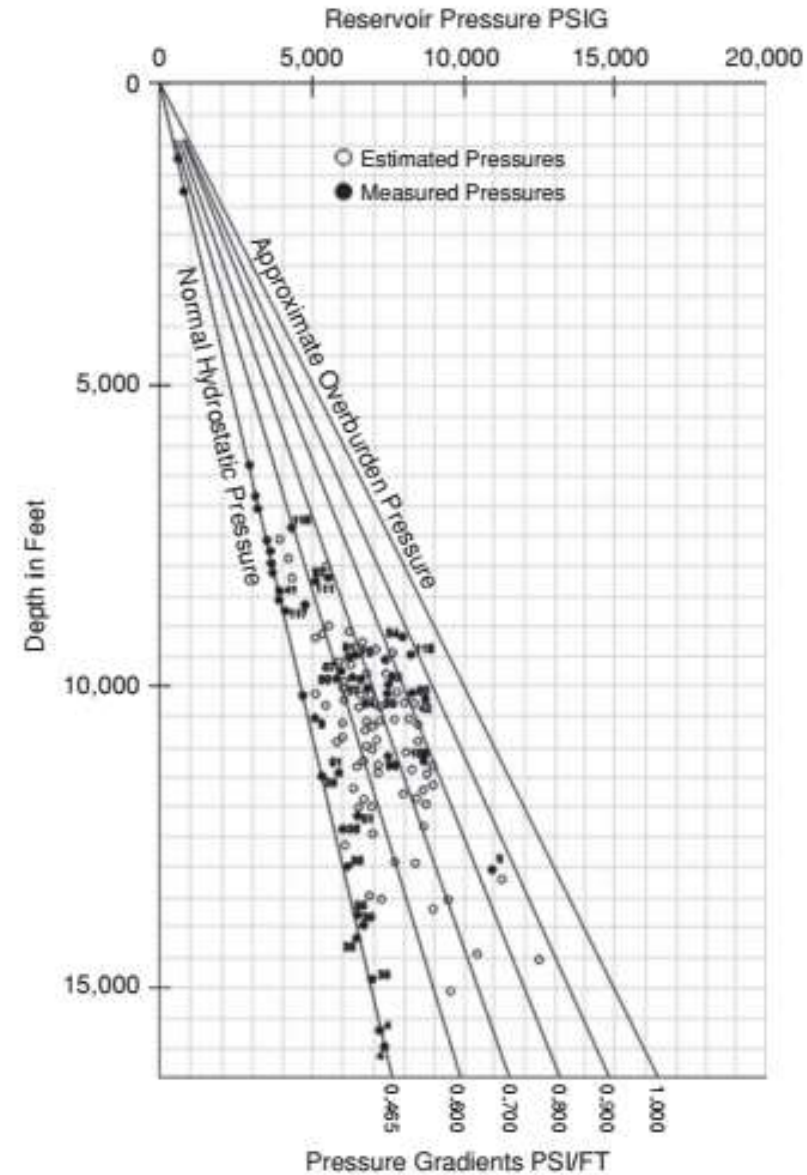
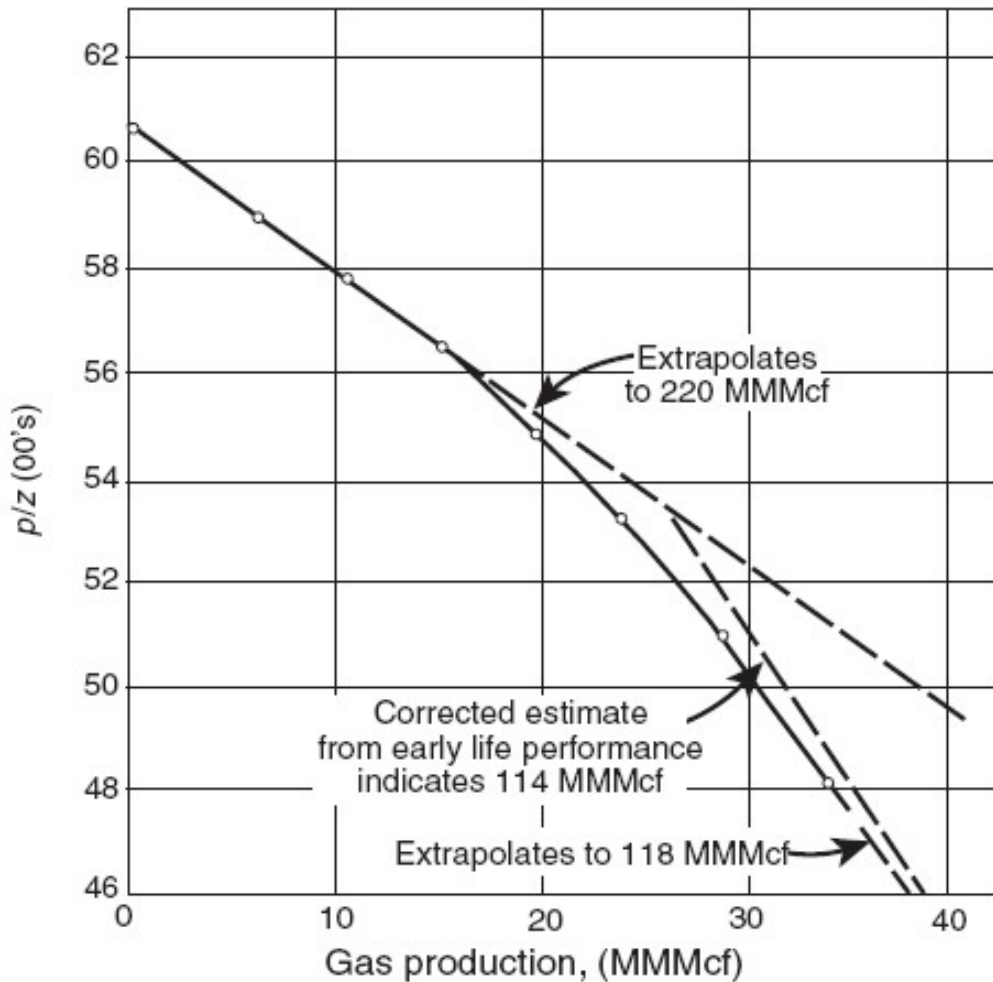
Abnormally Pressured Gas Reservoirs

- An “abnormally pressured” gas reservoir (sometimes called an “overpressured” or “geo-pressured” gas reservoir) is defined as a reservoir with pressures greater than a normal pressure gradient, i.e., over 0.5 psi/ft. A typical p/Z vs. G_p plot for an abnormally pressured gas reservoir will exhibit two straight lines as shown in Figure APGR.
 - (1) The first straight line corresponds to the “apparent” gas reservoir behaviour with an extrapolation that gives the “apparent gas-in-place G_{app} .”
 - (2) The second straight line corresponds to the “normal pressure behaviour” with an extrapolation that gives the “actual initial gas-in-place G .”
- Most early decisions are made based on the early life extrapolation of the p/Z plot; therefore, the effects of hydrocarbon PV change on reserve estimates, productivity, and abandonment pressure must be understood.

Abnormally Pressured Gas Reservoirs

- Hammerlindl (1971) pointed out that in abnormally high pressure volumetric gas reservoirs, two distinct slopes are evident when the plot of p/Z vs. G_p is used to predict reserves because of the formation and fluid compressibility effects as shown in Figure APGR. The final slope of the p/Z plot is steeper than the initial slope; consequently, reserve estimates based on the early life portion of the curve are erroneously high. The initial slope is due to gas expansion and significant pressure maintenance brought about by formation compaction, crystal expansion, and water expansion. At approximately normal pressure gradient, the formation compaction is essentially complete and the reservoir assumes the characteristics of a normal gas expansion reservoir. This accounts for the second slope.

Abnormally Pressured Gas Reservoirs



- Figure APGR: p/z versus cumulative production—North Ossum Field, Lafayette Parish, Louisiana NS2B reservoir (After Hammerlindl, 1971).

WALSH GENERALIZED MATERIAL BALANCE EQUATION

Walsh Generalized MBE

- Unlike the conventional material-balance equation (CMBE), the GMBE places no restrictions on initial fluid compositions as it uniquely accounts for volatilized-oil.
- This important because the restricted application of the CMBE to black-oil and dry-gas reservoirs thus not applicable to volatile-oil and gas-condensate systems
- Unique to the GMBE is the use of the volatile oil-gas ratio, R_v . This variable effectively describes the amount of volatilized oil in the reservoir gas-phase and is typically expressed in units of stb/scf or stb/mmscf. This variable is distinctly different from but analogous to the dissolved gas-oil ratio, R_s . The volatile oil-gas ratio is a function of the reservoir fluid composition. It also is a strong function of the separator configuration which seeks to maximize liquid dropout.

Walsh Generalized MBE

- For heavy- and black-oils, the volatile oil-gas ratio at the saturation pressure typically ranges from 0-10stb/mmscf; for volatile-oils, it ranges from 10-200 stb/mmscf; for near-critical fluids, it reaches maximum values and ranges from 150-400 stb/mmscf; for gas-condensates, it ranges from 50-250; for wet gases, it ranges from 20-100 stb/mmscf; and for dry gases, it approaches zero.

$$N_p(B_L - B_V R_s) + G_p(B_V - B_L R_v) \\ = G_i[R_{vi}(B_L - B_V R_s) - B_{Vi}(1 - R_s R_v) + (B_V - B_L R_v)]$$

$$N \left[B_{oi}(1 - R_v R_s) - (B_g - R_v B_o) R_{si} - (B_o - R_s B_g) \right] \\ + N_p \left[B_o(1 - R_v R_{ps}) + (R_{ps} - R_s) B_g \right] = \Delta W(1 - R_v R_s)$$

Forms of the General MBE

Eq. M10

$$NB_{oi} + NmB_{oi} - N[B_o + (R_{si} - R_s)B_g] + N_p[B_o + (R_{si} - R_s)B_g] + N_p(R_p - R_{si})B_g - \frac{NmB_{oi}B_g}{B_{gi}} = G_{inj}B_g + W_{inj}B_w + W_e - W_pB_w + (1 + m)NB_{oi}\left[\frac{c_wS_{wi} + c_f}{1 - S_{wi}}\right]\Delta p$$

$$N = \left(N_pB_o + (G_p - N_pR_s)B_g - (W_e - W_pB_w) - G_{inj}B_{ginj} - W_{inj}B_w \right) / \left((B_o - B_{oi}) + (R_{si} - R_s)B_g + mB_{oi}\left[(B_g/B_{gi}) - 1\right] + B_{oi}(1 + m)\left[(S_{wi}c_w + c_f)/(1 - S_{wi})\right]\Delta p \right)$$

$$N_p[B_t + (R_p - R_{si})B_g] + W_pB_w - G_{inj}B_g - W_{inj}B_w = N\left[(B_t - B_{ti}) + (1 + m)B_{ti}\left(\frac{c_wS_{wi} + c_f}{1 - S_{wi}}\right)\Delta p + \frac{mB_{ti}}{B_{gi}}(B_g - B_{gi})\right] + W_e$$

Eq. M11

End of Lecture 2

Water Influx – Lecture 3

Introduction

- Water-bearing rocks called **aquifers** surround nearly all hydrocarbon reservoirs. These aquifers may be substantially larger than the oil or gas reservoirs they adjoin as to appear infinite in size, and they may be so small in size as to be negligible in their effect on reservoir performance.
- As reservoir fluids are produced and reservoir pressure declines, a pressure differential develops from the surrounding aquifer into the reservoir. Following the basic law of fluid flow in porous media, the aquifer reacts by encroaching across the original hydrocarbon–water contact.
- In some cases, water encroachment occurs due to hydrodynamic conditions and recharge of the formation by surface waters at an outcrop.

Introduction

- In many cases, the pore volume of the aquifer is not significantly larger than the pore volume of the reservoir itself. Thus, the expansion of the water in the aquifer is negligible relative to the overall energy system, and the reservoir behaves volumetrically. In this case, the effects of water influx can be ignored.
- In other cases, the aquifer permeability may be sufficiently low such that a very large pressure differential is required before an appreciable amount of water can encroach into the reservoir. In this instance, the effects of water influx can be ignored as well.
- The objective of this lecture, however, concerns those reservoir–aquifer systems in which the size of the aquifer is large enough and the permeability of the rock is high enough that water influx occurs as the reservoir is depleted. A brief guide to various water influx calculation models and their application will be provided.

Classification of Aquifers

- Water Influx (encroachment), which is a response to hydrocarbon production from the reservoir, is attributed to:
 - expansion of the water in the aquifer;
 - compressibility of the aquifer rock;
 - artesian flow where the water-bearing formation outcrop is located structurally higher than the pay zone
- Reservoir–aquifer systems are commonly classified on the basis described in the following:
 - Degree of pressure maintenance
 - Outer boundary conditions
 - Flow regimes (steady state, pseudo-steady state & unsteady state)
 - Flow geometries

Degree of Pressure Maintenance

- The natural water drive is often qualitatively described as “active”, “partial” or “limited”
- The term “active” water drive refers to the water encroachment mechanism in which the rate of water influx equals the reservoir total production rate. Active water drive reservoirs are typically characterized by a gradual and slow reservoir pressure decline.
- If during any long period the production rate and reservoir pressure remain reasonably constant, the reservoir voidage rate must be equal to the water influx rate:

$$\left[\begin{array}{c} \text{water influx} \\ \text{rate} \end{array} \right] = \left[\begin{array}{c} \text{oil flow} \\ \text{rate} \end{array} \right] + \left[\begin{array}{c} \text{free gas} \\ \text{flow rate} \end{array} \right] + \left[\begin{array}{c} \text{water} \\ \text{production} \\ \text{rate} \end{array} \right]$$

$$e_w = Q_o B_o + Q_g B_g + Q_w B_w$$

$$e_w = \frac{dW_e}{dt} = B_o \frac{dN_p}{dt} + (\text{GOR} - R_s) \frac{dN_p}{dt} B_g + \frac{dW_p}{dt} B_w$$

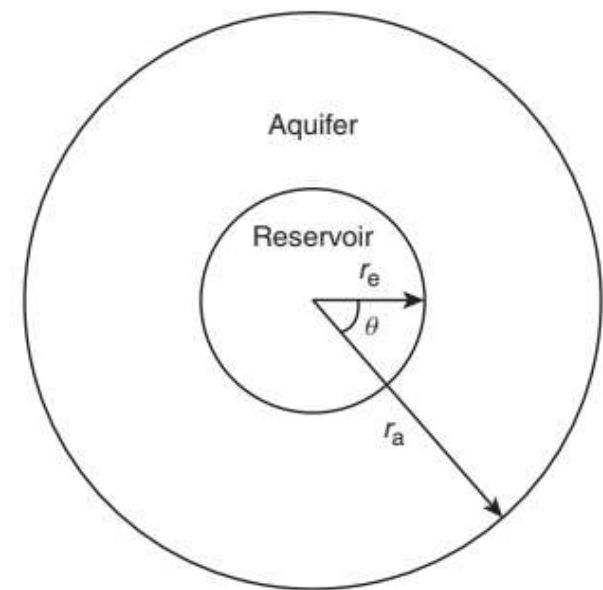
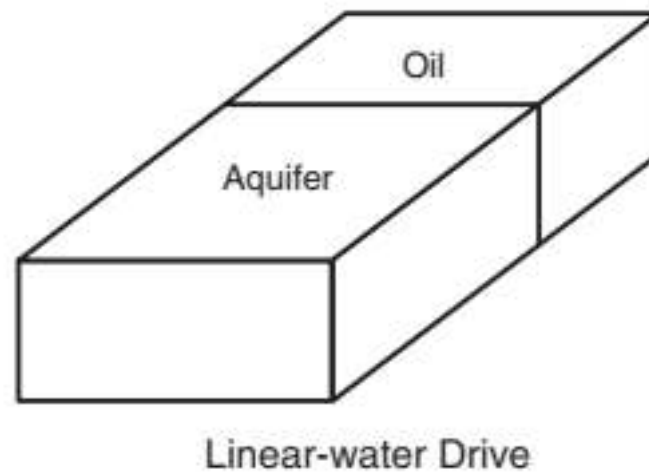
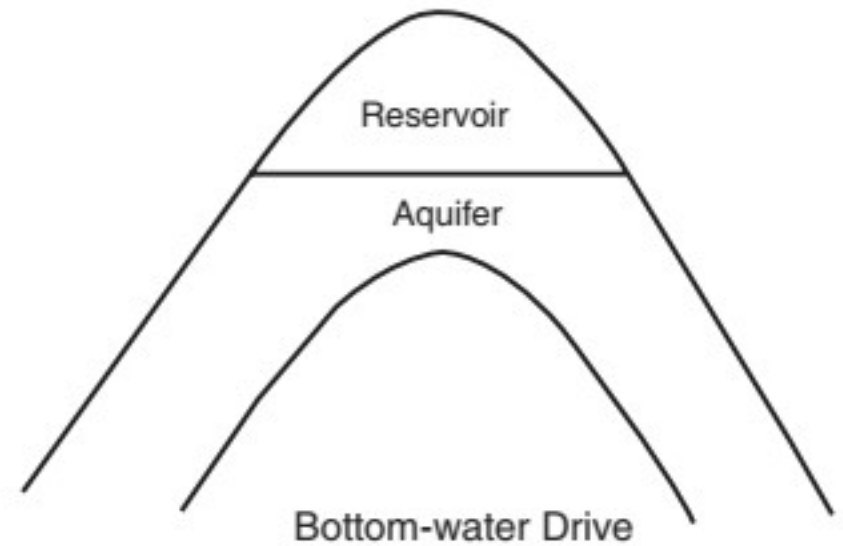
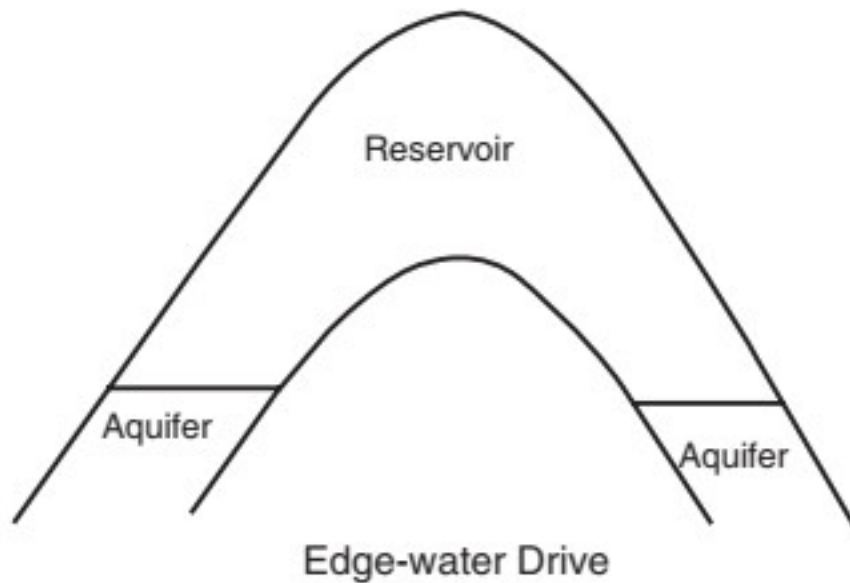
Outer boundary Conditions

- The aquifer can be classified as infinite or finite (bounded).
- Geologically all formations are finite but may act as infinite if the changes in the pressure at the oil–water contact are not “felt” at the aquifer boundary. Some aquifers outcrop and are infinite acting because of surface replenishment.
- In general, the outer boundary governs the behavior of the aquifer and can be classified as follows:
 - Infinite system indicates that the effect of the pressure changes at the oil/aquifer boundary can never be felt at the outer boundary. This boundary is for all intents and purposes at a constant pressure equal to initial reservoir pressure.
 - Finite system indicates that the aquifer outer limit is affected by the influx into the oil zone and that the pressure at this outer limit changes with time.

Flow Geometries

- Reservoir–aquifer systems on the basis of flow geometry can be classified as follows:
 - In edge-water drive, as shown in Figure aq.1, water moves into the flanks of the reservoir as a result of hydrocarbon production and pressure drop at the reservoir–aquifer boundary. The flow is essentially radial with negligible flow in the vertical direction.
 - Bottom-water drive occurs in reservoirs with large areal extent and a gentle dip where the reservoir–water contact completely underlies the reservoir. The flow is essentially radial and, in contrast to the edge-water drive, the bottom-water drive has significant vertical flow.
 - In linear–water drive, the influx is from one flank of the reservoir. The flow is strictly linear with a constant cross sectional area.

Flow Geometries



Radial aquifer geometries.

- Fig. aq.1. Flow geometries

Recognition of Water Influx

- Normally very little information is obtained during the exploration and development period of a reservoir concerning the presence or characteristics of an aquifer that could provide a source of water influx during the depletion period.
- Natural water drive may be assumed by analogy with nearby producing reservoirs, but early reservoir performance trends can provide clues.
- A comparatively low, and decreasing, rate of reservoir pressure decline with increasing cumulative withdrawals is indicative of fluid influx. Successive calculations of barrels withdrawn per psi change in reservoir pressure can supplement performance graphs.

Recognition of Water Influx

- However, if the reservoir limits have not been delineated by the developmental dry holes the influx could be from an undeveloped area of the reservoir not accounted for in averaging reservoir pressure.
- If the reservoir pressure is below the oil saturation pressure, a low rate of increase in produced GOR is also indicative of fluid influx.
- Early water production from edge wells is indicative of water encroachment. Such observations must be tempered by the possibility that the early water production is due to formation fractures, thin high-permeability streaks, or to coning in connection with a limited aquifer. The water production may be due to casing leaks.
- Calculation of increasing original oil-in-place from successive reservoir pressure surveys by using the material balance and assuming no water influx is also indicative of fluid influx.

Water Influx Models

- There are more uncertainties attached to this part of reservoir engineering than to any other. This is simply because one seldom drills wells into an aquifer to gain the necessary information about the porosity, permeability, thickness, and fluid properties. Instead, these properties have frequently to be inferred from what has been observed in the reservoir. Even more uncertain, however, is the geometry and areal continuity of the aquifer itself.
- Several models have been developed for estimating water influx that is based on assumptions that describe the characteristics of the aquifer. Due to the inherent uncertainties in the aquifer characteristics, all of the proposed models require historical reservoir performance data to evaluate constants representing aquifer property parameters since these are rarely known from exploration and development drilling with sufficient accuracy for direct application.

Water Influx Models

- The material balance equation can be used to determine historical water influx provided original oil-in-place is known from pore volume estimates. This permits evaluation of the constants in the influx equations so that future water influx rate can be forecast.
- The mathematical water influx models that are commonly used in the petroleum industry include:
 - pot aquifer;
 - Schilthuis steady state;
 - Hurst modified steady state;
 - van Everdingen and Hurst unsteady state:
 - edge-water drive; – bottom-water drive;
 - Carter–Tracy unsteady state;
 - Fetkovich method:
 - radial aquifer; – linear aquifer.

Water Influx Models – The pot aquifer model

- The simplest model that can be used to estimate the water influx into a gas or oil reservoir is based on the basic definition of compressibility. A drop in the reservoir pressure, due to the production of fluids, causes the aquifer water to expand and flow into the reservoir.
- Water influx = (aq. compressibility)(initial volume of water)(Δp)
- $W_e = c_t W_i (p_i - p)$ $c_t = c_w + c_f$ (3.1)

where:

W_e = cum. water influx, bbl; c_t = aquifer total compressibility, psi^{-1}

c_w = aquifer water compressibility, psi^{-1} ; c_f = aquifer rock compressibility, psi^{-1}

W_i = initial volume of water in the aquifer, bbl; p_i = initial reservoir pressure, psi

p = current reservoir pressure (pressure at oil–water contact), psi

Water Influx Models – The pot aquifer model

- Calculating the initial volume of water in the aquifer requires knowledge of aquifer dimensions and properties. These, however, are seldom measured since wells are not deliberately drilled into the aquifer to obtain such information. For instance, if the aquifer shape is radial, then:

$$W_i = \left[\frac{\pi (r_a^2 - r_e^2) h \phi}{5.615} \right] \quad (3.2)$$

where: r_a = radius of the aquifer, ft; r_e = radius of the reservoir, ft

h = thickness of the aquifer, ft; ϕ = porosity of the aquifer

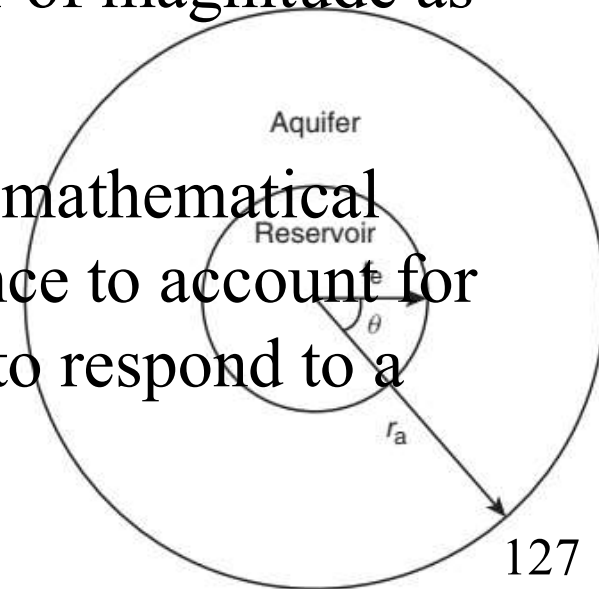
- Equation 3.2 suggests that water is encroaching in a radial form from all directions. Quite often, water does not encroach on all sides of the reservoir, or the reservoir is not circular in nature. To account for these cases, a modification to Equation 2.3.2 must be made in order to properly describe the flow mechanism. One of the simplest modifications is to include the fractional encroachment

Water Influx Models – The pot aquifer model

- One of the simplest modifications is to include the fractional encroachment angle f in the equation, as illustrated in Figure 2.2, to give: $W_e = c_t W_i f (p_i - p)$
- where the fractional encroachment angle f is defined by:

$$f = \frac{(\text{encroachment angle})^\circ}{360^\circ} = \frac{\theta}{360^\circ}$$

- The above model is only applicable to a small aquifer, i.e., pot aquifer, whose dimensions are of the same order of magnitude as the reservoir itself.
- Dake (1978) suggested that for large aquifers, a mathematical model is required which includes time dependence to account for the fact that it takes a finite time for the aquifer to respond to a pressure change in the reservoir.



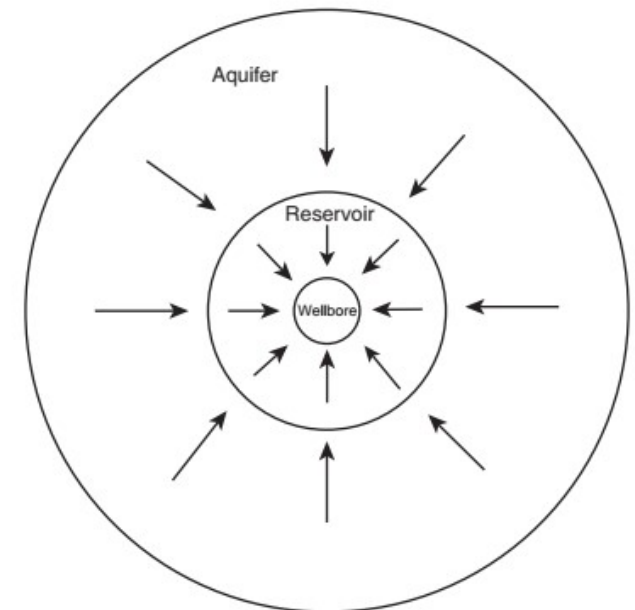
Water Influx Models –

The van Everdingen & Hurst unsteady-state model

- The mathematical formulations that describe the flow of a crude oil system into a wellbore are identical in form to those equations that describe the flow of water from an aquifer into a cylindrical reservoir, as shown schematically below.
- The dimensionless form of the diffusivity equation, is basically the general mathematical equation that is designed to model the transient flow behavior in reservoirs or aquifers. In a dimensionless form, the diffusivity equation is:

$$\frac{\partial^2 P_D}{\partial r_D^2} + \frac{1}{r_D} \frac{\partial P_D}{\partial r_D} = \frac{\partial P_D}{\partial t_D}$$

When an oil well is brought on production at a constant flow rate after a shut-in period, the pressure behavior is essentially controlled by the transient (unsteady-state) flowing condition. This flowing condition is defined as the time period during which the boundary has no effect on the pressure behavior.



Water Influx Models –

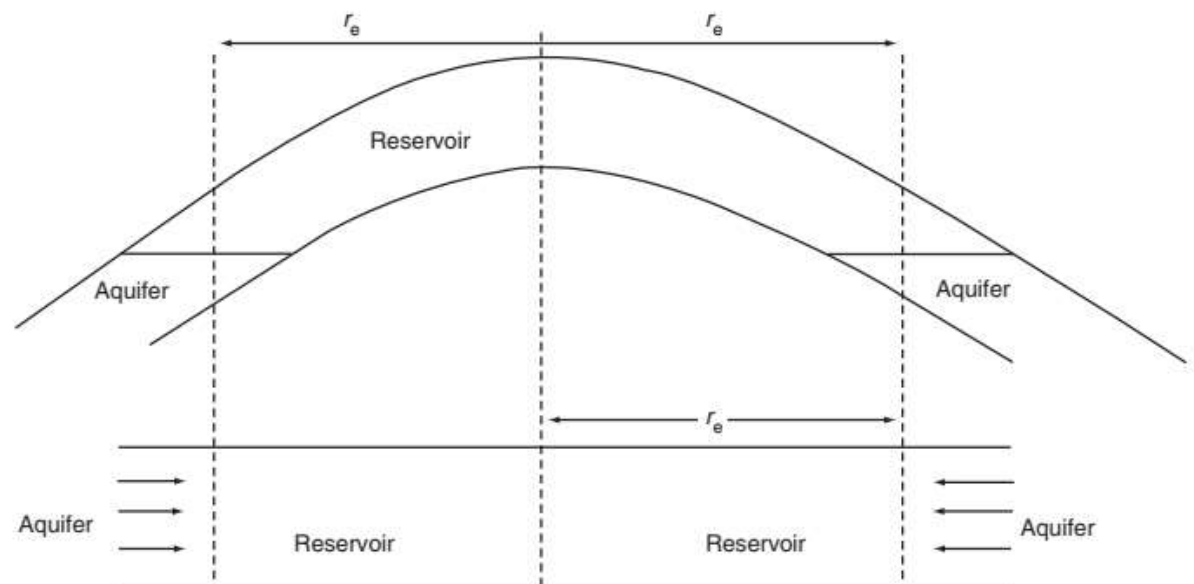
The van Everdingen & Hurst unsteady-state model

- Van Everdingen and Hurst (1949) proposed solutions to the dimensionless diffusivity equation for the following two reservoir–aquifer boundary conditions: (1) constant terminal rate; (2) constant terminal pressure.
- For the constant-terminal-pressure boundary condition, a boundary pressure drop is assumed constant over some finite time period, and the water influx rate is determined.
- In the description of water influx from an aquifer into a reservoir, there is greater interest in calculating the influx rate rather than the pressure. This leads to the determination of the water influx as a function of a given pressure drop at the inner boundary of the reservoir–aquifer system.

Water Influx Models –

The van Everdingen & Hurst unsteady-state model

- Van Everdingen and Hurst (1949) solved the diffusivity equation for the aquifer–reservoir system by applying the Laplace transformation to the equation. The authors' solution can be used to determine the water influx in the following systems:
 - edge-water drive system (radial system);
 - bottom-water drive system;
 - linear-water drive system.



Idealized radial flow model

Water Influx Models –

The van Everdingen & Hurst unsteady-state model

- **Edge-water drive:** The inner boundary is defined as the interface between the reservoir and the aquifer. The flow across this inner boundary is considered horizontal and encroachment occurs across a cylindrical plane encircling the reservoir. With the interface as the inner boundary, it is possible to impose a constant terminal pressure at the inner boundary and determine the rate of water influx across the interface.
- Initial Conditions: $p = p_i$ for all values of radius r
- Outer boundary conditions:
 - For an infinite aquifer:
$$p = p_i \quad \text{at } r = \infty$$
 - For a bounded aquifer
$$\frac{\partial p}{\partial r} = 0 \quad \text{at } r = r_a$$
- Assumptions
 - uniform thickness; constant permeability;
 - uniform porosity; constant rock compressibility;
 - constant water compressibility

Water Influx Models –

The van Everdingen & Hurst unsteady-state model

- **Edge-water drive:** The authors expressed their mathematical relationship for calculating the water influx in the form of a dimensionless parameter called dimensionless water influx W_{eD} . They also expressed the dimensionless water influx as a function of the dimensionless time t_D and dimensionless radius r_D ; thus they made the solution to the diffusivity equation generalized and it can be applied to any aquifer where the flow of water into the reservoir is essentially radial. The solutions were derived for the cases of bounded aquifers and aquifers of infinite extent. The two dimensionless parameters t_D and r_D are given by:

$$t_D = 6.328 \times 10^{-3} \frac{kt}{\phi \mu_w c_t r_e^2} \quad r_D = \frac{r_a}{r_e} \quad c_t = c_w + c_f$$

- Where: t = time, days; k = permeability of the aquifer, md; ϕ = porosity of the aquifer; μ_w = viscosity of water in the aquifer, cp; r_a = radius of the aquifer, ft; r_e = radius of the reservoir, ft; c_w = compressibility of the water, psi⁻¹; c_f = compressibility of the aquifer formation, psi⁻¹; c_t = total compressibility coefficient, psi⁻¹

Water Influx Models –

The van Everdingen & Hurst unsteady-state model

- The van Everdingen and Hurst unsteady-state model is given by:

$$W_e = B \Sigma \Delta p W_{eD} \quad (\text{M.20}) \quad \text{with } B = 1.119 \phi c_t r_e^2 h f$$

- Van Everdingen and Hurst presented the dimensionless water influx W_{eD} as a function of the dimensionless time t_D and dimensionless radius r_D that are given by:

$$t_D = 6.328 \times 10^{-3} \frac{kt}{\phi \mu_w c_t r_e^2}$$

$$c_t = c_w + c_f$$

$$r_D = \frac{r_a}{r_e}$$

- For a unit pressure drop, the cumulative water influx in an **infinite linear case** is simply proportional to $\sqrt{t}$ and does not require the estimation of the dimensionless time t_D .
- Thus, the Van Everdingen and Hurst dimensionless water influx W_{eD} in Equation M20 is replaced by the square root of time, to give:

$$W_w = B \sum \left[\Delta p_n \sqrt{(t - t_n)} \right]$$

Water Influx Models –

The Carter & Tracy Model

- The van Everdingen and Hurst methodology provides the exact solution to the radial diffusivity equation and therefore is considered the correct technique for calculating water influx.
- However, because superposition of solutions is required, their method involves tedious calculations. To reduce the complexity of water influx calculations, Carter and Tracy (1960) proposed a calculation technique that does not require superposition and allows direct calculation of water influx.
- The primary difference between the Carter–Tracy technique and the van Everdingen and Hurst technique is that Carter–Tracy assumes constant water influx rates over each finite time interval.

Water Influx Models –

The Carter & Tracy Model

- Using the Carter–Tracy technique, the cumulative water influx at any time, t_n , can be calculated directly from the previous value obtained at t_{n-1} , or:

$$(W_e)_n = (W_e)_{n-1} + [(t_D)_n - (t_D)_{n-1}] \times \left[\frac{B \Delta p_n - (W_e)_{n-1} (p_D')_n}{(p_D)_n - (t_D)_{n-1} (p_D')_n} \right]$$

- Where:
- B = the van Everdingen and Hurst water influx constant
- t_D = the dimensionless time
- n = the current time step; $n - 1$ = the previous time step
- Δp_n = total pressure drop, $p_i - p_n$, psi
- p_D = dimensionless pressure
- p_D' = dimensionless pressure derivative

Values of the dimensionless pressure p_D as a function of t_D and r_D are tabulated

Water Influx Models –

The Carter & Tracy Model

- Edwardson et al. (1962) developed the following approximation of p_D for an infinite-acting aquifer:

$$p_D = \frac{370.529\sqrt{t_D} + 137.582t_D + 5.69549(t_D)^{1.5}}{328.834 + 265.488\sqrt{t_D} + 45.2157t_D + (t_D)^{1.5}}$$

- The dimensionless pressure derivative can then be approximated by:

$$p_D' = \frac{E}{F} \quad \text{where} \quad \begin{aligned} E &= 716.441 + 46.7984(t_D)^{0.5} + 270.038t_D + 71.0098(t_D)^{1.5} \\ F &= 1296.86(t_D)^{0.5} + 1204.73t_D + 618.618(t_D)^{1.5} \\ &\quad + 538.072(t_D)^2 + 142.41(t_D)^{2.5} \end{aligned}$$

When the dimensionless time $t_D > 100$, the following approximation can be used for p_D :

$$p_D = \frac{1}{2}[\ln(t_D) + 0.80907]$$

with the derivative given by:

$$p_D' = \frac{1}{2t_D}$$

Water Influx Models –

The Fetkovich Model

- Fetkovich (1971) developed a method of describing the approximate water influx behavior of a finite aquifer for radial and linear geometries. In many cases, the results of this model closely match those determined using the van Everdingen and Hurst approach. The Fetkovich theory is much simpler, and, like the Carter–Tracy technique, this method does not require the use of superposition. Hence, the application is much easier, and this method is also often utilized in numerical simulation models.
- The Fetkovich model is based on the premise that the productivity index concept will adequately describe water influx from a finite aquifer into a hydrocarbon reservoir. That is, the water influx rate is directly proportional to the pressure drop between the average aquifer pressure and the pressure at the reservoir–aquifer boundary.

Water Influx Models –

The Fetkovich Model

- The method neglects the effects of any transient period. Thus, in cases where pressures are changing rapidly at the aquifer–reservoir interface, predicted results may differ somewhat from the more rigorous van Everdingen and Hurst or Carter–Tracy approaches.
- However, in many cases pressure changes at the waterfront are gradual and this method offers an excellent approximation to the two methods discussed above.
- This approach begins with two simple equations. The first is the productivity index (PI) equation for the aquifer, which is analogous to the PI equation used to describe an oil or gas well:

$$e_w = \frac{dW_e}{dt} = J (\bar{p}_a - p_r)$$

- where: e_w = water influx rate from aquifer, bbl/day; p_a = average aquifer pressure, psi; J = productivity index for the aquifer, bbl/day/psi; p_r = inner aquifer boundary pressure, psi

Water Influx Models –

The Fetkovich Model

- The second equation is an aquifer material balance equation for a constant compressibility, which states that the amount of pressure depletion in the aquifer is directly proportional to the amount of water influx from the aquifer, or: $W_e = c_t W_i f(p_i - p_a)$
- Equation 3.25 suggests that the maximum possible water influx occurs if $p_a = 0$, or: $W_{ei} = c_t W_i f p_i$
- Combining Equation 3.36 with 3.35 gives:

$$\bar{p}_a = p_i \left(1 - \frac{W_e}{c_t W_i p_i} \right) = p_i \left(1 - \frac{W_e}{W_{ei}} \right)$$

- Equation 3.35 provides a simple expression to determine the average aquifer pressure p_a after removing W_e bbl of water from the aquifer to the reservoir, i.e., cumulative water influx. Differentiating Equation 3.37 with respect to time gives:

$$\frac{dW_e}{dt} = - \frac{W_{ei}}{p_i} \frac{d\bar{p}_a}{dt}$$

Water Influx Models –

The Fetkovich Model

- Fetkovich combined Equation 3.38 with 3.34 and integrated to give the following form:

$$W_e = \frac{W_{ei}}{p_i} (p_i - p_r) \exp \left(\frac{-Jp_i t}{W_{ei}} \right)$$

- Equation 3.39 has no practical applications since it was derived for a constant inner boundary pressure. To use this solution in the case in which the boundary pressure is varying continuously as a function of time, the superposition technique must be applied. Rather than using superposition, Fetkovich suggested that, if the reservoir–aquifer boundary pressure history is divided into a finite number of time intervals, the incremental water influx during the n th interval is:
$$(\Delta W_e)_n = \frac{W_{ei}}{p_i} [(\bar{p}_a)_{n-1} - (\bar{p}_r)_n] \left[1 - \exp \left(-\frac{Jp_i \Delta t_n}{W_{ei}} \right) \right]$$
- where $(p_a)_{n-1}$ is the average aquifer pressure at the end of the previous time step.

Water Influx Models –

The Fetkovich Model

- This average pressure is calculated from Equation 3.37 as::

$$(\bar{p}_a)_{n-1} = p_i \left(1 - \frac{(W_e)_{n-1}}{W_{ei}} \right)$$

- The average reservoir boundary pressure $(p_r)_n$ is estimated from:

$$(\bar{p}_r)_n = \frac{(p_r)_n + (p_r)_{n-1}}{2}$$

- The productivity index J used in the calculation is a function of the geometry of the aquifer. Fetkovich calculated the productivity index from Darcy's equation for bounded aquifers. The Fetkovich method can be extended to infinite-acting aquifers by requiring that the ratio of water influx rate to pressure drop is approximately constant throughout the productive life of the reservoir. The productivity index J of the aquifer is given by the following expressions:

Water Influx Models – The Fetkovich Model

<i>Type of outer aquifer boundary</i>	<i>J for radial flow (bbl/day/psi)</i>	<i>J for linear flow (bbl/day/psi)</i>	<i>Equation</i>
Finite, no flow	$J = \frac{0.00708khf}{\mu_w [\ln(r_D) - 0.75]}$	$J = \frac{0.003381 kwh}{\mu_w L}$	[2.3.43]
Finite, constant pressure	$J = \frac{0.00708 khf}{\mu_w [\ln(r_D)]}$	$J = \frac{0.001127 kwh}{\mu_w L}$	[2.3.44]
Infinite	$J = \frac{0.00708 khf}{\mu_w \ln(a/r_e)}$ $a = \sqrt{0.0142 kt / (\phi \mu_w c_t)}$	$J = \frac{0.001 kwh}{\mu_w \sqrt{0.0633 kt / (\phi \mu_w c_t)}}$	[2.3.45]

where:

w = width of the linear aquifer, ft; L = length of the linear aquifer, ft

r_D = dimensionless radius, r_a/r_e ; k = permeability of the aquifer, md

t = time, days; θ = encroachment angle; h = thickness of the aquifer; $f = \theta/360$

End of Lecture 3

Immiscible Fluid Displacement – Lecture 4

Introduction

- It is important to understand the fundamental processes that occur when reservoir fluids are displaced immiscibly by gas or water.
- The displacement process is affected by the wettability of the rock, and the mobility ratio between the displaced & the displacing fluids.
- The total efficiency of the displacement process is measured in terms of the effectiveness of water or gas in displacing the reservoir fluids, and the proportion of the reservoir actually contacted by the displacing fluids.
- The fractional flow equations are presented, followed with the presentation of the Buckley-Leverett equation, and the Welge method for estimating average water saturation in a water displacement process.

Displacement Efficiency

- In addition to the concepts of Rock Wettability, Capillary Pressure, Relative Permeability and Mobility Ratio, the efficiency of fluid displacement is estimated in terms of the Fluid (Microscopic) Displacement Efficiency, Vertical Sweep Efficiency, Volumetric Displacement Efficiency and the total Recovery Efficiency.
- Fluid (or microscopic) displacement efficiency, is defined as the volume of oil displaced from the invaded region divided by the volume of oil initially in place in the invaded region. Fluid displacement efficiency is affected by rock wettability, capillary pressure, relative permeability, and mobility ratios of the fluids. If reservoir pressure is maintained at initial conditions and saturation within the invaded region is reduced to residual oil saturation, the displacement efficiency can be estimated with the expression:

$$E_D = 1 - \frac{S_{or}}{S_{oi}} \quad (4.01)$$

Displacement Efficiency

- Volumetric (or macroscopic) displacement efficiency, is the fraction of the reservoir volume swept by the displacing fluid. It is composed of two parts: areal sweep efficiency, & vertical sweep efficiency. Areal sweep efficiency is the fraction of the reservoir area contacted by the displacing fluid. Vertical sweep efficiency is the fraction of the vertical reservoir cross-section contacted by the displacing fluid.
- Volumetric displacement efficiency is the product of areal and vertical sweep efficiencies:
$$E_V = E_A \times E_I \quad (4.02)$$
- Total recovery efficiency, is the volume of oil displaced divided by the initial volume of oil in place in the swept portion of the reservoir. Thus, total recovery efficiency is given by:

$$E_R = E_D \times E_V \quad (4.03)$$

Fractional Flow Equations

- The fractional flow equation is used to calculate the flow rate of a fluid as a fraction of the total fluid flow rate when only two fluids are flowing in the reservoir.
- Diffuse flow condition is assumed which means that fluid saturations at any point in the linear displacement path are uniformly distributed with respect to thickness. The sole reason for making this assumption is that it permits the displacement to be described, mathematically, in one dimension and this provides the simplest possible model of the displacement process.
- Fractional flow of a fluid in a reservoir is primarily dependent on its relative permeability, but can be affected by capillary and gravity forces. The fractional flow equations presented in this lecture are based on the assumption of linear flow.

Fractional Flow Equation for Oil Displaced by Water

- The total reservoir liquid flow rate, is the sum of the water flow rate, and the oil flow rate, expressed as:

$$q_t = q_w + q_o \quad (4.1)$$

In RB/D where:

$$q_w = -1.127 \times 10^{-3} \frac{k_w A}{\mu_w} \left[\frac{\partial p_w}{\partial l} + 0.00694 \rho_w \sin \alpha \right] \quad (4.2)$$

$$q_o = -1.127 \times 10^{-3} \frac{k_o A}{\mu_o} \left[\frac{\partial p_o}{\partial l} + 0.00694 \rho_o \sin \alpha \right] \quad (4.3)$$

N.B.: The equations are given in field units. The density here are in lbm/ft³. Properties are at reservoir conditions

- The fractional flow of water, as a fraction of the total flow rate and the fractional flow of oil, as a fraction of the total flow rate are:

$$f_w = \frac{q_w}{q_t} \quad (4.4) \quad f_o = \frac{q_o}{q_t} \quad (4.5) \quad (1 - f_w)q_t = q_o$$

- Using the equations above and that $p_c = p_o - p_w$ gives:

$$f_w = \frac{1}{\left(1 + \frac{\mu_w}{k_w} \times \frac{k_o}{\mu_o}\right)} + \frac{1.127 \times 10^{-3} \frac{k_o A}{\mu_o q_t} \left[\frac{\partial p_c}{\partial l} + 0.00694 (\rho_o - \rho_w) \sin \alpha \right]}{\left(1 + \frac{\mu_w}{k_w} \times \frac{k_o}{\mu_o}\right)} \quad (4.6)$$

Fractional Flow Equation for Oil Displaced by Water

- Eq. (4.6) is the complete form of the fractional flow equation for water flowing linearly in an oil-water reservoir. A quick examination of Eq. (4.6) shows that it has all the factors that affect the flow of water in an oil-water reservoir. These factors are fluid properties (viscosities, densities), rock properties (effective permeabilities, saturations, capillary pressure), total flow rate, and structural inclination of the reservoir (dip angle, α).
- If it is assumed that fluid properties and total flow rate are constant, then fractional flow of water is only a function of water saturation in the reservoir. A plot of shown in Figure 4.1 is commonly called the fractional flow curve. The fractional flow curve can be used to calculate the flow rate of water in the reservoir at any water saturation.

Fractional Flow Equation for Oil Displaced by Water

- The effect of capillary pressure can be assumed to be negligible. In such cases, Eq. (4.6) reduces to the following form:

$$f_w = \frac{1}{\left(1 + \frac{\mu_w}{k_w} \times \frac{k_o}{\mu_o}\right)} + \frac{7.821 \times 10^{-6} \frac{k_o A}{\mu_o q_t} (\rho_o - \rho_w) \sin \alpha}{\left(1 + \frac{\mu_w}{k_w} \times \frac{k_o}{\mu_o}\right)} \quad (4.7)$$

- Equation (4.6) can be simplified further by assuming that both capillary and gravity effects are negligible to give:

$$f_w = \frac{1}{\left(1 + \frac{\mu_w}{k_w} \times \frac{k_o}{\mu_o}\right)} \quad (4.8)$$

- In terms of relative permeabilities, Eq. 4.8 becomes:

$$f_w = \frac{1}{\left(1 + \frac{\mu_w}{k_{rw}} \times \frac{k_{ro}}{\mu_o}\right)} \quad (4.9)$$

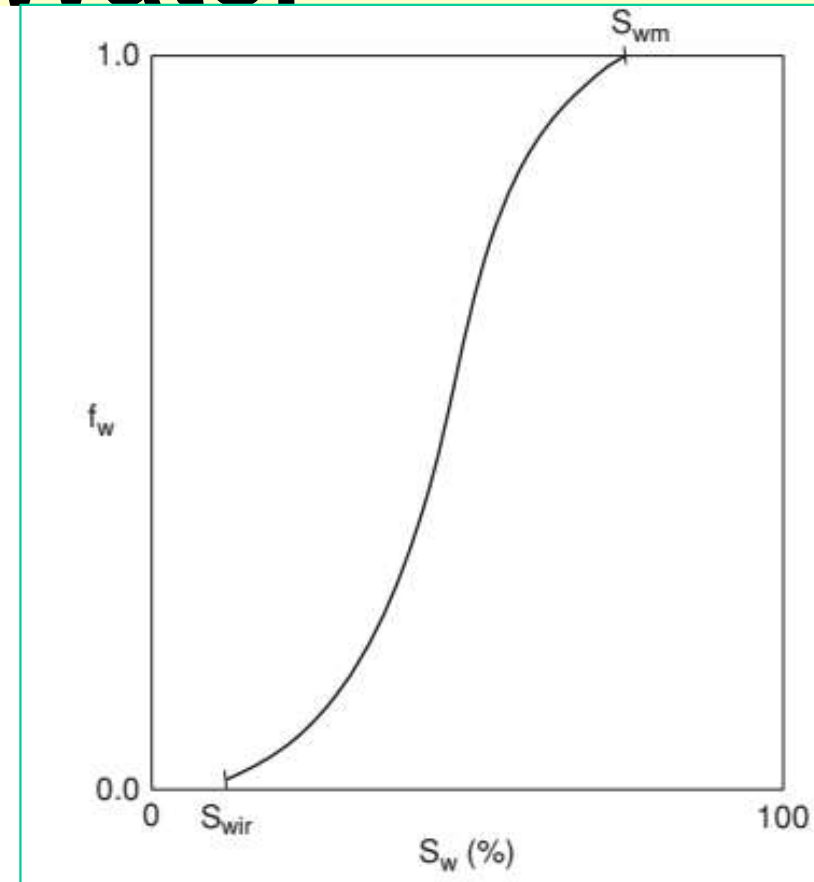


Fig. 4.1. Fractional flow curve for water with an unfavourable mobility ratio ($M > 1$) oil-water system.

Fractional Flow Equation for Oil Displaced by Water (Producing watercut at the surface, f_{ws})

- For horizontal flow (as discussed already) the fractional flow in the reservoir is:

$$f_w = \frac{1}{\left(1 + \frac{\mu_w}{k_{rw}} \times \frac{k_{ro}}{\mu_o}\right)} \quad (4.9)$$

- The producing watercut at the surface, f_{ws} , is (4.9(b))

$$f_{ws} = \frac{q_w / B_w}{q_w / B_w + q_o / B_o}$$

- where the rates are expressed in rb/d. Combining the above equation with (4.9) leads to an expression for the surface watercut as:

$$f_{ws} = \frac{1}{1 + \frac{B_w}{B_o} \left(\frac{1}{f_w} - 1 \right)} \quad (4.9(c))$$

Fractional Flow Equation for Oil Displaced by Gas

- For reservoirs in which oil is displaced by gas, a comparable set of equations can be written for fractional flow of gas. The fractional flow equation for gas flow is:

$$f_g = \frac{1}{\left(1 + \frac{\mu_g}{k_g} \times \frac{k_o}{\mu_o}\right)} + \frac{1.127 \times 10^{-3} \frac{k_o A}{\mu_o q_t} \left[\frac{\partial p_c}{\partial l} + 0.00694(\rho_o - \rho_g) \sin \alpha \right]}{\left(1 + \frac{\mu_g}{k_g} \times \frac{k_o}{\mu_o}\right)} \quad (4.10)$$

- Equation (4.10) shows that the flow of a gas in a reservoir is affected by fluid properties, rock properties, total flow rate, and structural inclination of the reservoir. If it is assumed that fluid properties and total flow rate are constant, then fractional flow of gas is only a function of gas saturation in the reservoir. A fractional flow curve of similar to Figure 4.1 can be generated for gas flow in a reservoir.

Fractional Flow Equation for Oil Displaced by Gas

- If the effects of capillary pressure are assumed to be negligible, Eq. (4.10) reduces to:

$$(4.11) \quad f_g = \frac{1}{\left(1 + \frac{\mu_g}{k_g} \times \frac{k_o}{\mu_o}\right)} + \frac{7.821 \times 10^{-6} \frac{k_o A}{\mu_o q_t} (\rho_o - \rho_g) \sin \alpha}{\left(1 + \frac{\mu_g}{k_g} \times \frac{k_o}{\mu_o}\right)}$$

- If the effects of both capillary pressure and gravity are assumed to be negligible, Eq. (4.10) simplifies to the form:

$$f_g = \frac{1}{\left(1 + \frac{\mu_g}{k_g} \times \frac{k_o}{\mu_o}\right)} \quad (4.12)$$

- In terms of relative permeability, Eq. (4.12) is written as:

$$f_g = \frac{1}{\left(1 + \frac{\mu_g}{k_{rg}} \times \frac{k_{ro}}{\mu_o}\right)} \quad (4.13)$$

Waterflood Saturation Distribution

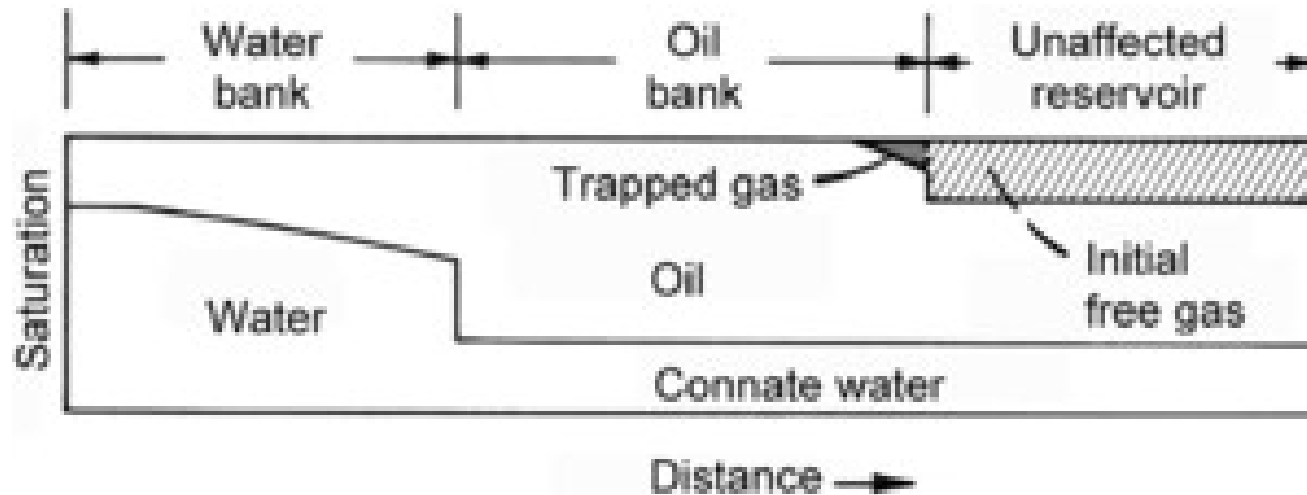


Fig. 4.x1 Saturation Profile during a waterflood

These saturation changes cause the relative permeability values and pressures to change as a function of time at each position in the rock.

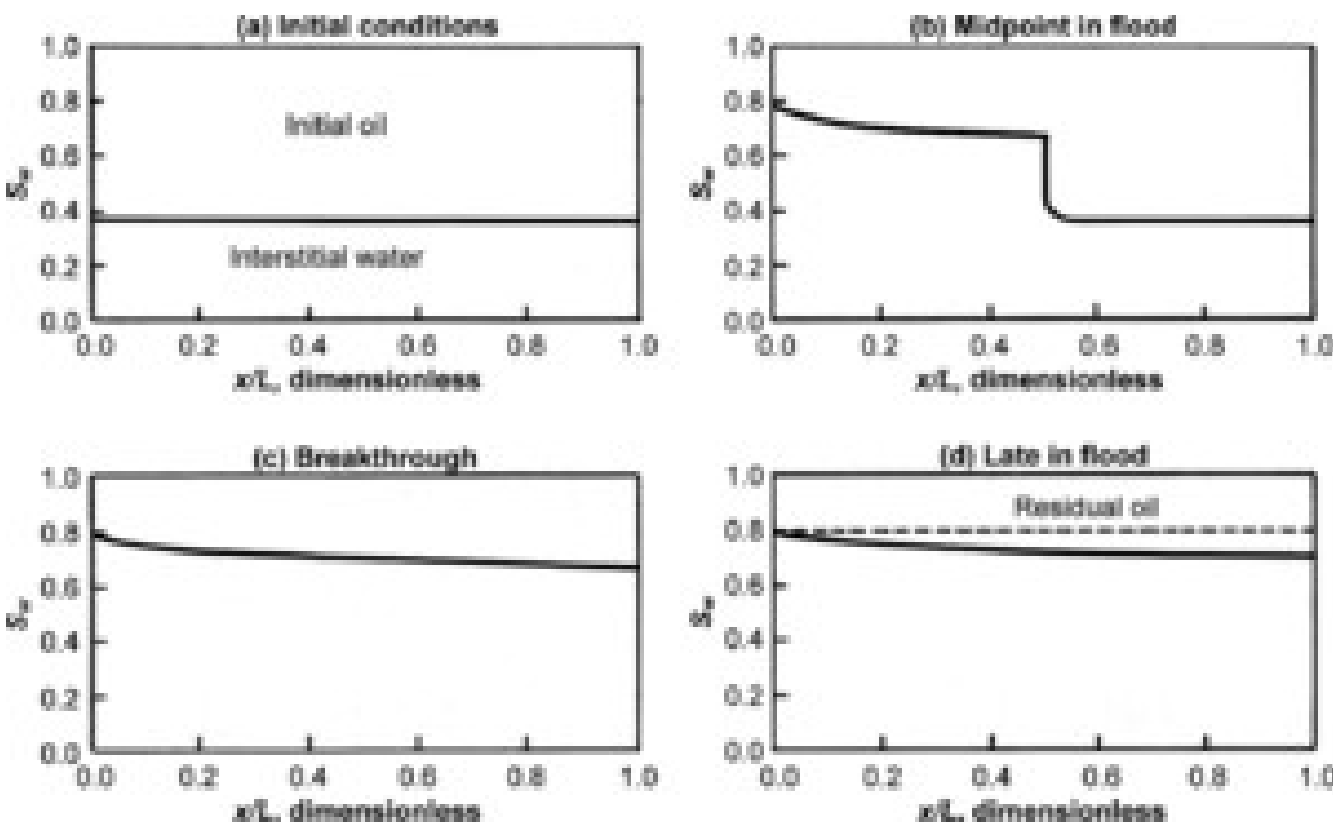


Fig. 4.x2 Saturation distribution during different stages of a waterflood. L = length, ft; x = x-direction length, ft; and x/L is dimensionless and varies from 0 to 1..

Significance of the Fractional Flow Equation & the Buckley-Leverett Equation

- The fractional flow equation is used to calculate the fraction of the total flow which is water, at any point in the reservoir, assuming the water saturation at that point is known.
- Precisely how to determine when a given water saturation plane reaches a particular point in the linear system requires the application of the displacement theory presented in the following slides.
- In 1942 Buckley and Leverett presented what is recognized as the basic equation for describing immiscible displacement in one dimension. **For water displacing oil, the equation determines the velocity of a plane of constant water saturation travelling through a linear system.**

The Buckley-Leverett Equation

- The Buckley-Leverett equation is based on the principle of conservation of mass for linear flow of a fluid (water or gas) through a reservoir at constant total flow rate.
- To illustrate the derivation of the Buckley-Leverett equation, the case of water displacing oil is used. Note that the same equation can be developed representing the case for gas displacing oil.
- Consider a volume element of a linear reservoir model shown in Figure 4.2. Let the thickness of the element be represented as Δx and located at a distance x , from the inlet face of the linear model.

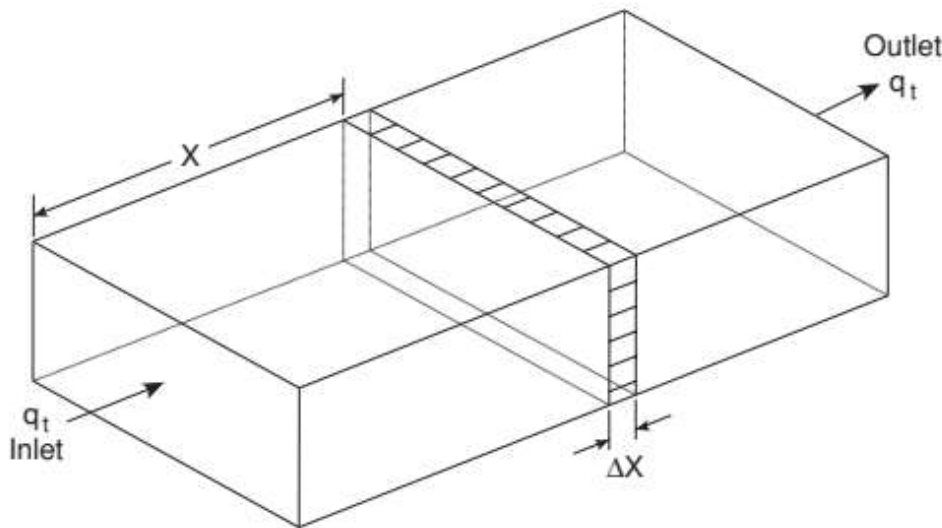


Fig. 4.2. Linear reservoir model.

The Buckley-Leverett Equation

- A volumetric balance in terms of the water phase (assuming density of water is constant) for the element of the reservoir model can be written as:

$$\left[\text{Vol. of water flowing into element in time, } \Delta t \right] - \left[\text{Vol. of water flowing out of element in time, } \Delta t \right] = \left[\text{Accumulation of water in element, } \Delta x \right] \quad (4.14)$$

- Equation (4.14) can be expressed algebraically as:

$$[f_w q_t \Delta t]_x - [f_w q_t \Delta t]_{x+\Delta x} = \left[\frac{\phi A \Delta x}{5.615} \Delta S_w \right] \quad (4.15)$$

- where q_t is in RB/D; Δt in time interval in days; A in ft². Rearranging Eq. (4.15) gives
$$\left[\frac{\phi A}{5.615 q_t} \frac{\Delta S_w}{\Delta t} \right] = - \frac{[f_w]_{x+\Delta x} - [f_w]_x}{\Delta x} \quad (4.16)$$
- Taking limits as and yields the continuity equation:

$$\frac{\phi A}{5.615 q_t} \left(\frac{\partial S_w}{\partial t} \right)_x = - \left(\frac{\partial f_w}{\partial x} \right)_t \quad (4.17)$$

- The fractional flow of water is a function of water saturation only if fluid properties and total flow rate are constant.

The Buckley-Leverett Equation

- By application of chain rule, $f_w = f_w(S_w)$ can be expressed as:

$$\left(\frac{\partial f_w}{\partial x}\right)_t = \left(\frac{\partial f_w}{\partial S_w}\right)_t \left(\frac{\partial S_w}{\partial x}\right)_t \quad (4.18) \quad \frac{\phi A}{5.615 q_t} \left(\frac{\partial S_w}{\partial t}\right)_x = -\left(\frac{\partial f_w}{\partial x}\right)_t \quad (4.17)$$

- Substituting Eq. (4.18) into Eq. (4.17) and re-arranging gives:

$$\left[\frac{\partial S_w}{\partial t}\right]_x = -\frac{5.615 q_t}{\phi A} \left[\left(\frac{\partial f_w}{\partial S_w}\right)_t \left(\frac{\partial S_w}{\partial x}\right)_t \right] \quad (4.19)$$

- Eq. (4.19) gives water saturation as a function of time at a given location. A more useful equation expressing water saturation as a function of location at a given time can be developed from Eq. (4.19). For any displacement, the distribution of water saturation is a function of both location and time. This is represented as:

$$S_w = S_w(x, t) \quad \text{and the total derivative of } S_w \text{ is then:}$$

- The total derivative of S_w is then:

$$dS_w = \left(\frac{\partial S_w}{\partial x}\right)_t dx + \left(\frac{\partial S_w}{\partial t}\right)_x dt \quad (4.20)$$

The Buckley-Leverett Equation

- Since the focus is on a fixed water saturation, then $dS_w = 0$. And Eq. (4.20) becomes:

$$0 = \left(\frac{\partial S_w}{\partial x} \right)_t dx + \left(\frac{\partial S_w}{\partial t} \right)_x dt \quad (4.21)$$

- By re-arrangement, Eq. (4.21) becomes:

$$(4.22) \quad \left(\frac{dx}{dt} \right)_{S_w} = - \frac{\left(\frac{\partial S_w}{\partial t} \right)_x}{\left(\frac{\partial S_w}{\partial x} \right)_t}$$

- Substituting Eq. (4.22) into Eq. (4.19) gives:

$$(4.23) \quad \left(\frac{dx}{dt} \right)_{S_w} = \frac{5.615 q_t}{\phi A} \left(\frac{\partial f_w}{\partial S_w} \right)_t$$

- Since the total flow rate is assumed to be constant, then fractional flow of water is independent of time. Hence,

$$\left(\frac{\partial f_w}{\partial S_w} \right)_t = \frac{df_w}{dS_w}$$

- Equation (4.23) then becomes:

$$(4.24) \quad \left(\frac{dx}{dt} \right)_{S_w} = \frac{5.615 q_t}{\phi A} \frac{df_w}{dS_w}$$

- Equation (4.24) is the Buckley-Leverett equation. It is also called the frontal advance equation.

Significance of the Fractional Flow Equation & the Buckley-Leverett Equation

- For incompressible displacement, the velocity of a plane of constant water saturation traveling through a linear system was given by:

$$\left(\frac{dx}{dt} \right)_{S_w} = \frac{5.615 q_t}{\phi A} \frac{df_w}{dS_w} \quad (4.24) \quad v_{S_w} = \frac{dx}{dt} \Big|_{S_w} = \frac{q_t}{A\phi} \frac{df_w}{dS_w} \Big|_{S_w}$$

- This is the equation of Buckley-Leverett which implies that, for a constant rate of water injection ($q_t = q_i$), the velocity of a plane of constant water saturation is directly proportional to the derivative of the fractional flow equation evaluated for that saturation.
- Integration of Eq. (4.24) yields a useful form of the Buckley-Leverett equation:

$$x = \frac{5.615 q_i t}{\phi A} \left(\frac{df_w}{dS_w} \right)_{S_w} \quad (4.25) \quad x_{S_w} = \frac{W_i}{A\phi} \frac{df_w}{dS_w} \Big|_{S_w}$$

- where W_i is the cumulative water injected and it is assumed, as an initial condition, that $W_i = 0$ when $t = 0$.

The Buckley-Leverett Equation

- Therefore, at a given time after the start of injection ($W_i = \text{constant}$) the position of different water saturation planes can be plotted, using Eqn. (4.25), merely by determining the slope of the fractional flow curve for the particular value of each saturation.
- A similar equation can be written for gas displacement of oil as follows:

$$x = \frac{5.615q_it}{\phi A} \left(\frac{df_g}{dS_g} \right)_{S_g} \quad (4.26) \quad x = \frac{5.615q_it}{\phi A} \left(\frac{df_w}{dS_w} \right)_{S_w} \quad (4.25)$$
- Equation (4.25) can be used to calculate the distribution of water saturation as a function of time in a linear reservoir under water injection or aquifer influx. **The distance travelled by a given saturation in a specified time interval is proportional to the slope of the fractional flow curve at that saturation assuming the total flow rate and reservoir properties are constant.**
- Using this approach, the distribution of water saturation in the reservoir as a function of time can be calculated by determining the slope of the fractional flow curve at that saturation.

The Buckley-Leverett Equation

- There is a mathematical difficulty encountered in applying this technique which can be appreciated by considering the typical fractional flow curve shown in Fig. 4.1 in conjunction with Eq. (4.25).
- Since there is frequently a point of inflexion in the fractional flow curve then the plot of df_w/dS_w versus S_w will have a maximum point, as shown in Fig. 4.1(c).
- Using Eqn. (4.25) to plot the saturation distribution at a particular time will therefore result in the solid line shown in Fig. 4.3. This bulbous saturation profile is physically impossible since it indicates that multiple water saturations can co-exist at a given point in the reservoir. What actually occurs is that the intermediate values of the water saturation, which as shown in Fig. 4.1(a) have the maximum velocity, will initially tend to overtake the lower saturations resulting in the formation of a saturation discontinuity or shock front.

The Buckley-Leverett Equation

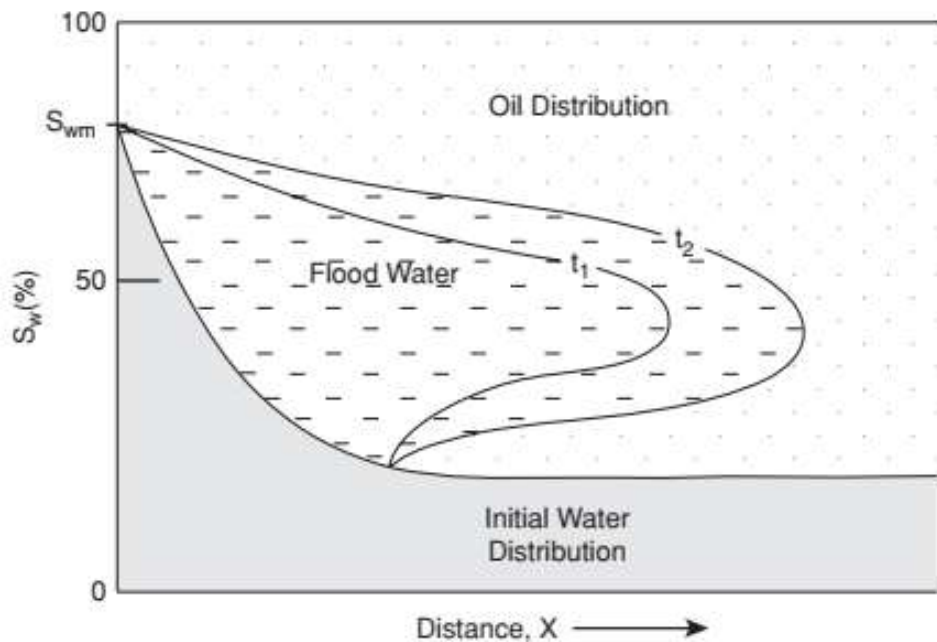


Fig. 4.3 Saturation distribution based on frontal advance equation

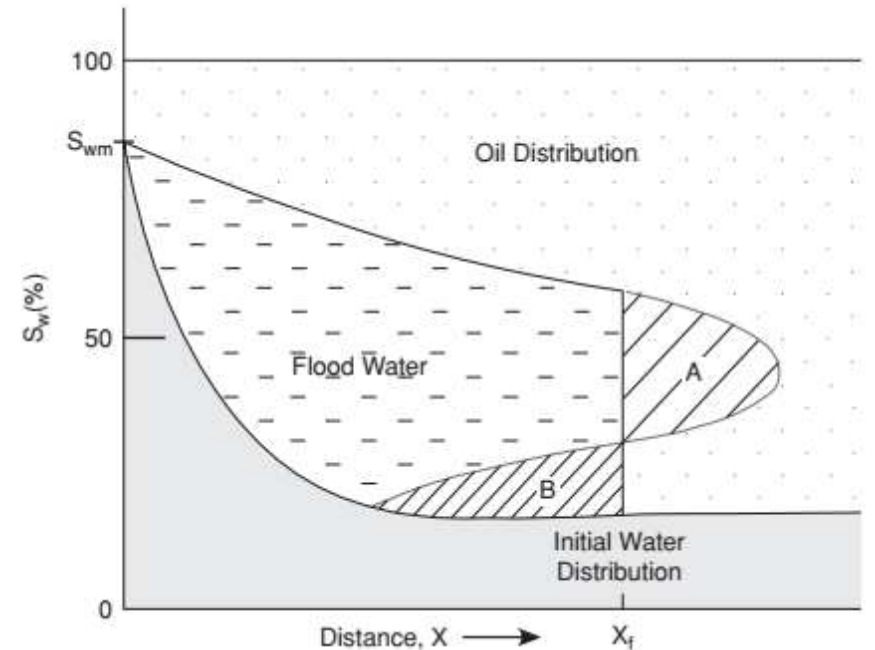


Fig. 4.4 Location of the flood front as determined by material balance

Fig. 4.1c: Saturation derivative of a typical fractional flow curve

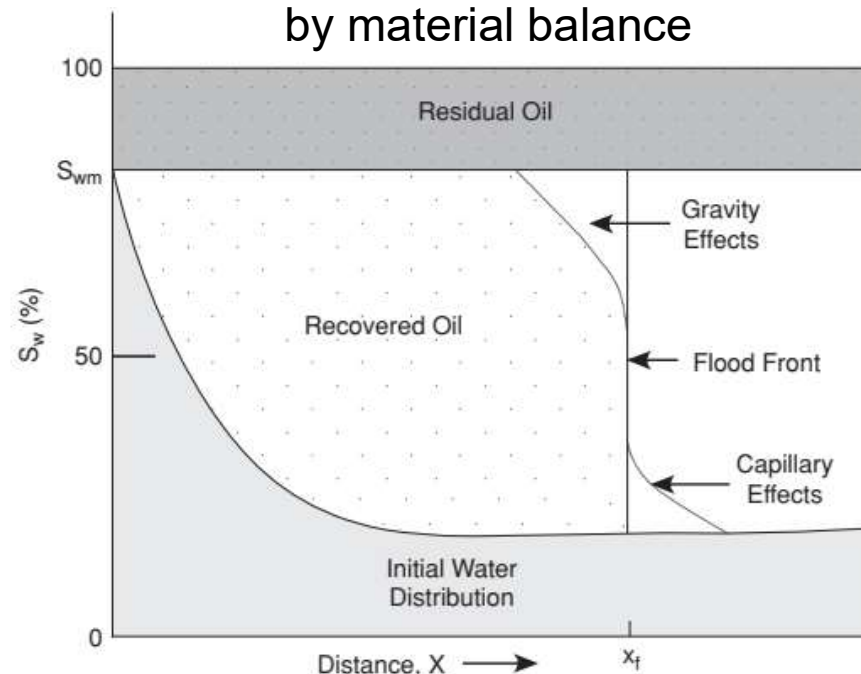
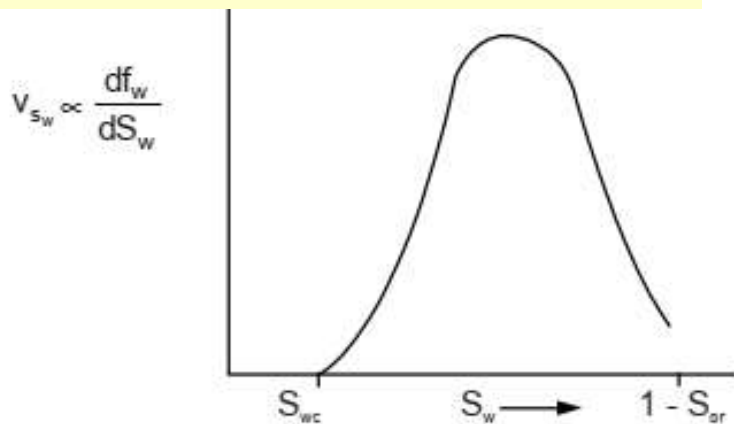
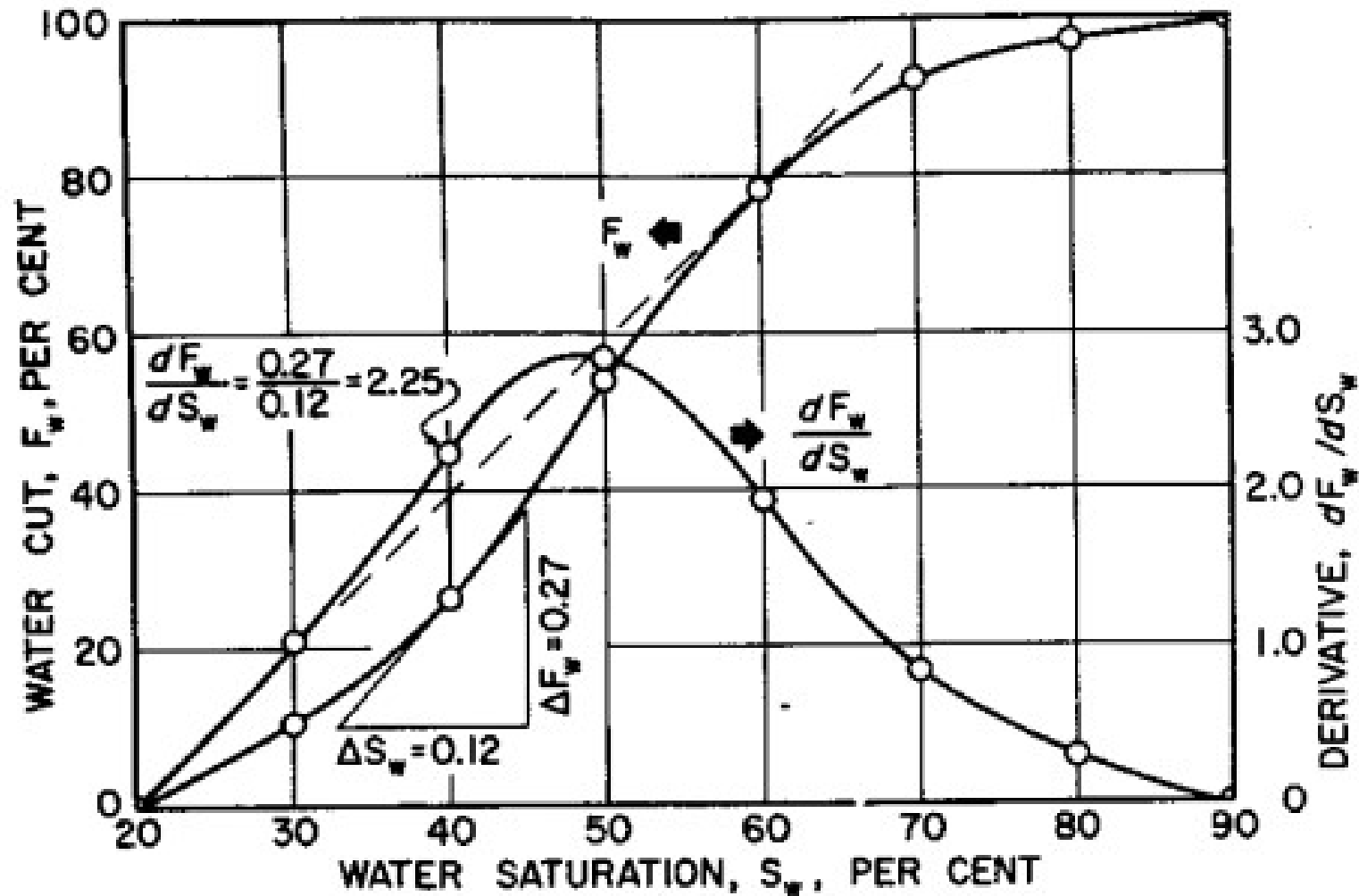


Fig. 4.5 Location of the flood front with capillary and gravity effects.

The Buckley-Leverett Equation

- Because of this discontinuity the mathematical approach of Buckley-Leverett, which assumes that S_w is continuous and differentiable, will be inappropriate to describe the situation at the front itself. Behind the front, however, in the saturation range: $S_{wf} < S_w < 1 - S_{or}$
- where S_{wf} is the shock front saturation, Eqs. (4.24) and (4.25) can be applied to determine the water saturation velocity and position.
- To draw the correct water saturation profile using the Buckley Leverett technique requires the determination of the vertical dashed line, shown in Fig. 4.4, such that the shaded areas A and B are equal. The dashed line then represents the shock front saturation discontinuity. This is because capillary and gravity effects were assumed to be negligible.
- If capillary and gravity effects are considered, the distribution of water saturation at the flood front is more gradual as represented in Figure 4.5.

The Buckley-Leverett Equation



- Fig. 4.6 Illustrating how to obtain the derivative (df_w/dS_w) from its Fractional flow curve

The Buckley-Leverett Equation

- The higher values of saturation catch up with the lower leading to the development of a shock-front saturation discontinuity, S_{wf} . The magnitude of this saturation is dependent on the mobility ratio which is unfavourable ($M > 1$) for the displacement shown in Fig. 4.7(a). However, for a favourable mobility ratio, $M < 1$, piston-like displacement occurs such that $S_{wf} = 1 - S_{or}$, the flood-out saturation, as shown in Fig. 4.7(b).

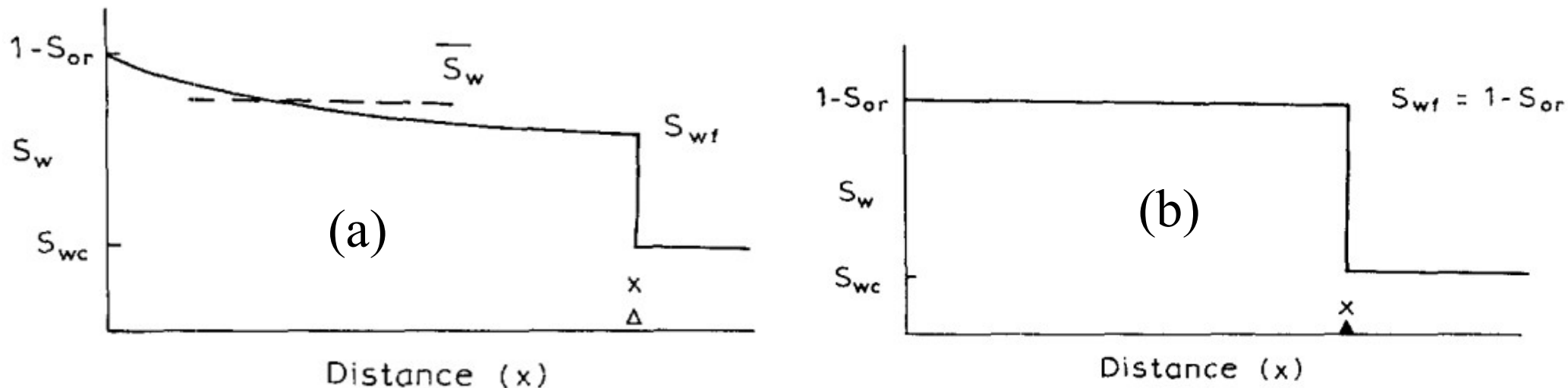


Fig. 4.7 Water saturation distribution as a function of distance, prior to breakthrough in the producing well (a) Shock front development in immiscible displacement ($M > 1$) (b) Saturation distribution for piston-like displacement ($M < 1$)

The Welge Method

- Welge's paper published in 1952 provided engineers with a simple method of applying Buckley-Leverett's theory, including the shock-front effect, in a simple fashion to calculate oil recovery as a function of the cumulative water injected.
- Welge's aim was to calculate the average water saturation, $\bar{S}_w$, (Fig. 4.7(a)), in the core plug as the flood progressed. The reason for doing so was that the difference between this and the initial saturation must be equal to the oil recovered from the plug, that is

$$N_{pd} = \bar{S}_w - S_{wc} \quad (PV) \quad (4.27)$$

- Since saturations are always expressed in pore volumes (fractions of the pore volume) then so too is the oil recovery, N_{pd} - dimensionless pore volumes. In fact, in waterdrive recovery calculations it is conventional to work in pore volumes irrespective of whether they are being performed for a core flooding experiment or a reservoir problem since the recovery is always related directly to saturation changes.

The Welge Method

- In the case of a reservoir flood, the pore volume recovery can be readily expressed as a real volume provided the dimensions of the system are known.
- Welge's method consists of integrating the saturation distribution over the distance from the injection point to the front, thus obtaining the average water saturation behind the front $\bar{S}_w$ as shown in Fig. 4.7.
- The situation depicted is at a fixed time, before water breakthrough in the producing well, corresponding to an amount of water injection W_i . At this time the maximum water saturation, $S_w = 1 - S_{or}$, has moved a distance x_1 , its velocity being proportional to the slope of the fractional flow curve evaluated for the maximum saturation which, as shown in Figs. 4.1 and 4.1(c), is small but finite.

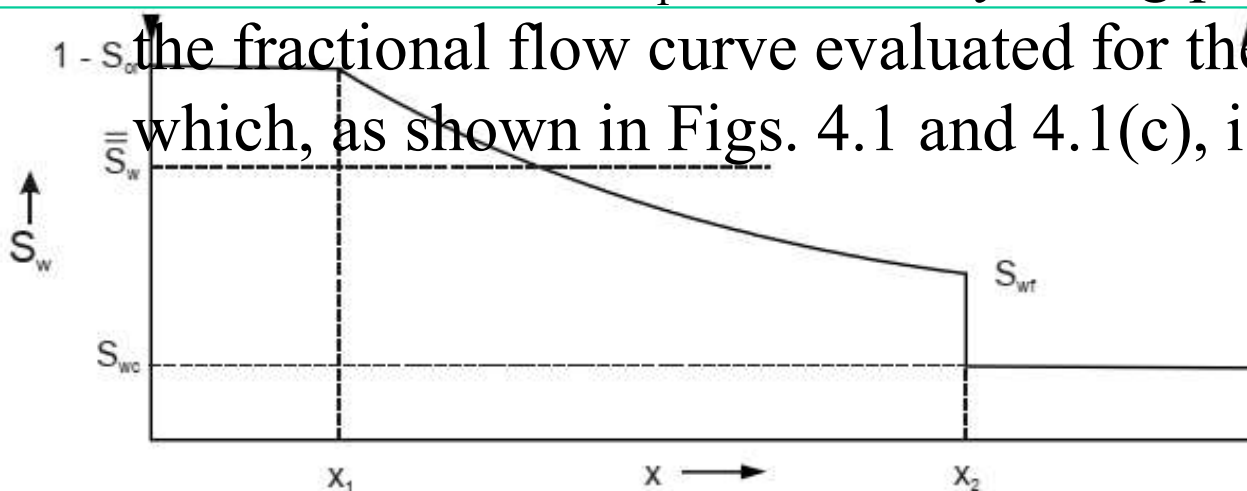


Fig. 4.7 Water saturation distribution as a function of distance, prior to breakthrough in the producing well

The Welge Method

- The flood front saturation S_{wf} is located at position x_2 measured from the injection point. Applying the simple material balance

$$W_i = x_2 A \phi (\bar{S}_w - S_{wc}) \quad \text{or} \quad \bar{S}_w - S_{wc} = \frac{W_i}{x_2 A \phi}$$

- and using Eq. (4.25) which is applicable up to the flood front at x_2 , then

$$\bar{S}_w - S_{wc} = \frac{W_i}{x_2 A \phi} = \frac{1}{\left. \frac{df_w}{dS_w} \right|_{S_{wf}}} \quad (4.28)$$

- Water Saturation at the Flood Front** - Water saturation at the flood front, can be determined graphically using the Welge method by drawing a straight line from initial water saturation tangent to the fractional flow curve as shown in Figure 4.8. If the initial water saturation is greater than the irreducible water saturation, the tangent line is drawn from the initial water saturation as shown in Figure 4.9.

The Welge Method

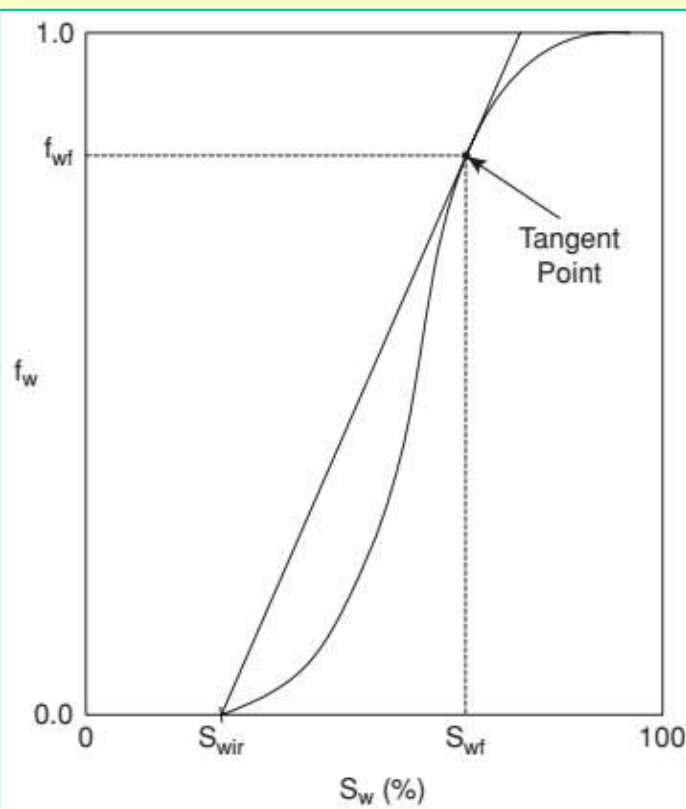


Fig. 4.8 Fractional flow curve with application of the Welge method

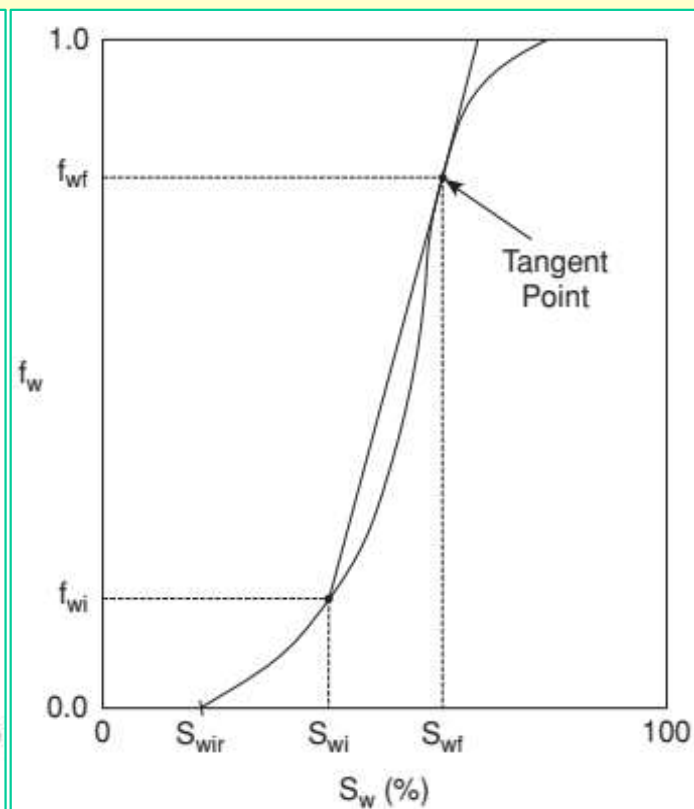


Fig. 4.9 Fractional flow curve with application of Welge method for $S_{wi} > S_{wir}$.

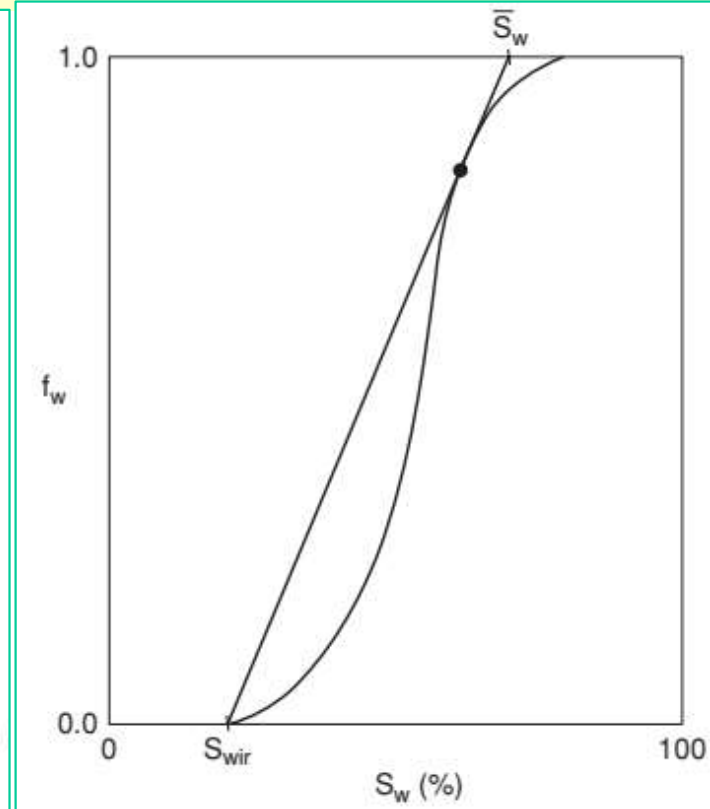


Fig. 4.10 Fractional flow curve with application of Welge method for average water saturation.

The Welge Method

- **Average Water Saturation behind the Flood Front** - By extending the tangent line drawn to the fractional flow curve as shown in Figures 4.8 or 4.9 to the point where $f_w = 1.0$, the average water saturation, behind the flood front can be determined as shown in Figure 4.10. At water breakthrough $\bar{S}_w = \bar{S}_{wbt}$, where $\bar{S}_{wbt}$ is the average water saturation in the reservoir at water breakthrough.
- **Average Water Saturation after Water Breakthrough** - The average water saturation after water breakthrough is determined as shown in Figure 4.11 by drawing a tangent line to the fractional flow curve at water saturation, S_{w2} , greater than S_{wf} but less than the maximum water saturation, S_{wm} (that is, $S_{wf} < S_{w2} < S_{wm}$). The water saturation, S_{w2} , is the saturation at the outlet end of the linear system after water breakthrough with the corresponding fractional flow of water denoted as f_{w2} . By extending the tangent line to the point where $f_w = 1.0$, the average water saturation, $\bar{S}_{wabt}$ in the system after water breakthrough is determined.

The Welge Method

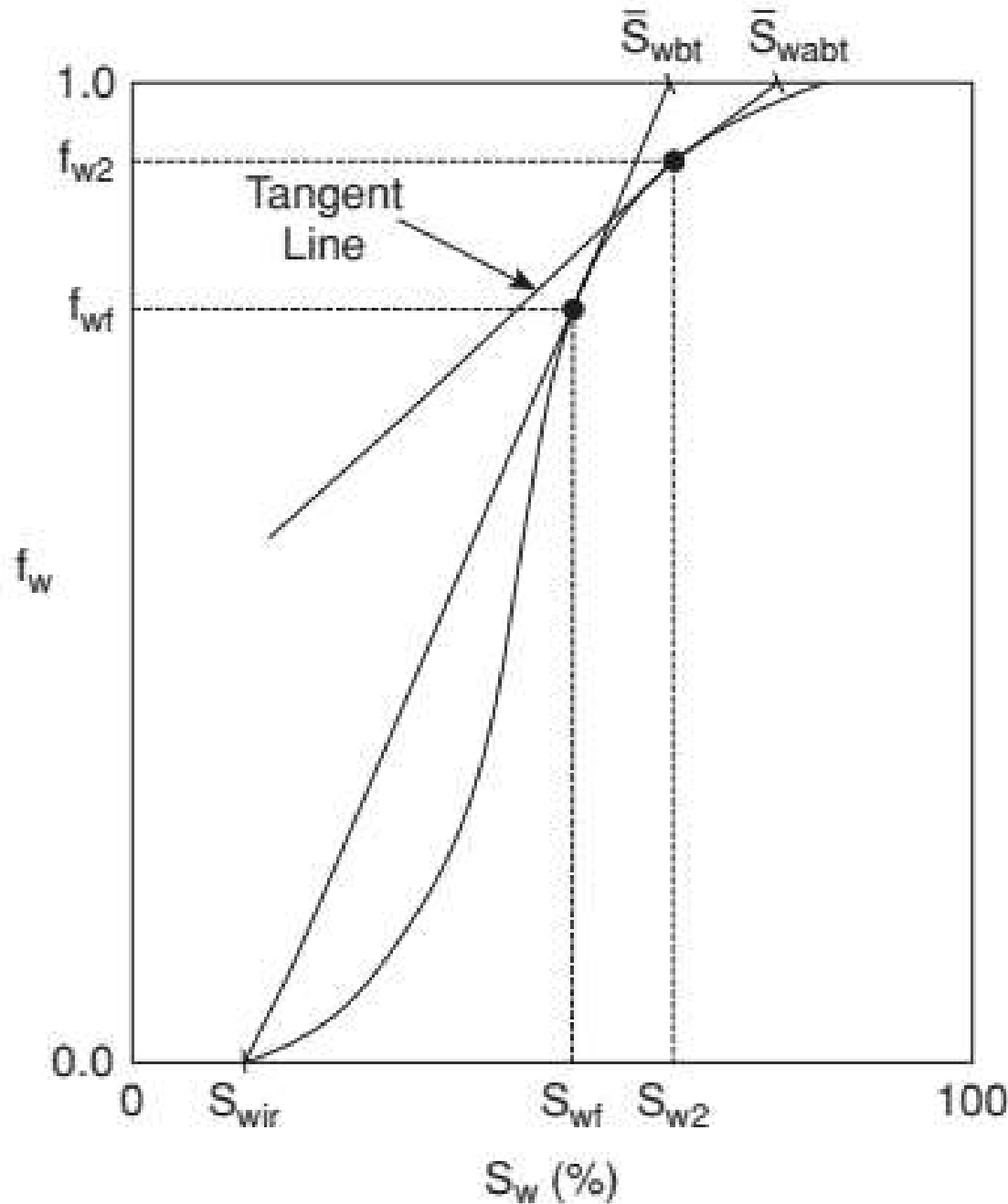


Fig. 4.11 Application of Welge method for average water saturation after water breakthrough

Oil Recovery Calculations

- **Before water breakthrough (bt) in the producing well**, Eq. (4.25) can be applied to determine the positions of planes of constant water saturation, for $S_{wf} < S_w < 1 - S_{or}$, as the flood moves through the reservoir, and hence the water saturation profile.

$$x = \frac{5.615 q_i t}{\phi A} \left(\frac{df_w}{dS_w} \right)_{S_w}$$

- **At the time of Water Breakthrough and subsequently** this equation is used in a different manner, to study the effect of increasing the water saturation at the producing well. In this case $x = L$, the length of the reservoir block, which is a constant, and Eq. (4.25) can be expressed as:

$$\frac{W_i}{LA\phi} = \frac{1}{\left. \frac{df_w}{dS_w} \right|_{S_{we}}} = W_{id} \quad (4.29)$$

- in which S_{we} is the current value of the water saturation in the producing well, Fig. 4.12, and W_{id} the dimensionless number of pore volumes of water injected ($1PV = LA\phi$).

Oil Recovery Calculations

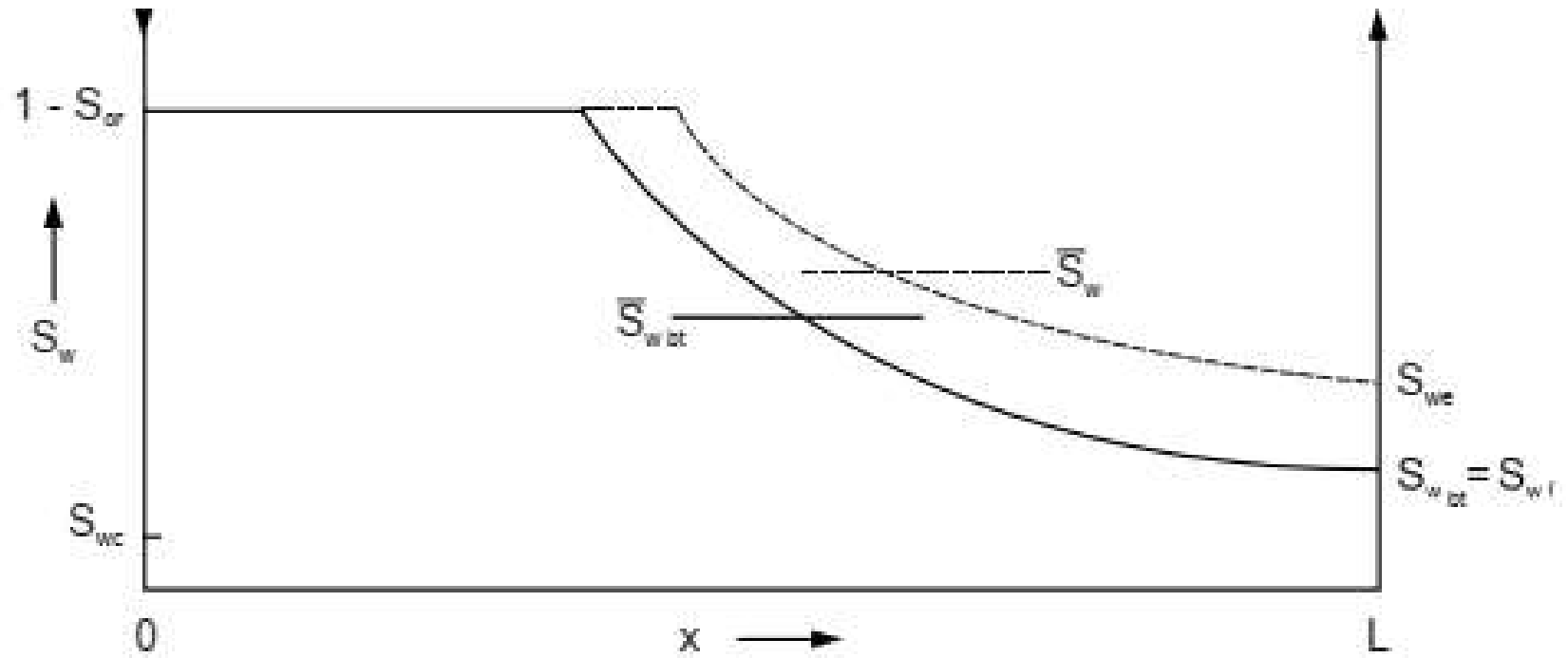


Fig. 4.12 Water saturation distributions at breakthrough and subsequently in a linear waterflood

Oil Recovery Calculations

- Before breakthrough occurs the oil recovery calculations are trivial. For incompressible displacement the oil recovered is simply equal to the volume of water injected, there being no water production during this phase.
- At the time of breakthrough the flood front saturation, $S_{wf} = S_{wbt}$ reaches the producing well and the reservoir watercut increases suddenly from zero to $f_{wbt} = f_w \big|_{S_{wf}}$ a phenomenon frequently observed in the field and one which confirms the existence of a shock front. At this time Eq. (4.25) can be interpreted in terms of Eq. (4.29) to give:

$$N_{pbt} = W_{ibt} = q_{id} t_{bt} = (\bar{S}_{wbt} - S_{wc}) = \frac{1}{\left. \frac{df_w}{dS_w} \right|_{S_{wbt}}} \quad (4.30)$$

$$t_{bt} = \frac{W_{ibt}}{q_{id}} \quad (4.31)$$

- in which all volumes are expressed, for convenience, as dimensionless pore volumes. In particular, the dimensionless injection rate facilitates the calculation of the time at which breakthrough occurs as Eq. (4.31)

Oil Recovery Calculations – After Breakthrough

- After breakthrough, L remains constant in Eq. (4.29) and S_{we} and f_{we} , the water saturation and fractional flow at the producing well, gradually increase as the flood moves through the reservoir, as shown in Fig. 4.12. During this phase the calculation of the oil recovery is somewhat more complex. There is the graphical technique and alternatively the is to use Welge's equation directly, which is equivalent to drawing the tangent, to give the oil recovery equation.
- **The graphical approach:** Choosing S_{we} as the independent variable; allow its value to increase in increments of, say, 5% above the saturation at breakthrough. Each point on the fractional flow curve, for $S_{we} > S_{wbt}$, has co-ordinates $S_w = S_{we}, f_w = f_{we}$ and, applying Eq. (4.32), Fig. 4.13 demonstrates that the tangent to fractional flow curve intersects the line $f_w = 1$ to give the current value of the average water saturation in the reservoir block, $\bar{S}_w$

Oil Recovery Calculations – After Breakthrough

- After breakthrough, The graphical approach continued: For each new value of S_{we} the corresponding value of $\bar{S}_w$ is determined graphically and the oil recovery calculated as: $N_{pd} = \bar{S}_w - S_{wc}$
- The reciprocal of the slope of the fractional flow curve, for each value of S_{we} , gives W_{id} , the number of pore volumes of water injected, Eq. (4.29). This allows a time scale to be attached to the recovery since, $W_{id} = q_{id} t$

$$\bar{S}_w = S_{we} + (1 - f_{we}) \frac{1}{\left. \frac{df_w}{dS_w} \right|_{S_{we}}} \quad (4.32)$$

$$\bar{S}_w = S_{we} + (1 - f_{we}) W_{id}$$

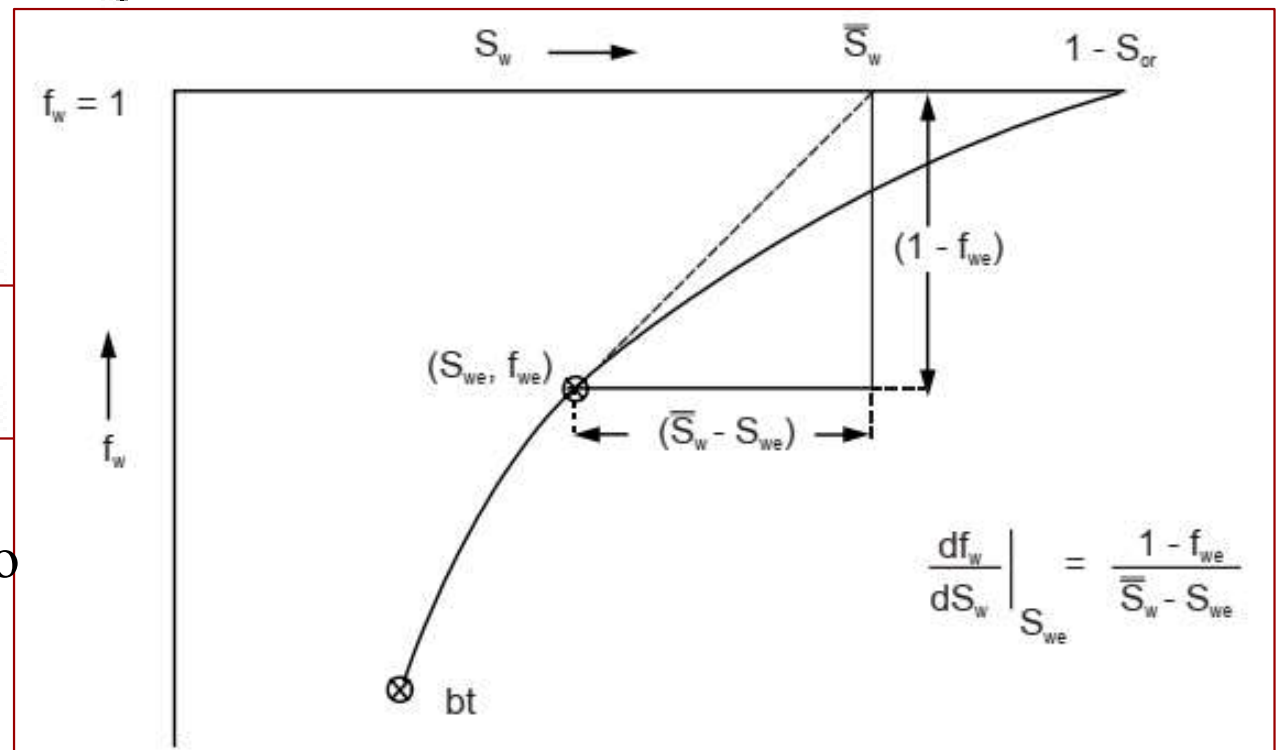


Fig. 4:13 Application of the Welge graphical technique to determine the oil recovery after water breakthrough

Oil Recovery Calculations – After Breakthrough

- After breakthrough, **Direct use of Welge's equation** is equivalent to drawing the tangent, to give the oil recovery equation.

$$N_{pd} = (S_{we} - S_{wc}) + (1 - f_{we})W_{id} \quad (PV) \quad (4.34)$$

- All the parameters on the right can be read directly from the fractional flow curve except the cumulative water influx, W_{id} (PV), which comes directly from the Buckley-Leverett theory as stated in Eq. (4.29):

$$W_{id} = \frac{W_i}{LA\phi} = \frac{1}{\frac{df_w}{dS_w}} \quad (PV) \quad \frac{W_i}{LA\phi} = \frac{1}{\left. \frac{df_w}{dS_w} \right|_{S_{we}}} = W_{id}$$

- This latter method is illustrated in Example 2 in which N_{pd} and W_{id} are evaluated numerically.

The Welge Method – Example 1

- The relative permeability ratio of a linear reservoir as a function of water saturation is given in Table 4.1. Plot the fractional flow curve. Using the fractional flow curve, calculate the following:
 1. Average water saturation and fractional flow of water at water breakthrough.
 2. Displacement efficiency at water breakthrough.
 3. Average water saturation and fractional flow of water after water breakthrough.
- Assume capillary pressure and gravity effects are negligible. Other reservoir and fluid properties are as follows:
 - Porosity, ϕ 0.25
 - Initial water saturation, S_w 0.20
 - Viscosity of water, μ_w 0.95 cp
 - Viscosity of oil, μ_o 1.60 cp

The Welge Method – Example 1

- Step 1: Calculate the fractional flow of water. Since capillary & gravity effects are negligible, calculate the f_w as a function of S_w using Eq. (4.9) as shown
- Step 2: Plot the fractional flow curve. The data in this Table are used to plot the fractional flow curve as shown in Figure 4.13.
- Step 3: Determine S_{wf} , f_{wf} and $\bar{S}_{wbt}$. Draw a straight line from the initial water saturation tangent to the fractional flow curve as shown in Figure 4.13. Extend this line to the point where . Read values for S_{wf} , f_{wf} and $\bar{S}_{wbt}$ as 0.62, 0.88 and 0.68 respectively as shown in Figure 4.13.
- Step 4: Calculate displacement efficiency, at water breakthrough. By definition,

$$f_w = \frac{1}{\left(1 + \frac{\mu_w}{k_{rw}} \times \frac{k_{ro}}{\mu_o}\right)}$$

Table 4.1

Sw	kro/krw	fw
0.2	∞	0
0.25	110.2	0.015
0.3	34.4	0.047
0.35	12.2	0.121
0.4	5.6	0.231
0.45	2.8	0.376
0.5	1.4	0.546
0.55	0.75	0.692
0.6	0.35	0.828
0.65	0.12	0.933
0.7	0.02	0.988
0.75	0	1

$$E_D = \frac{0.68 - 0.20}{1 - 0.20} = 0.60$$

$$E_D = \frac{S_{wbt} - S_{wi}}{1 - S_{wi}}$$

The Welge Method – Example 1

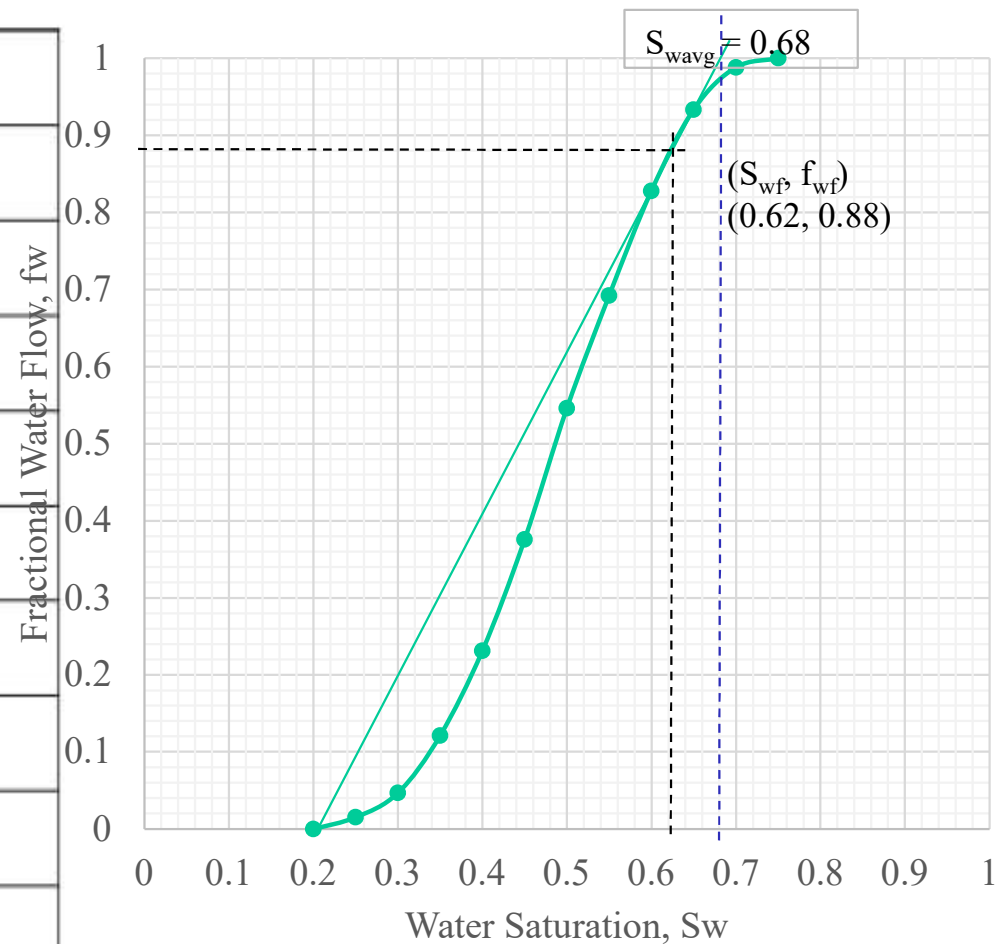
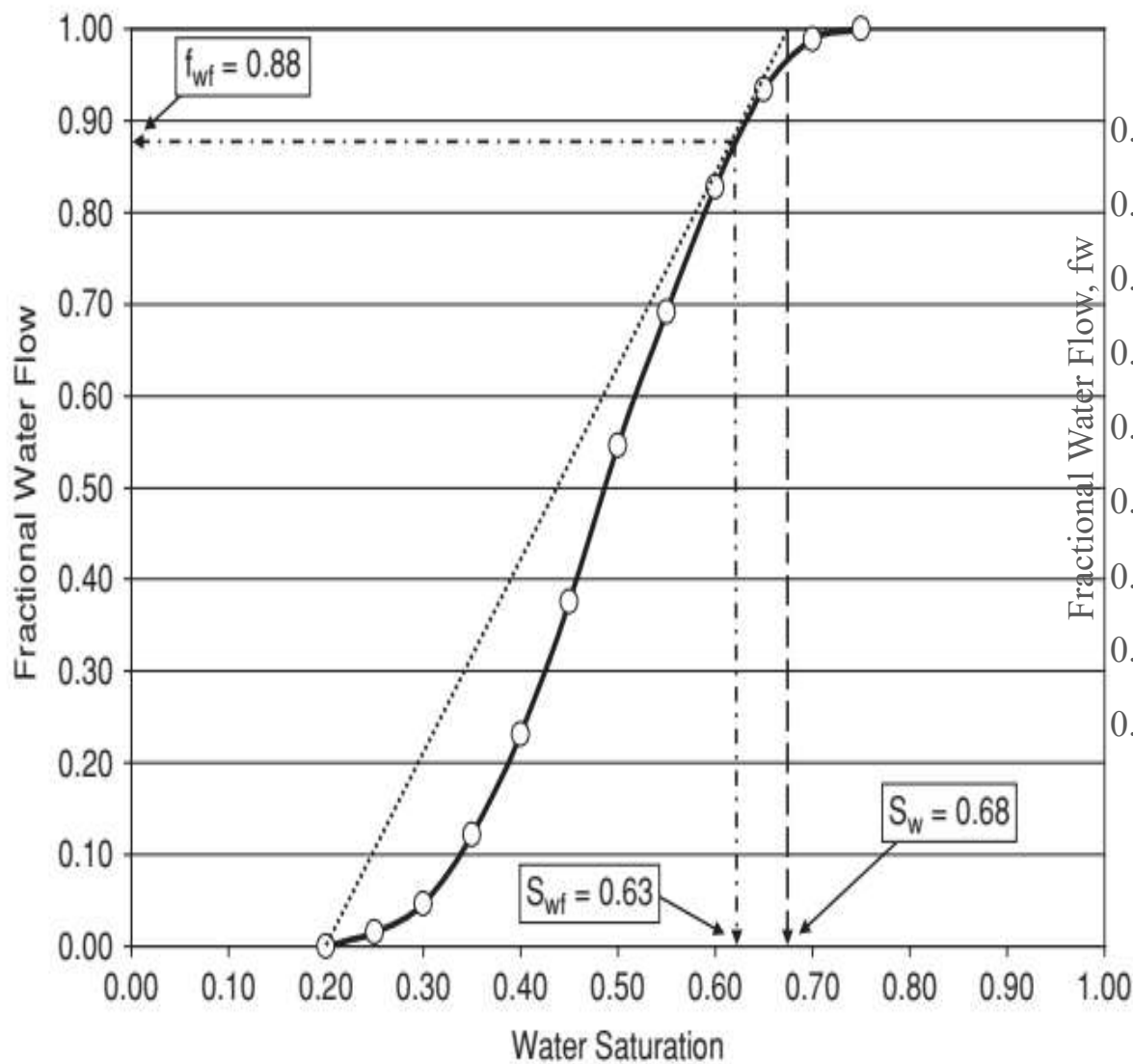
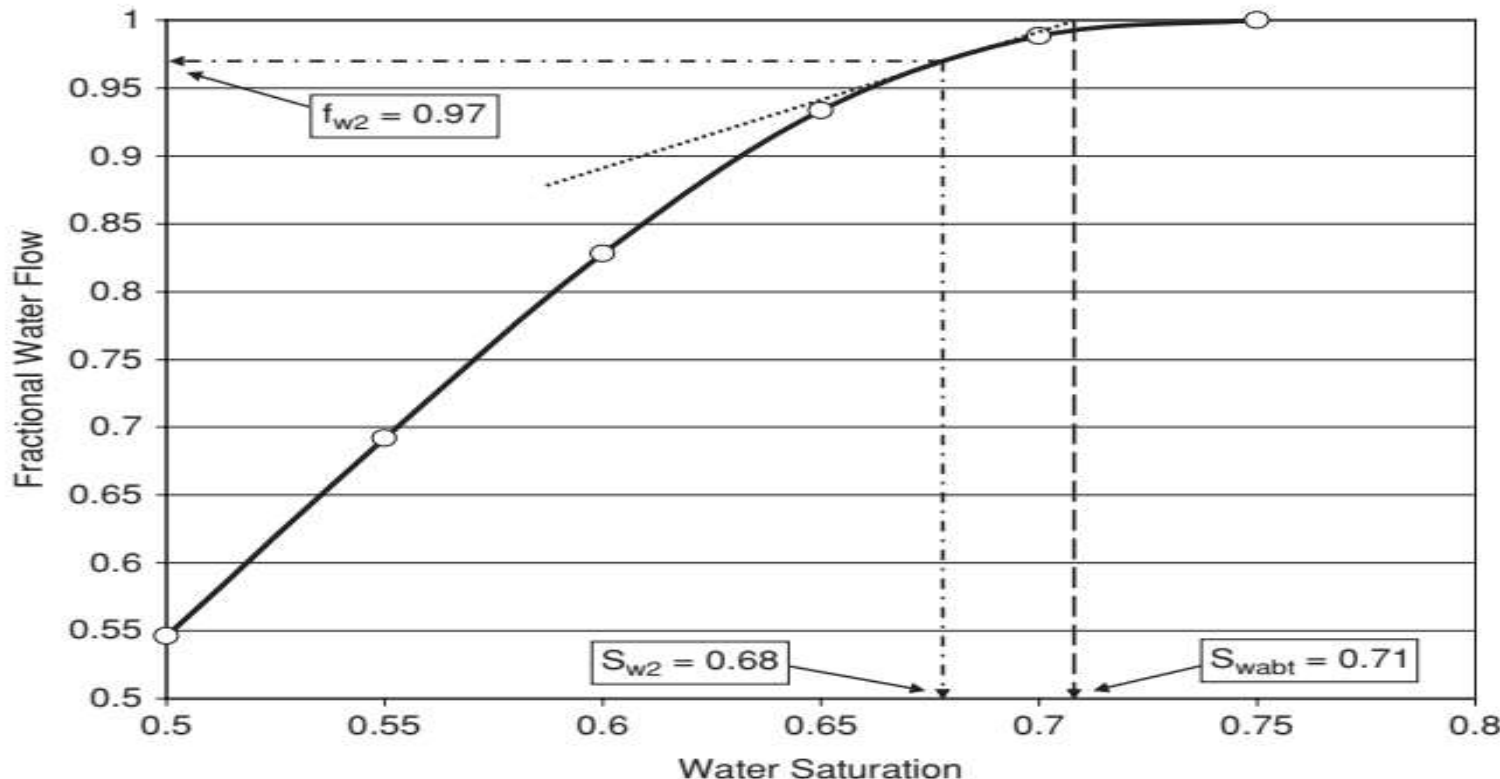


Fig. 4.13: Fractional flow curve

The Welge Method – Example 1

- Step 5. Determine fractional water flow and water saturations after water breakthrough. Expand the fractional flow curve as shown in Figure 4.14. Draw tangents at saturations greater than S_{wf} . Read the fractional flow of water and average saturations as shown in Figure 4.14. For instance, at $S_{w2} = 0.68$, $f_{w2} = 0.97$, and $\bar{S}_{wabt} = 0.71$. Note that several average water saturations after water breakthrough can be determined by drawing additional tangents at saturations greater than S_{wf} .



$$\left. \frac{df_w}{dS_w} \right|_{S_{we}} = \frac{1 - f_{we}}{\bar{S}_w - S_{we}}$$

$$\begin{aligned} \frac{df_w}{dS_w} \Big|_{S_{we}} &= (1 - 0.97) / (0.71 - 0.68) = 1 \\ W_{id} &= 1/1 = 1 \end{aligned}$$

The Welge Method – Example 1

- Step 5. Alternatively, using Equation (4.32) for $S_{we} = 0.68$ and $f_{we} = 0.97$

$$N_{pd} = (S_{we} - S_{wc}) + (1 - f_{we})W_{id} \quad (PV)$$

- Taking S_{we} , the water saturation at the producing end of the block, to rise in increments of 4% (for $S_{we} \geq S_{wbt}$) the corresponding values of W_{id} are calculated (see Table 4.2 below).

S _{we}	f _{we}	ΔS _{we}	Δf _{we}	Δf _{we} / ΔS _{we}	S* _{we}	W _{id}
0.62	0.88	0.04	0.06505	1.6262	0.64	0.6149
0.66	0.94505					
		0.04	0.04044	1.0110	0.68	0.9892
0.70	0.98549					

- In this table, values of $\Delta f_{we} / \Delta S_{we}$ have been calculated rather than determined graphically as suggested in the text. The values of S^*_{we} in column 6 are the mid points of each saturation increment, at which discrete values of W_{id} have been calculated using Eq. (4.29).

The Welge Method – Example 1

- Step 5. The oil recovery as a function of both W_{id} and time can now be determined using Eq. (4.34), as listed in the Table 4.3 below.

S^*_{we}	$S^*_{we} - S_{wc}$	f^*_{we}	$1 - f^*_{we}$	W_{id} (PV)	N_{pd} (PV)
0.64	0.44	0.9139	0.0861	0.6149	0.493
0.68	0.48	0.9689	0.0311	0.9892	0.511

- The values of f^*_{we} in column 3 of the table above have been obtained from the fractional flow curve (Fig. 4.13), for the corresponding values of S^*_{we} .

The Welge Method – Example 2

- Water is being injected at a constant rate of 1000 b/d/well in a direct line drive in a reservoir with the following rock & fluid properties.
 - Porosity, ϕ 0.18
 - Initial water saturation, S_{wc} 0.20; Residual oil saturation, S_{or} 0.20
 - Viscosity of water, μ_w 5 cp; Viscosity of oil, μ_o 0.5 cp
- The relative permeabilities for oil and water are presented in Table 4.2 and the flood pattern geometry is as follows.
 - Dip angle = 0° ; Distance between injection wells = 625 ft
 - Reservoir thickness = 40 ft; Distance between injectors and producers = 2000 ft
- Assuming that diffuse flow conditions prevail and that the injection project starts simultaneously with oil production from the reservoir:
 1. determine the time when breakthrough occurs
 2. determine the cumulative oil production as a function of both the cumulative water injected and the time.

The Welge Method – Example 2

- Step 1: Calculate the fractional flow of water. Since capillary & gravity effects are negligible, calculate the f_w as a function of S_w using Eq. (4.9) as shown
- Step 2: Plot the fractional flow curve. The data in this Table are used to plot the fractional flow curve as shown in Figure 4.15.
- Step 3: Determine S_{wf} , f_{wf} and S_{wbt} as 0.45, 0.70 and 0.56 respectively as shown in Figure 4.15. At water breakthrough, $N_{pd} = W_{id} = (0.56 - 0.2) = 0.36$
- Step 4: Calculate the water breakthrough time. Combining Equations (4.29) & (4.31) with dimensional consistency gives: $W_{id} * (L * A * \phi) = q * t_{bt}$

$$t = \frac{W_{id} \times (\text{one pore volume})}{q_i \times 5.615 \times 365} \frac{(\text{cu.ft})}{(\text{cu.ft / year})}$$

$$t = \frac{W_{id} \times 625 \times 40 \times 2000 \times .18}{1000 \times 5.615 \times 365} \text{ (years)} = 4.39 \frac{W_{id}}{1} = 1.58 \text{ years}$$

$$f_w = \frac{1}{\left(1 + \frac{\mu_w}{k_{rw}} \times \frac{k_{ro}}{\mu_o}\right)}$$

Table 4.4

Sw	krw	kro	kro/krw	fw
0.2	0	0.8	∞	0
0.25	0.002	0.61	305	0.0317
0.3	0.009	0.47	52.222	0.1607
0.35	0.02	0.37	18.5	0.3509
0.4	0.033	0.29	8.636	0.5366
0.45	0.051	0.22	4.314	0.6986
0.5	0.075	0.16	2.173	0.8215
0.55	0.1	0.12	1.2	0.8929
0.6	0.132	0.08	0.614	0.9422
0.65	0.17	0.05	0.294	0.9714
0.7	0.208	0.03	0.13	0.9872
0.75	0.251	0.01	0.04	0.996
0.8	0.3	0	0	1

The Welge Method – Example 2

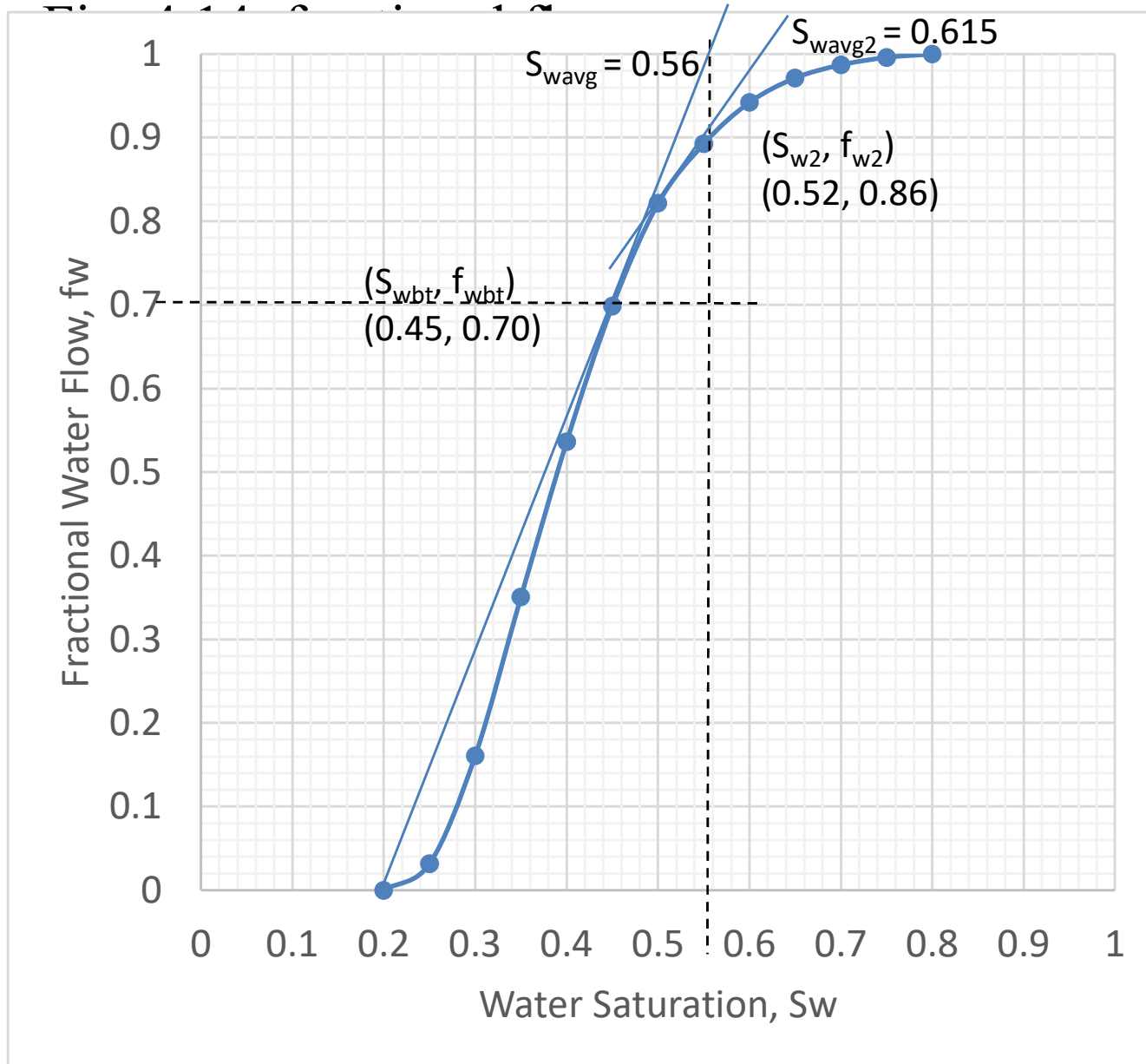
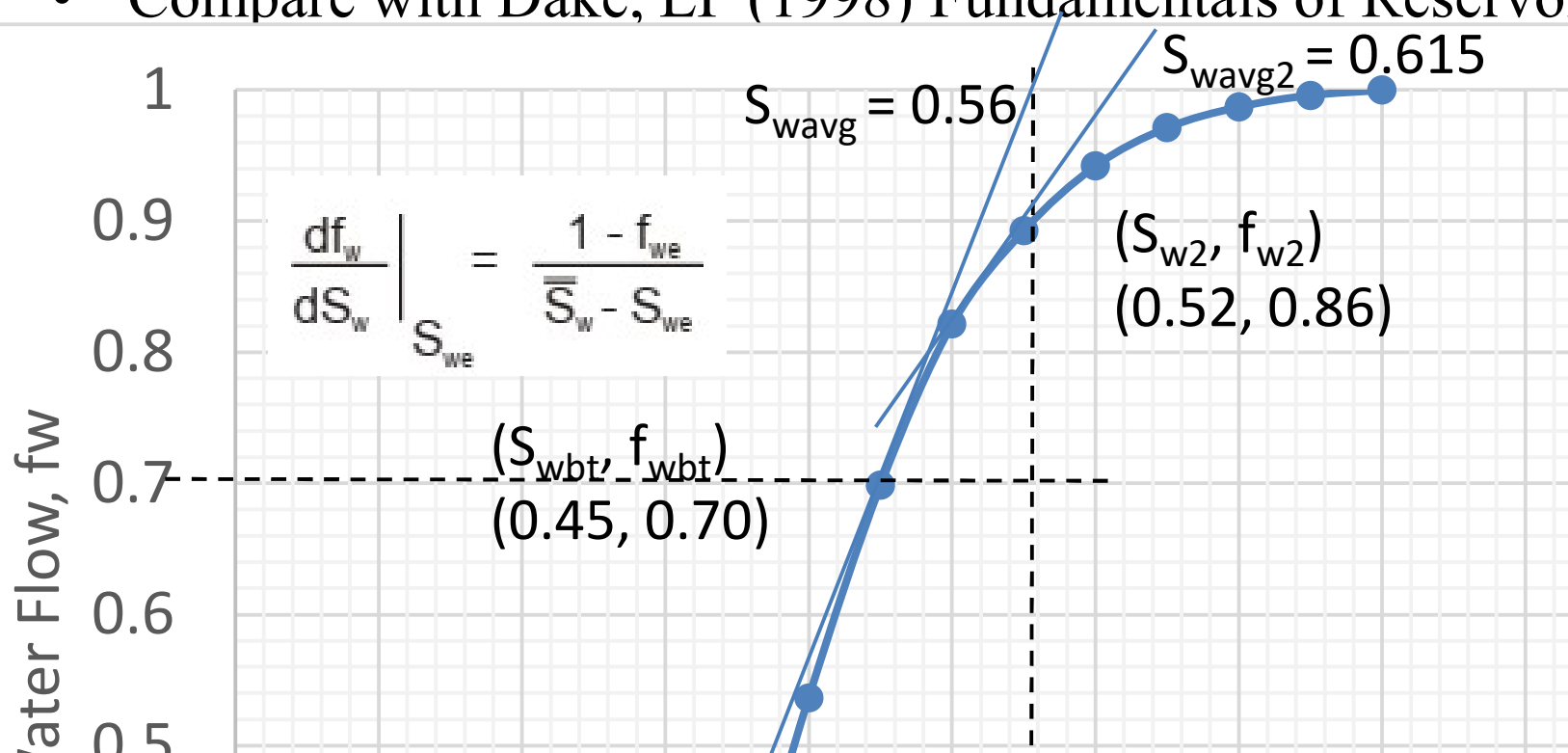


Fig. 4.15: Fractional flow curve

The Welge Method – Example 2

- Step 5. Determine the cumulative oil production as a function of both the cumulative water injected and the time. Expand the fractional flow curve as shown in Figure 4.15. Draw tangents at saturations greater than S_{wf} . Read off $S_{w2}=0.52$, $f_{w2}=0.86$ and $\bar{S}_{wa}=0.615$
- $df_w/dS_w|_{S_{we}=0.52} = (1-0.86)/(0.615-0.52) = 1.474$.
- Therefore $W_{id}=1/1.474=0.6786$; $N_{pd}=(0.52-0.2)+(1-0.86)*0.6786=0.415$
- Compare with Dake, LP (1998) Fundamentals of Reservoir Engineering



Corresponding
Time
 $t = 4.39 * W_{id}$
 $= 2.979$ years

END OF LECTURE

Secondary Recovery Methods – Lecture 5

Introduction

- Secondary recovery methods are defined as processes that are used to increase hydrocarbon recovery from the reservoir beyond primary recovery. Typical secondary recovery methods are considered to be intervention methods implemented during the primary recovery period to improve projected low hydrocarbon recovery from the primary process. Focus would be on waterflooding and gasflooding.
- Waterflooding of hydrocarbon reservoirs is generally an immiscible displacement process since water is virtually immiscible with hydrocarbons even at high pressures. Gasflooding can be an immiscible or miscible displacement process, depending on pressure and temperature of the reservoir, the type of injection gas, the composition of the injection gas, the composition of the reservoir fluid, the nature of reservoir heterogeneities, and other factors. Waterflooding and gasflooding are presented in this lecture as immiscible displacement processes.

Introduction

- Secondary recovery methods are presented in descriptive terms under the categories of well patterns, well surveillance, reservoir performance monitoring, and project management.
- The application of prediction methods such as the Stiles method, Dykstra-Parsons method, Craig-Geffen-Morse method, and other methods for the calculation of recoveries from waterflooding are not here. The Welge method for gasflooding is also not presented.
- For flood design and calculation of flood performance, these analytical methods have been rendered virtually obsolete by faster and more robust methods based on reservoir simulation.
- However, fundamental understanding of immiscible flooding processes can be gained by using very simple reservoir models to explore key factors (such as rock and fluid properties, reservoir heterogeneities, well pattern, injection rates, production rates, etc.) that influence HC recovery under secondary recovery processes

Waterflooding is the most common secondary recovery method used worldwide to improve oil recovery from reservoirs. The performance of a waterflood project in terms of oil recovery depends on many factors, such as rock wettability, rock and fluid properties, formation heterogeneities, flood patterns, composition of injection water, water injection rates, fluid production rates, etc. These factors should be considered in the design and installation of a waterflood project.

WATERFLOODING

Waterdrive

- Waterdrive in macroscopic reservoir sections occurs on the scale of flooding in hillsides rather than core plugs. At this level, there are three factors which govern the oil recovery efficiency: mobility ratio, heterogeneity and gravity.
- Precisely how they interact requires careful consideration and, can sometimes provide surprising results in enhancing or downgrading oil recovery by waterdrive.
- Correct evaluation of the influence of the three factors amounts to paying rigorous attention to detail in determining the vertical sweep efficiency across sand sections, E_V .
- The vertical sweep is implicitly input to areal simulation models in the form of thickness averaged or pseudo-relative permeabilities. The simulator is then required to solve for the areal sweep, E_A , through the successful tracking of areal fluid movement.

Planning a Waterflood

- Engineered waterdrive is the principal form of secondary recovery practiced by the industry for the obvious reason that in many producing areas water is in plentiful supply and is therefore inexpensive. Not only that, but it is usually more stable than the alternative form of secondary flooding-gas drive. Waterdrive serves two purposes in maintaining the reservoir pressure which energizes the system and in displacing the oil towards the production wells.
- Fields that have been selected for waterdrive usually have moderate to high **permeabilities**. The high flow rates also have the favourable effect of leading to relatively short project duration.
- Fields selected for development by waterdrive invariably have **low oil viscosity**, typically less than one centipoise. Considering Darcy's law, it can be seen that this condition has the same effect as high permeabilities in promoting high flow rates thereby accelerating field developments.

Planning a Waterflood

- Perhaps of greater significance, however, is the favourable influence of the low viscosity in enhancing the efficiency of water-oil displacement, at least on the microscopic scale. The parametric group which dictates the efficiency is known as the end-point mobility ratio (or simply the mobility ratio)

$$M = \frac{k'_{rw}}{\mu_w} / \frac{k'_{ro}}{\mu_o}$$

- The incorporation of the maximum end-point relative permeabilities means that, by direct application of Darcy's law, the mobility ratio represents:

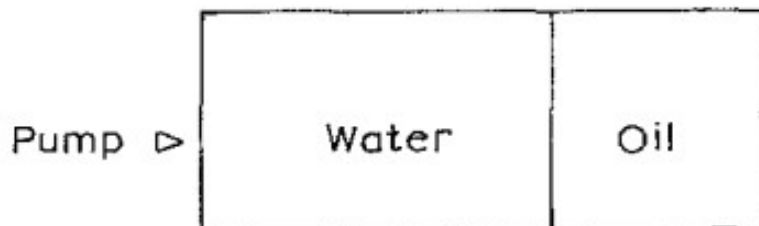
$$M = \frac{\text{maximum velocity of the displacing phase (water)}}{\text{maximum velocity of the displaced phase (oil)}}$$

- The significance of $M < 1$ is that, in a one-dimensional flooding experiment in a homogeneous core plug, the displacement will be completely stable. That is, it is the water that is being injected and pushes the oil but since $M < 1$ the water cannot travel faster than the oil and therefore displaces it in a perfect piston-like manner.

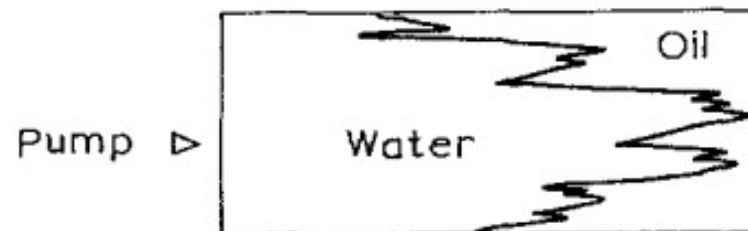
Planning a Waterflood

- This is the most favourable form of displacement and means that the total volume of movable oil: $MOV = PV(1 - S_{or} - S_{we})$ can be recovered by the injection of an equivalent volume of water. Consequently, the flooding is both rapid and efficient.
- Conversely, if $M > 1$, on account of high oil viscosity, then the waterflood is inefficient. The water can now travel faster than the oil and, since it is the water that is doing the pushing, it does so and channels through the oil in an unstable fashion. Water breaks through prematurely at the end of the plug and if $M = 40$, for instance, it might require the circulation of about 100 pore volumes of water to recover the movable oil volume.

(a) $M \leq 1$



(b) $M > 1$



**1-D waterdrive
experiment in a
homogeneous
core plug**

Planning a Waterflood

- A further advantage in selecting low oil viscosity reservoirs for waterflooding ($M < 1$) is that at abandonment, the areal sweep of water will be very high.
- It must be stressed that the piston-like displacement associated with favourable mobility ratio displacement only occurs on the microscopic, one-dimensional scale of a core flooding experiment. In the flooding of macroscopic reservoir sections account must also be taken of the heterogeneity and gravity in accounting for the overall flooding efficiency.
- Other considerations include: **low volatility oil** because gas oil ratios are of a tolerable level and decline in reservoir pressure can be allowed without risking failing below the bubble point; **large deep oil accumulations which are significantly overpressured** at initial conditions as overpressure serves as a source of free energy.

Planning a Waterflood

- A further advantage in selecting low oil viscosity reservoirs for waterflooding ($M < 1$) is that at abandonment, the areal sweep of water will be very high.
- It must be stressed that the piston-like displacement associated with favourable mobility ratio displacement only occurs on the microscopic, one-dimensional scale of a core flooding experiment. In the flooding of macroscopic reservoir sections account must also be taken of the heterogeneity and gravity in accounting for the overall flooding efficiency.

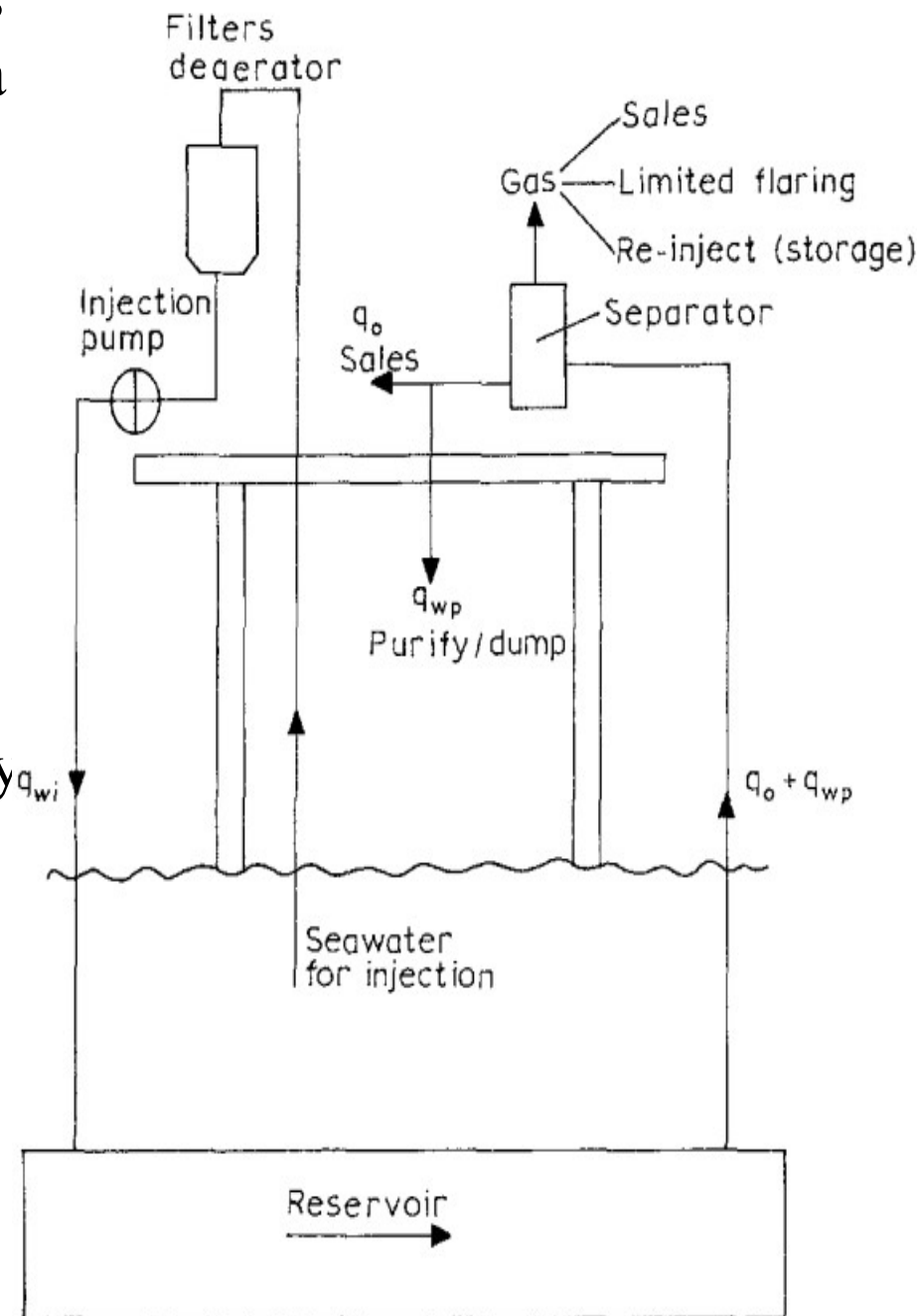
Engineering Design of Waterflood Projects

- Production Plateau Rate: While this is not purely a technical decision as Economists and Management tend to dominate decision making, the reservoir engineer is involved in making sure that the initial rates will not cause any reservoir damage, although this is extremely difficult to prove, and ensure that the plateau rate chosen is commensurate with the well numbers and injection / production facilities installed.
- Number of production / injection wells: Onshore, there is no great pressure in making the decision: wells can usually be drilled, as required. Offshore, however, the correct decision must be made in advance and, for fields developed from a fixed platform, the well deck containing the drilling slots and wellhead equipment designed accordingly. Where the wells should be drilled depends on the initial results of the RFT surveys which enable producers and injectors to be appropriately located in communicating regions of the field.

Engineering Design of Waterflood Projects

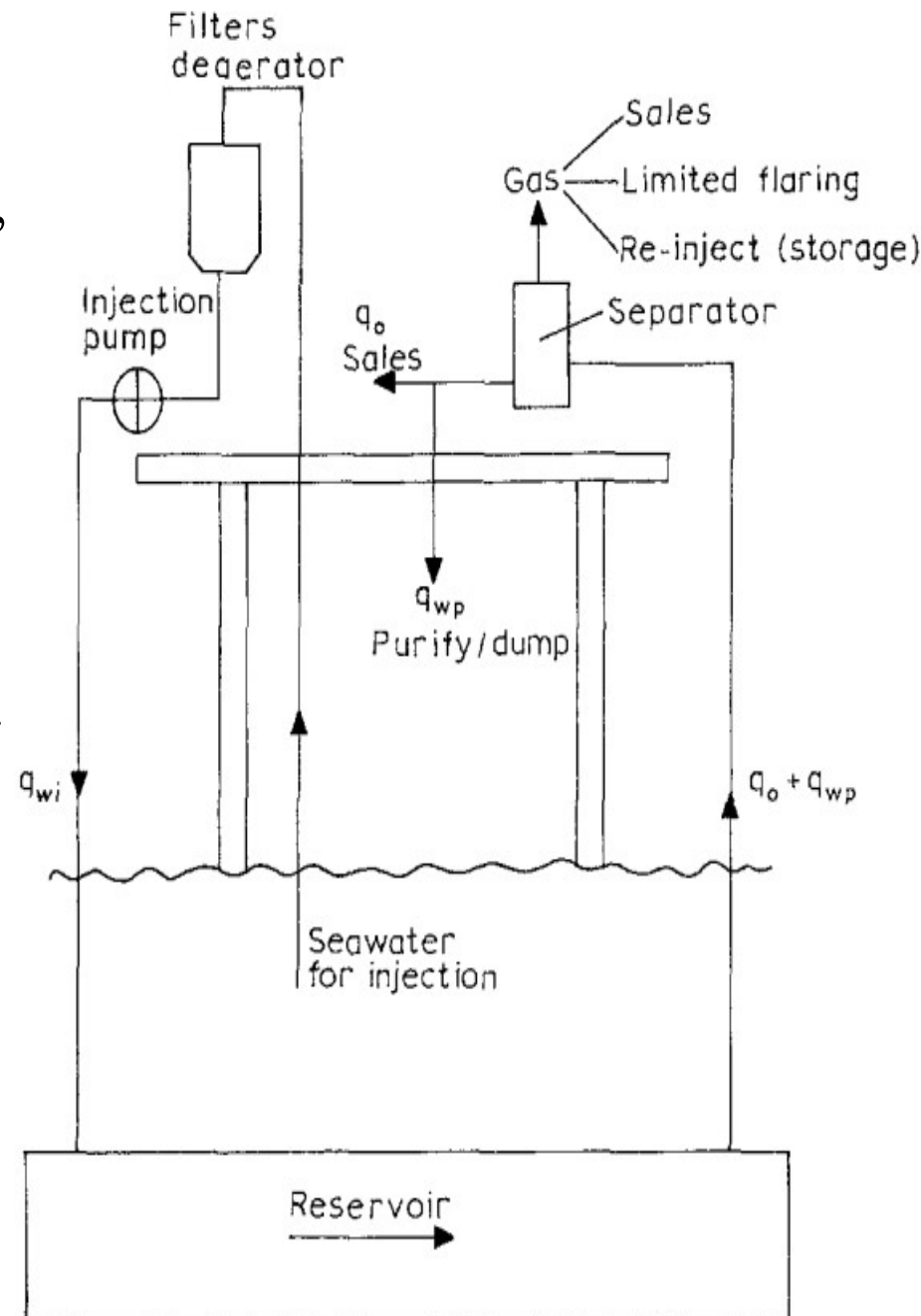
- Surface production/injection facilities: This is the most crucial aspect in the design of a waterdrive, especially offshore, & requires close liaison between the reservoir engineers, who specify the required capacities of the production/injection facilities, and the project engineers who design and install the equipment. What is vital is that the engineer is fully aware of the need to specify the capacities of the individual components and ensure that they are compatible with one another and with the reservoirs being produced. The capacities that concern the reservoir engineer are

- Water injection
- Separator train
- Water disposal



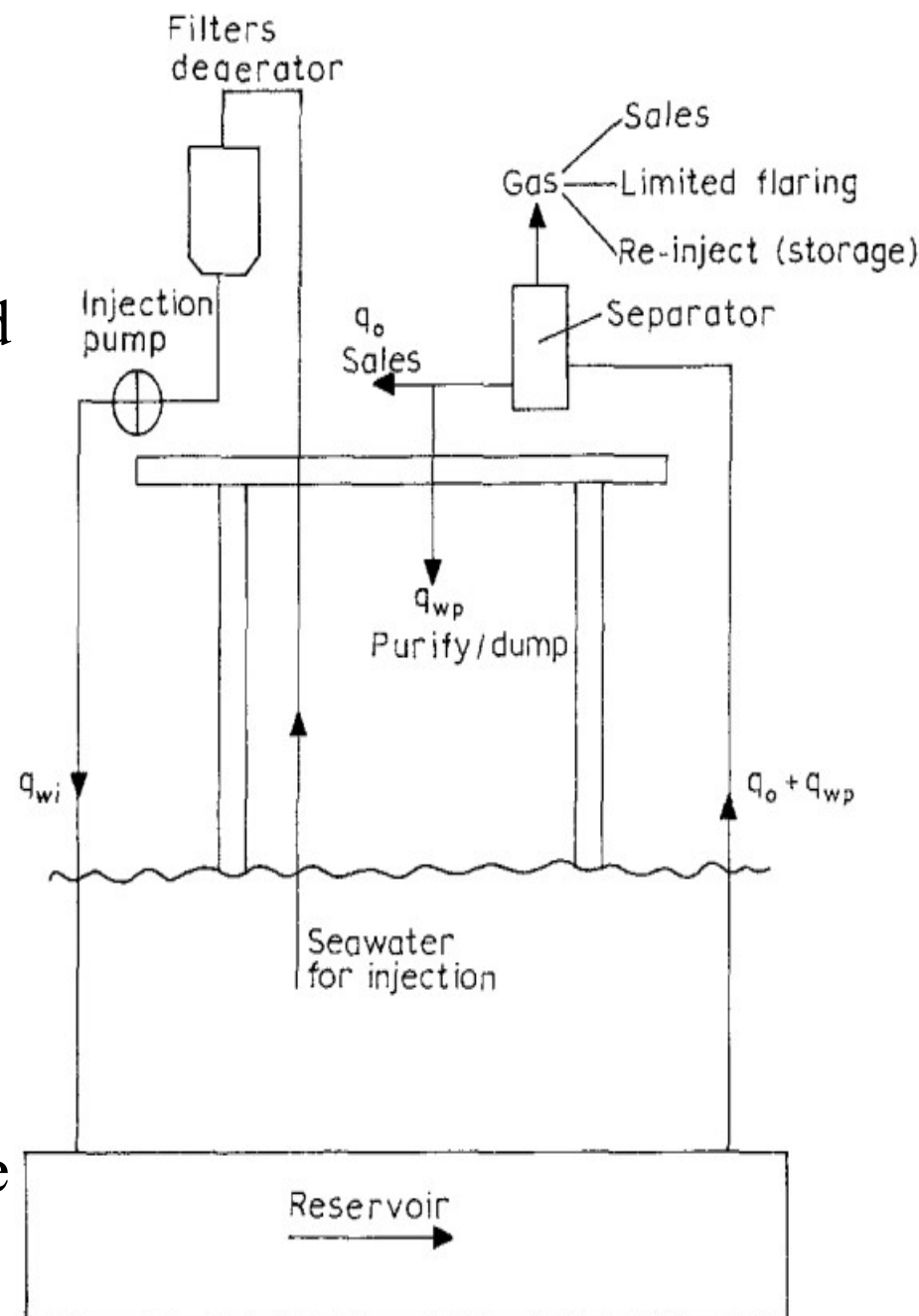
Engineering Design of Waterflood Projects

- Surface production/injection facilities: The seawater to be injected is pumped to the surface from a depth of usually 100-200 ft, at which its plankton content is measured to be at an acceptably low level. After removal of corrosive oxygen (deaerator) the water is filtered to remove solids (organic debris) that might otherwise plug the formation and reduce injectivity. What constitutes an acceptable minimum size of solids particles depends largely on the nature of the formation and, in particular, the pore throat size. Standards, of course, will vary from field to field and must be investigated separately. The water is then injected into the reservoir by a battery of injection pumps.



Engineering Design of Waterflood Projects

- Surface production/injection facilities: On the production side of the platform, the capacities of interest are those of the separator train and the oily water disposal equipment. Water, oil and gas are separated in several stages at pressures and temperatures designed to optimize the volume of stabilized crude oil. The water leaving the separators still has a fairly high content of dispersed oil which must be reduced to some low, environmentally acceptable level, usually about 50 ppm, before dumping the water back into the sea. Earlier equipment to remove the oil from the water was rather bulky and temperamental but now many platforms are equipped with compact and efficient hydro cyclones to perform this task.



Waterflood Patterns

- Waterflood patterns are based on the arrangement of injectors and producers in a waterflooded reservoir. The patterns are generally designed to improve areal sweep efficiency.
- It is important to note that waterflood patterns evolved from conversion of onshore oil fields developed with regular well spacing from primary production to secondary production by waterflooding. The criteria for converting these oil fields to waterflooding were to minimize drilling of new wells, maximize water injection, and improve oil recovery which depended on the density of existing wells, reservoir properties, and projected rate of oil recovery.
- Generally, pattern flooding is not efficient for development of deepwater reservoirs or reservoirs found in similar environments with high development costs. In these high cost environments, the ratio of water injectors to producers, and the location of these wells should be optimized through the use of reservoir simulation models

Waterflood Patterns

- Some of the common waterflood patterns are the direct line drive, the staggered line drive, the five-spot, and the nine-spot patterns. The direct line drive pattern is shown in Figure 5.1. For this pattern, the injectors and producers offset each other directly. This could lead to early water breakthrough if the distances between the injectors and producers are small.

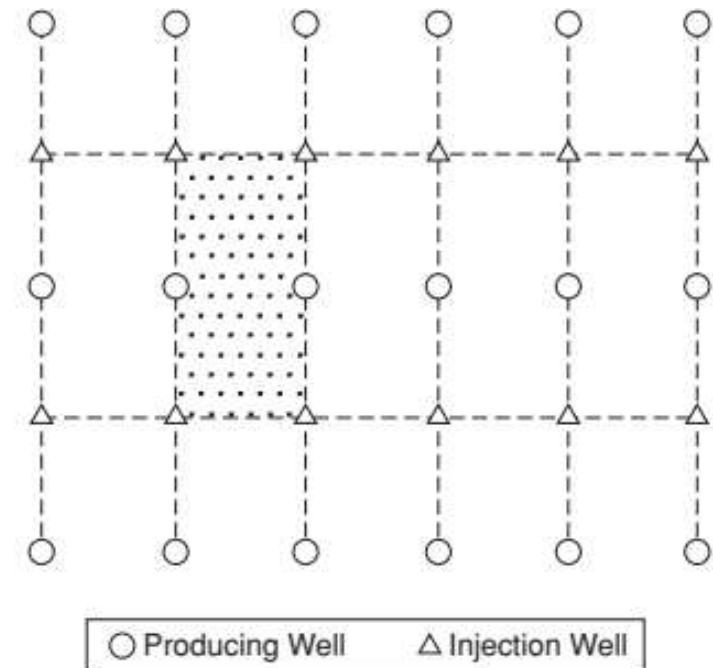


Fig. 5.1: Direct line pattern drive

Waterflood Patterns

- The impact of early water breakthrough on sweep efficiency for the direct line drive pattern can be reduced somewhat by using the staggered line drive pattern, as shown in Figure 5.2. As seen in Figure 5.2, the alternate rows of injectors and producers are displaced so that the wells do not offset each other directly. Note that the ratio of injectors to producers in both patterns is one.

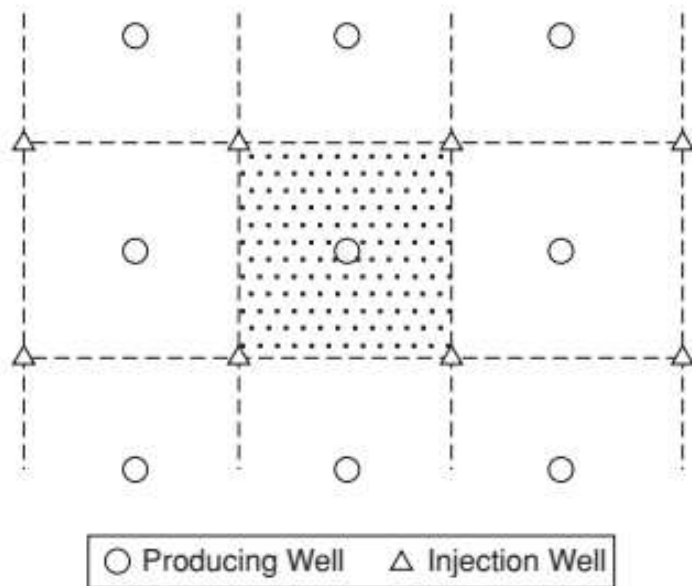


Fig. 5.2: Staggered line pattern drive

Waterflood Patterns

- The five-spot pattern is shown in Figure 5.3. The five-spot pattern has been used widely in many field applications and model studies. This pattern generally results in good areal sweep efficiency. The ratio of injectors to producers in a five-spot pattern is also one.

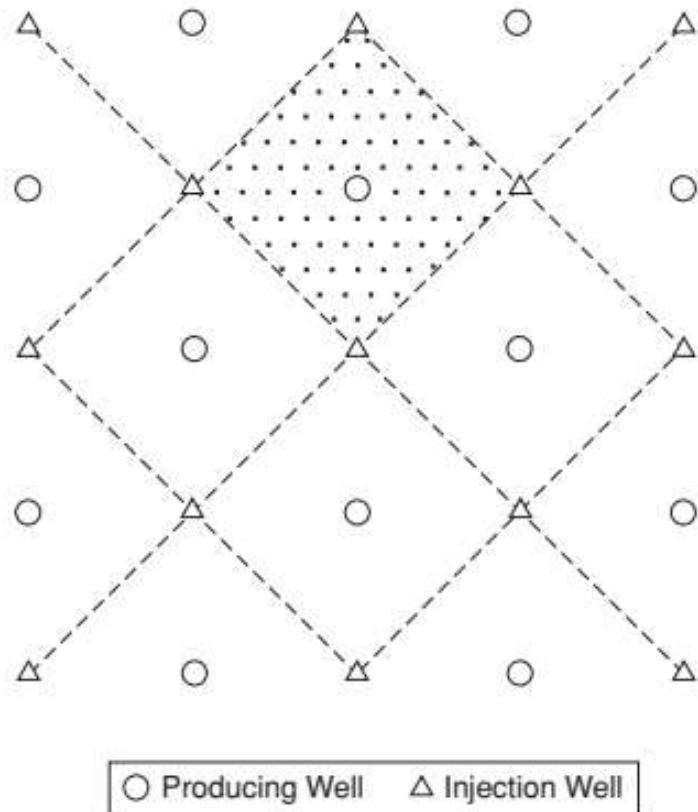


Fig. 5.3: Normal five-spot pattern

Waterflood Patterns

- The nine-spot pattern is shown in Figure 5.4. This pattern is used in low permeability reservoirs where high water injection volumes are desired. The ratio of injectors to producers in a nine-spot pattern is three.

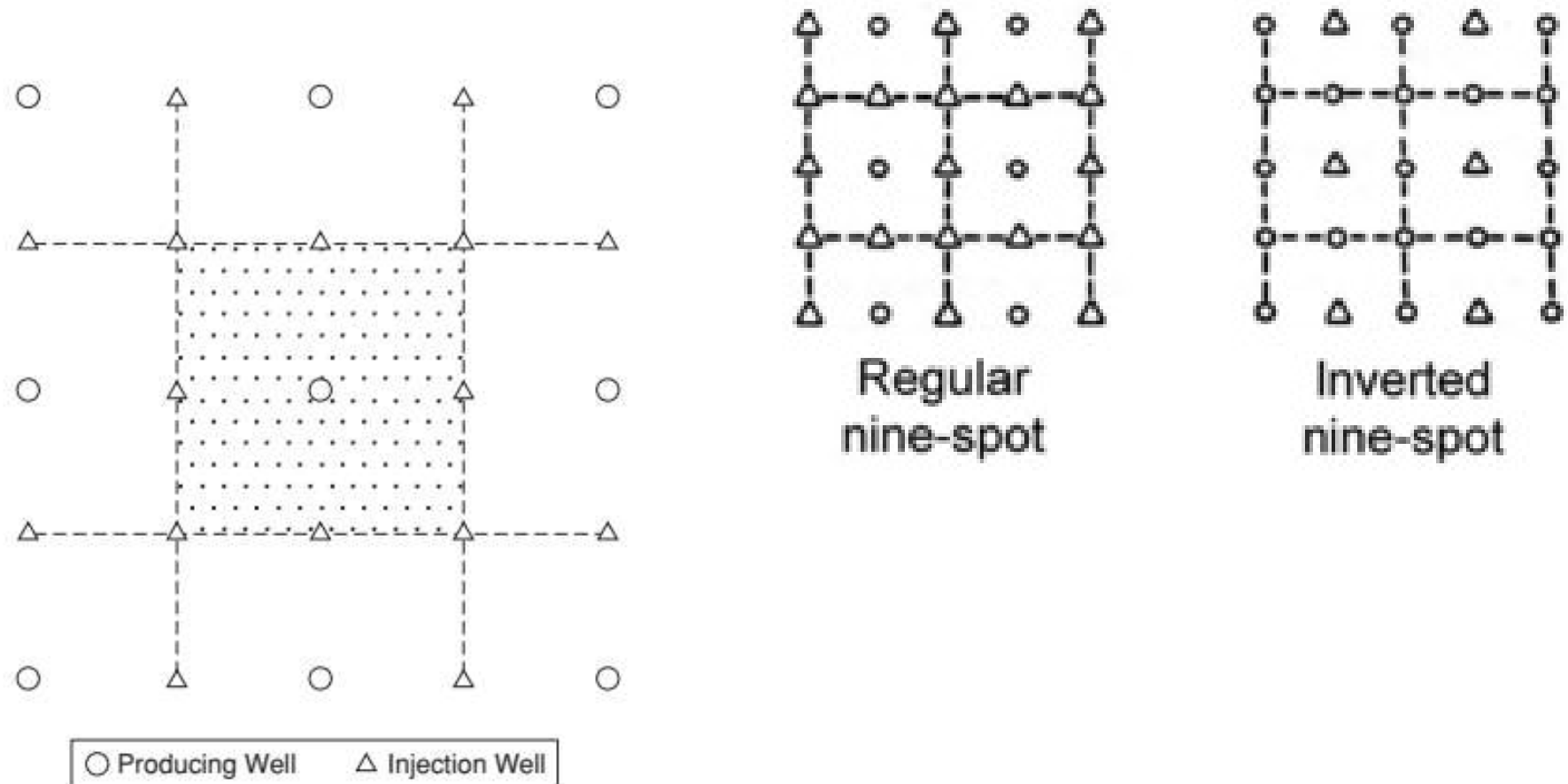


Fig. 5.4: Normal nine-spot pattern

Waterflood Patterns

- Peripheral flooding is developed when injectors are located around part or the entire boundary of the reservoir, as shown in Figure 5.5. If the injectors are located along one side of the reservoir or the middle of the reservoir, it is called a line flood. Peripheral flooding is generally used in reservoirs where the objectives are to maintain reservoir pressure, minimize water production, and improve oil recovery

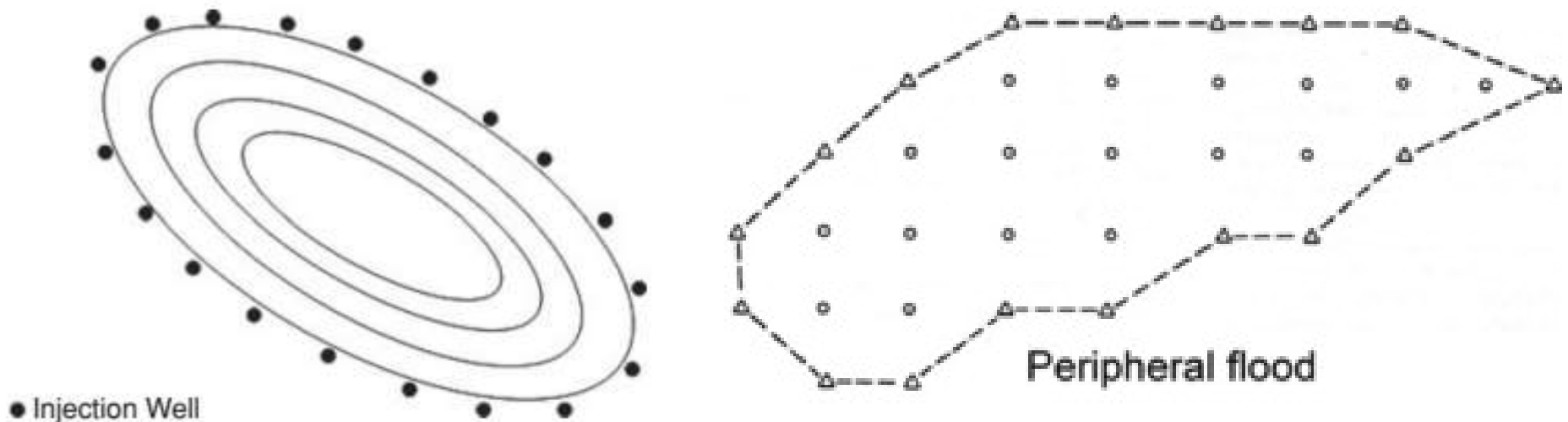


Fig. 5.5: Peripheral flood pattern

Waterflood Patterns

- There are numerous variations of these basic flood patterns, such as inverted five-spot and inverted nine-spot patterns. It is not necessary to replicate these other patterns here since the engineer can devise flood patterns that are best suited to the development of a particular reservoir with modern reservoir simulation tools.

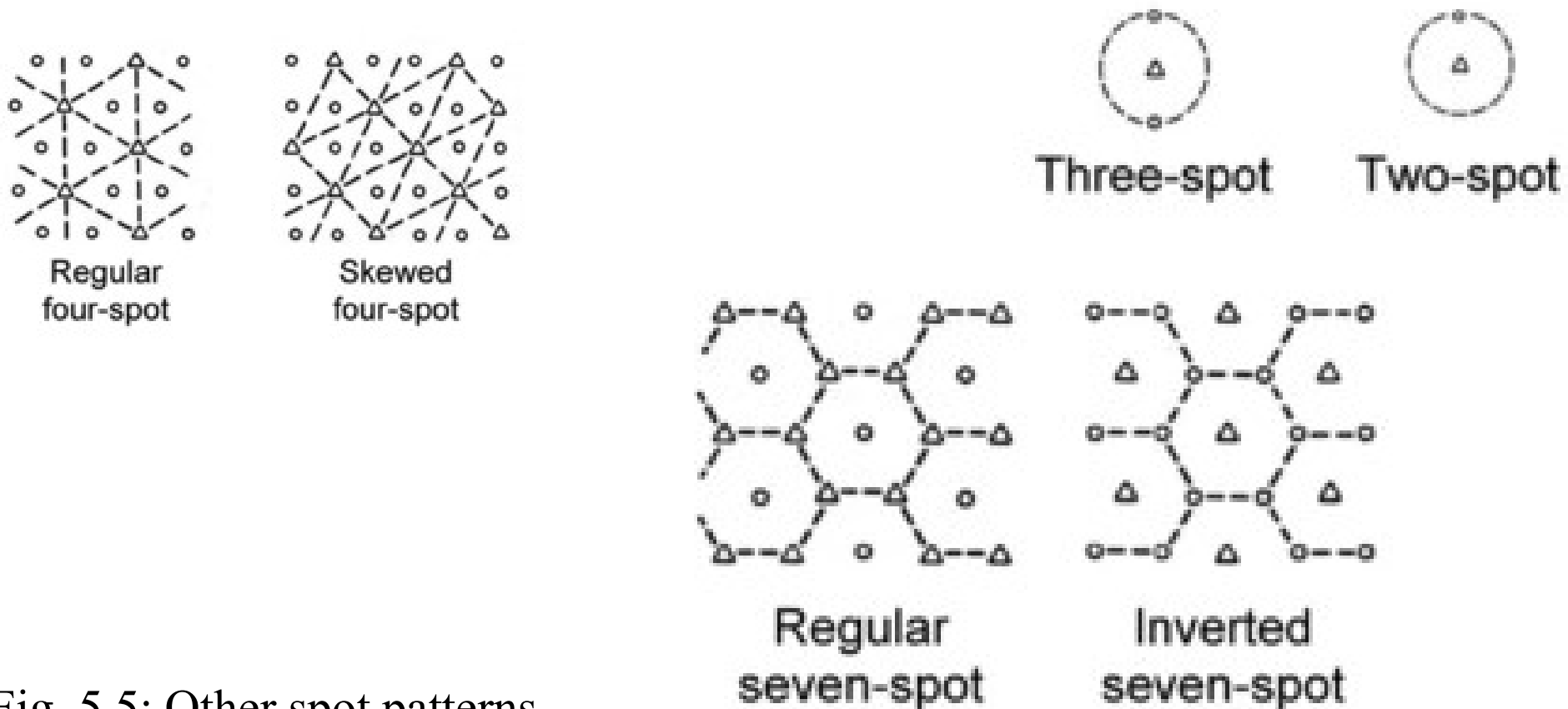


Fig. 5.5: Other spot patterns

Recommended Steps in Waterflood Design

1. Construct a geologic model of the reservoir or project area.
 - a) Identify and include all faults and other structural features that may affect fluid flow in a geologic frame work model of the reservoir.
 - b) Identify and include all reservoir heterogeneities, such as permeability barriers, reservoir unconformities, etc. in the geologic model.
 - c) Perform characterization of the geologic model to include areal and vertical variation of reservoir properties such as facies, porosity, permeability, saturations.
2. Analyze rock/fluid properties data.
 - a) Determine mineralogy of reservoir rocks.
 - b) Conduct studies on compatibility of injection water with reservoir rocks.
 - c) Determine PVT properties of reservoir fluids, including saturation pressures and oil viscosity.

Recommended Steps in Waterflood Design

3. Construct reservoir flow model with data from Steps 1 and 2.

- a) If the reservoir had prior production history, history-match reservoir model to obtain the current depleted state of the reservoir prior to waterflooding.
- b) At the completion of step 3a, determine gas cap size if a gas cap is present and extent of aquifer influx if reservoir has an active aquifer.
- c) Compare pressure distribution in model after step 3a to actual pressure data. Identify state of reservoir depletion.
- d) Explore fluid saturation distribution in the model after step 3a to identify potential undepleted areas that would be targets for waterflooding in reservoir.

4. Run prediction cases.

- a) Run a base prediction case assuming continuation of current depletion strategy.
- b) Run several predictive cases assuming the reservoir is waterflooded with different numbers and locations of water injectors and producers.
- c) Compare results from step 4a and step 4b. If the waterflood cases indicate substantial improvement in total oil recovery, then proceed to Step 5.

Recommended Steps in Waterflood Design

5. Optimize waterflood design.

- a. Choose the cases from step 4b with the “best” reservoir performance and optimize the numbers and locations of injectors and producers.
- b. Optimize injection and production rates for each case.
- c. Rank the cases by incorporating project economics.

6. Perform sensitivity analyses.

- a. Select two or three cases from step 5 and perform sensitivity analyses on key reservoir and operational variables of the waterflood design.
- b. Repeat economic analyses of the entire project based on results from step 6a.

7. Conduct a pilot waterflood project.

- a. For large waterflood projects, it is recommended that a pilot project should be installed. The pilot waterflood project would provide data, such as water injection rates, that could be used to adjust the waterflood design and improve predictions before implementation of the project over the entire reservoir.

Waterflood Management

- In many cases, proper management of a waterflood project is as important as designing the waterflood. A well-designed waterflood can become an engineering and economic failure if several important surveillance practices are not implemented in the management of the project. Monitoring and surveillance practices are presented for injection wells, injection water, production wells, production water, facilities, and the reservoir.
- **A. Injection Wells.** One of the key measures of a successful waterflood project is achieving and maintaining the desired levels of water injection rates. Water injection wells may exhibit reduced injectivity over time due to scale buildup and plugging of the formation from solid deposits. It is important to test the injection wells periodically to ensure that injectivity had not declined substantially. The most common test that can be used to gauge well injectivity is the fall-off test.

Waterflood Management

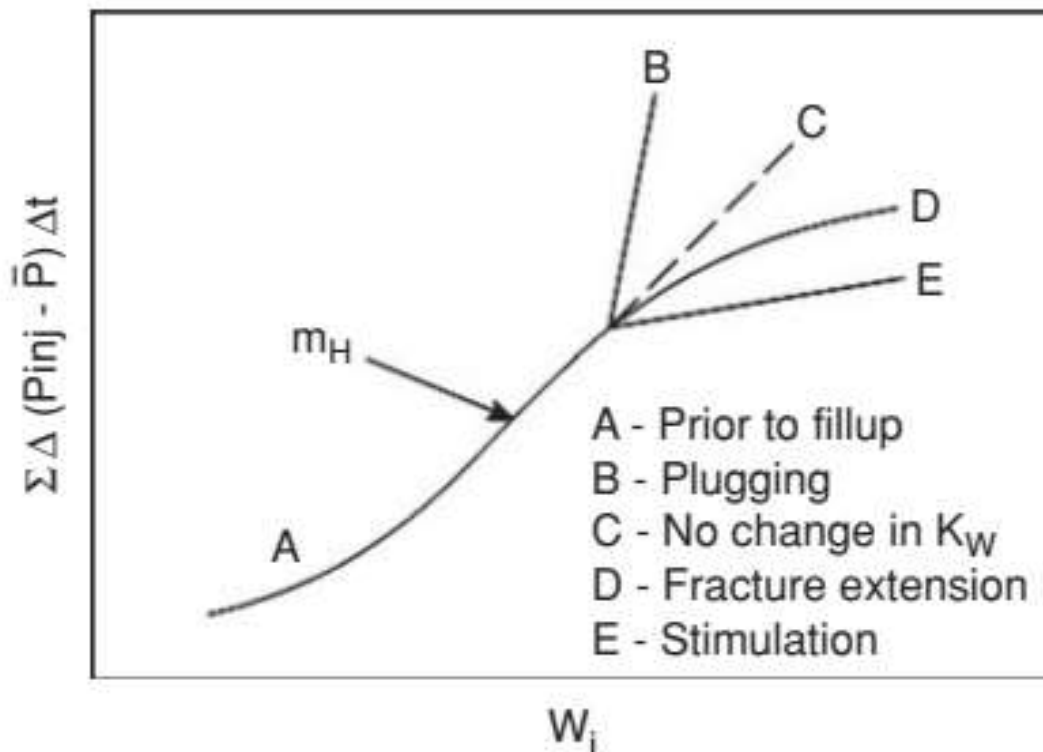
- **A. Injection Wells.** The condition of an injection well can be monitored periodically by preparing a Hall plot on the performance of the injector. The Hall plot can be prepared from data routinely collected as part of monitoring the performance of the injector. These data include cumulative water injection and injection pressure versus time. The Hall method is based on the radial form of the Darcy flow equation expressed as:

$$q_w = \frac{0.00708 k_w h (p_{inj} - \bar{p})}{\mu_w \ln \left(\frac{r_e}{r_w} \right)} \quad (5.1)$$

- In Eq. (5.1), q_w = water injection rate, BWPD; k_w = effective permeability to water, md; h = formation thickness, ft; p_{inj} = bottom-hole injection pressure, psia; $\bar{p}$ = average reservoir pressure, psia; μ_w = water viscosity, cp; r_e = external radius, ft; and r_w = wellbore radius, ft.

Waterflood Management

- **A. Injection Wells.** The parameter, would not be constant if permeability is reduced due to plugging or increased due to stimulation or fracturing. A characteristic Hall plot is shown in Figure 5.6. Note the changes in the shape of the curve due to effects of various injection conditions as indicated in Figure 5.6:



- Fig. 5.6 Characteristic Hall plot

Waterflood Management

- **B. Injection Water.** The source of injection water should be sufficient to supply as much water as needed during the early stages of the waterflood project. Later on, as the project matures, produced water could be recycled and used to make up the required volumes of injection water.
- The quality of injection water should be determined by performing complete water analysis on all sources of water which should evaluate the compatibility of source water with reservoir formation water and swelling clays or other minerals in the reservoir rock, the type of injection facility, and the extent of water treatment required before the water can be used safely for injection. Injection water, if not treated, could cause corrosion, solid precipitation, and promote growth of bacteria. The major cause of corrosion in injection water is dissolved gases in the form of oxygen, carbon dioxide, and hydrogen sulfide. Oxygen in injection water is very corrosive, and may cause precipitation of ferrous hydroxide, if the injection water contains iron₂₁₈

Waterflood Management

- **C. Production Wells.** Production wells must be monitored regularly in a waterflood project just as injection wells. Each production well should be tested frequently (depending on the availability of test equipment) to measure fluid production rates. The measured production rates should be plotted on various graphs to monitor the response of the well to water injection. Typical graphs commonly used are plots of fluid production rates versus time, percent oil cut versus cumulative oil produced, or water-oil ratio versus cumulative oil produced. There are numerous other types of plots that can be used to monitor well performance. These plot types can be tailored to suit the particular needs of the project. If the production well appears to be damaged, a pressure buildup test should be conducted on the well to determine the extent of damage.
- Profile (fluid entry) logs should be run periodically on production wells. These profile logs are extremely important in monitoring the intervals that are contributing to production.

Waterflood Management

- **D. Produced Water.** Water analysis should be performed regularly on produced water as a means of monitoring the progress of the waterflood, and tracking potential problems with solids deposition caused by mixing of source water and formation water. At early stages of the waterflood project, produced water consists mostly of formation water. Later, as production wells are affected by water breakthrough, produced water will contain increasing amounts of injected water. The proportion of injected water in produced water is sometimes used as an indicator in monitoring the location of the flood front.
- In many waterflood projects, some of the produced water is recycled and used as injection water. This reduces the amount of produced water to be disposed, thereby reducing the environmental impact of the project. Produced water recycled for re-injection should be treated to remove harmful substances, and improve its quality, as was the case for the source injection water. In some operations, produced water is also treated to remove harmful substances before disposal.

Waterflood Management

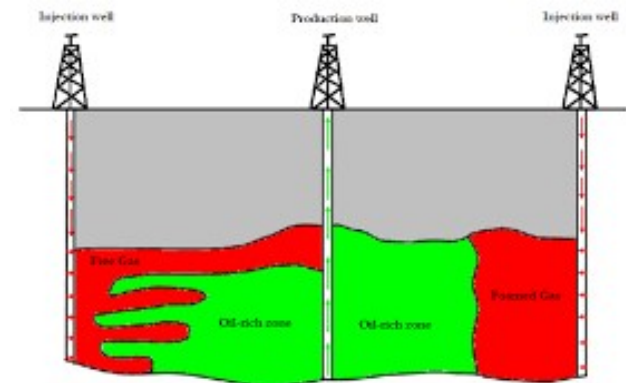
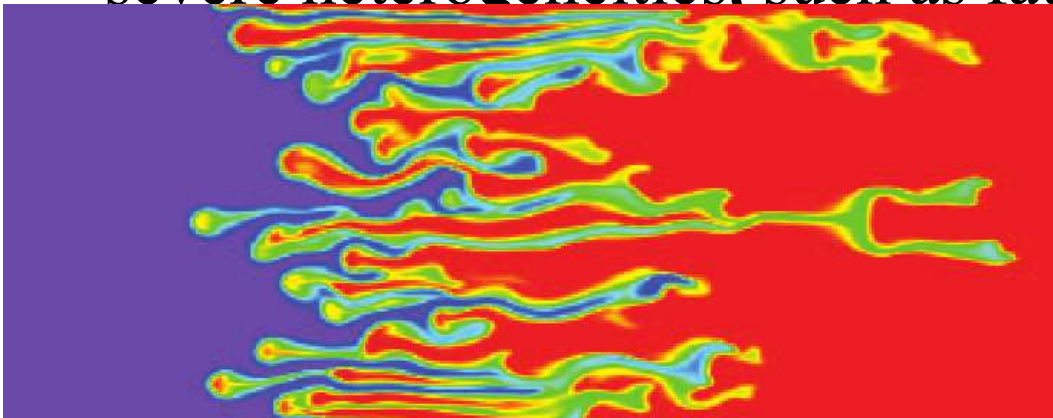
- **E. Facilities.** The facilities used in a waterflood project should be able to withstand potential threat from massive corrosion caused by the water. Protection of injection and production facilities from corrosion begins with careful water analysis, and selection of equipment with proper metallurgy. It is recommended that major waterflood projects should retain the services of a corrosion prevention department. The corrosion engineers should routinely monitor all facilities used in the project for corrosion, and implement preventative measures to forestall onset of corrosion in these facilities.

Gasflooding is presented in this chapter as an immiscible displacement process. Immiscible displacement occurs in a displacement process where a distinct interface (or boundary) exists between the displaced fluid and the displacing fluid. This includes displacement processes that are described as near-miscible. The key factors that determine whether a gasflooding process is immiscible or miscible are reservoir pressure, reservoir temperature, composition of injection gas, composition of reservoir fluid, and reservoir heterogeneities such as faults and permeability barriers. The impact of each factor can be determined with laboratory measurements and modeling of the displacement process.

GASFLOODING

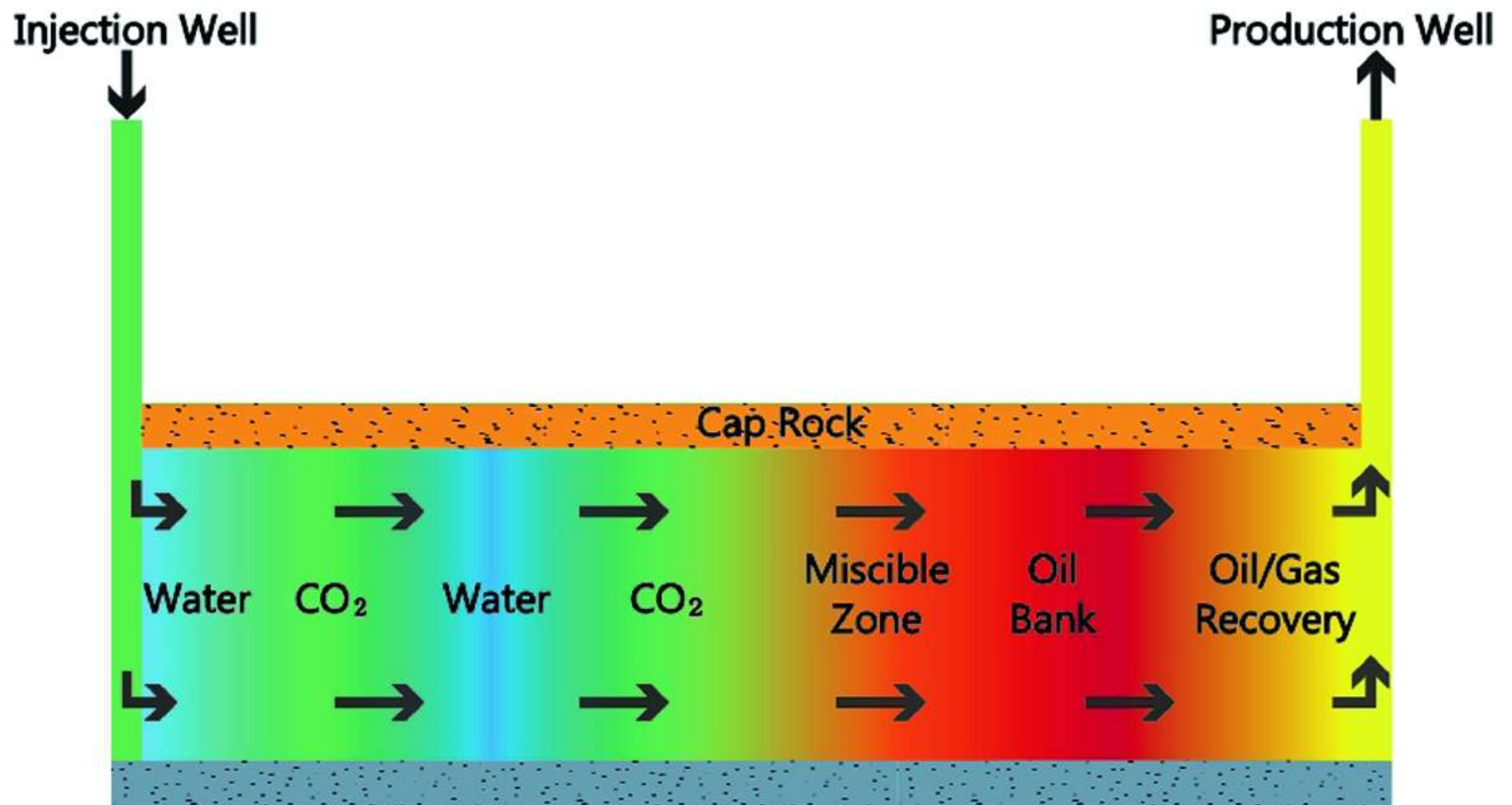
Gas Flooding

- Gasflooding is sometimes used instead of waterflooding because the fluid displacement (microscopic) efficiency of gas is superior in some applications. Plus, some of the injection gas (such as HC gas & carbon dioxide) can dissolve in the reservoir fluid to reduce its viscosity & density, thereby improving its mobility in the reservoir.
- On the other hand, the volumetric sweep efficiency (composed of areal and vertical sweep efficiencies) of gas displacement processes is generally low due to gravity over-ride, and viscous fingering caused by higher mobility ratios of gas in comparison to liquids. This effect can be compounded in low permeability reservoirs, or reservoirs with severe heterogeneities, such as faults & permeability barriers.



Gas Flooding

- Mobility ratios of gas displacement processes can be reduced by alternately injecting water and gas. This is described as a WAG (water-alternating-gas) displacement process. Volumetric sweep efficiency can be improved by application of WAG displacement process. This has been shown to improve oil recovery based on field results.



Applications of Gas Flooding

- The primary applications of gas injection into a HC reservoir are pressure maintenance, gas cycling, and fluid displacement. In many gas injection projects, these three primary functions are achieved at the same time to some degree. Pressure maintenance by gas injection is routinely implemented in many reservoirs.
- Reservoirs that are good candidates for pressure maintenance may have experienced steep, rapid pressure decline during primary depletion accompanied with the development of a secondary gas cap, or enlargement of a primary gas cap. For such reservoirs, gas is injected into the gas cap at sufficient volumes to reduce or stop the decline of reservoir pressure. In some cases, gas injection for pressure maintenance is initiated early close to start of production to prevent any decline in reservoir pressure. This practice is used, in some cases, to keep reservoir pressure from falling below bubble point for oil reservoirs, or dew point for gas reservoirs.

Applications of Gas Flooding

- Gas cycling is used to improve recovery from condensate reservoirs by vaporizing fluids that may have condensed in the reservoir with injected gas. Gas cycling can be used also to improve hydrocarbon recovery from secondary gas caps in oil reservoirs.
- In many steeply dipping reservoirs, gas is injected at the crest of the structure to assist in displacing oil towards downdip wells. The volumetric sweep efficiency can be high for such gravity stable processes.
- Gasflooding in horizontal reservoirs can lead to gravity over-ride, where the injected gas rises to the top of the reservoir because it is less dense than oil. Severe cases of gravity over-ride can lead to low volumetric sweep efficiencies leaving large portions of the reservoir unswept by injected gas. The effects of gravity over-ride & viscous fingering can be mitigated by using mobility control measures such as WAG as part of the injection strategy.

Gas Flooding Design

- The first key factor in gasflood design is obtaining a fluid property model that represents the interactions that occur between the reservoir fluids and the injected gas. These interactions can be both physical and chemical in nature. For instance, lean hydrocarbon gas is capable of physically vaporizing light hydrocarbons from reservoir oil. Carbon dioxide can dissolve chemically in reservoir oil to reduce its density and viscosity, and improve its ability to flow. These changes can be measured in the laboratory by conducting PVT studies, slim-tube tests, and core floods with the injection gas and the reservoir fluids. The data from these studies can be used to modify the parameters of an equation of state (EOS) which can then be used to simulate and reproduce the phase behavior resulting from the interaction between the injected gas and reservoir fluids.
- The second key factor in gasflood design is proper geologic modeling and characterization of the reservoir that is the candidate for gasflooding.

Gasflood Management

- **A. Gas Injectors.** It is essential to maintain the gas injectors in good conditions so that the required gas volumes are injected. This is particularly important for pressure maintenance projects. Gas injectivity may decline with time due to plugging of perforations from deposition of wax, asphaltene, & hydrates. Loss of gas injectivity can be remediated by stimulating the well with acids, solvents, or other agents for removal of well damage. It is important to run injection profile logs periodically on gas injectors to evaluate the distribution of the injected gas over the perforated interval.
- If the profile injection logs indicate unfavorable distribution of injection gas across the perforated interval, then the gas injector should undergo remedial work to direct injected gas to the intended formation intervals. If the injected gas is carbon dioxide or flue gas, the injector should be completed with equipment that can resist corrosion, and the well should be inspected regularly for corrosion.

Gasflood Management

- **B. Producers.** Producers in gasflood projects should be monitored regularly to prevent excessive production of injected gas which could result from early gas breakthrough due to channeling, or gravity override. If injected gas is allowed to flow to producers without contacting unswept portions of the reservoir, the volumetric sweep efficiency will be reduced resulting in poor hydrocarbon recovery. Gas overrides or channeling is particularly detrimental to gas cycling or gas displacement projects. This problem can be remedied at the producers by isolating intervals showing high gas production with packers, or by squeezing them with cement. As recommended for injectors, producers should be under surveillance regularly for corrosion, especially if the injection gas is carbon dioxide or flue gas. In most gasfloods containing carbon dioxide or other corrosive injectants, producers are outfitted with subsurface and surface equipment that are resistant to corrosion.

Gasflood Management

- **C. Processing Facilities.** All facilities used in the injection of gas and processing of produced fluids are grouped under the umbrella of processing facilities. These are gas injection facilities, production separators and stock tanks, flow lines, gas plants (for processing, separation and recycling of injection gas), and storage tanks. These facilities must be fitted to resist corrosion and inspected regularly to avoid the initiation of corrosion in the equipment, especially if the injection gas contains carbon dioxide or other corrosive agents.
- Reservoirs under gasflood require extensive program of monitoring and surveillance. The program depends on the type of the gasflood project. The progression of a gasflood project is rapid because of the high mobility of gas. Consequently, it is highly recommended that the gasflood project should be managed with a reservoir model.

END OF LECTURE 5

Enhanced Oil Recovery – Lecture 6

Presentation Outline

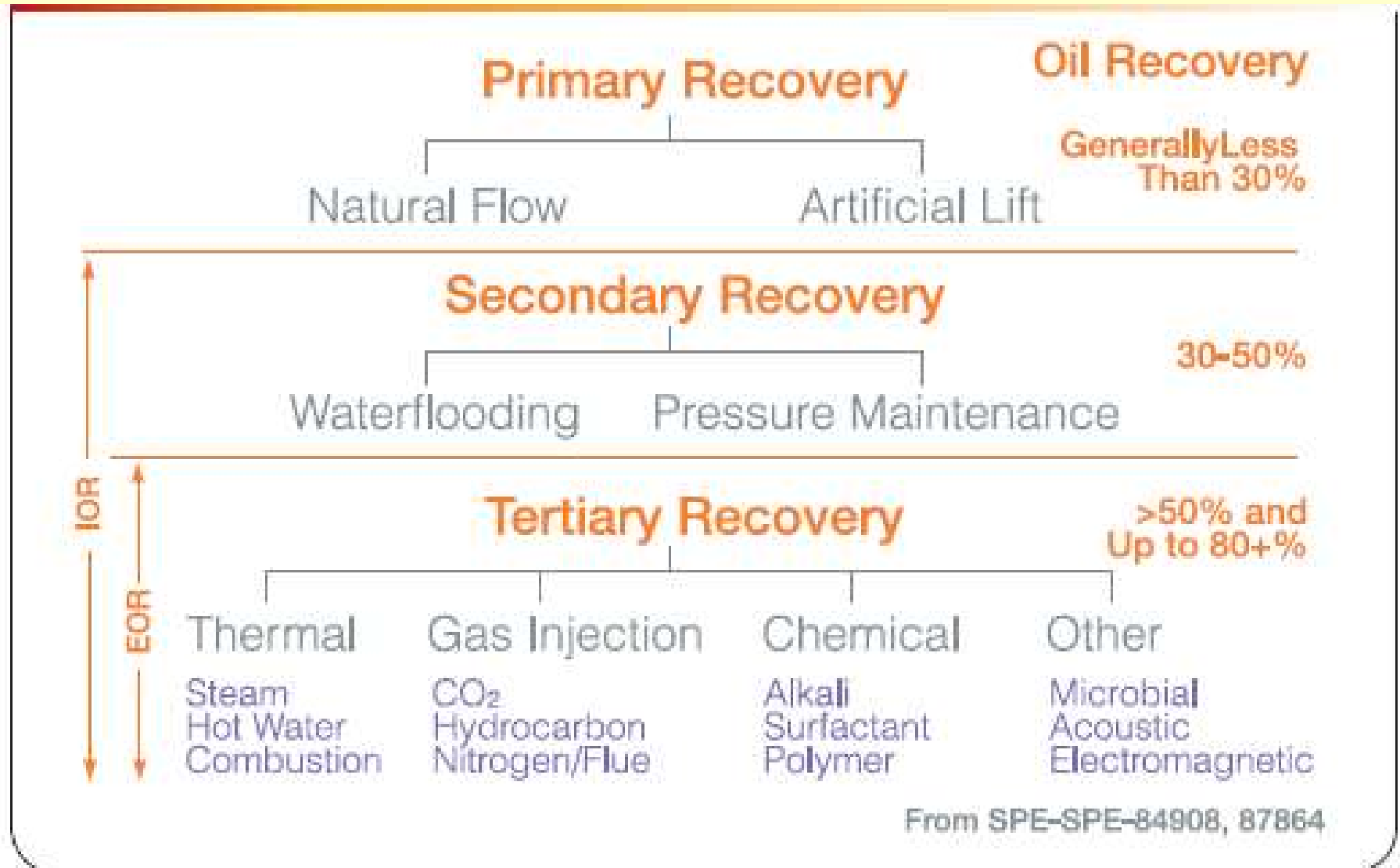
- Background
- EOR Technical Screening Guides
- EOR Technology / Processes
 - Miscible Gas Injection Processes
 - Chemical EOR
 - Thermal EOR
- Implementation of EOR Projects
- Conclusion

BACKGROUND

What is EOR?

- Primary and secondary recovery (conventional recovery) targets mobile oil in the reservoir while tertiary recovery or EOR targets immobile oil (that oil which cannot be produced due to capillary and viscous forces).
- Primary, secondary and tertiary (EOR) recovery methods follow a natural progression of oil production from the start to a point where it is no longer economical to produce from the HC reservoir. EOR processes attempt to recover oil beyond secondary methods, or what is left.

What is EOR?

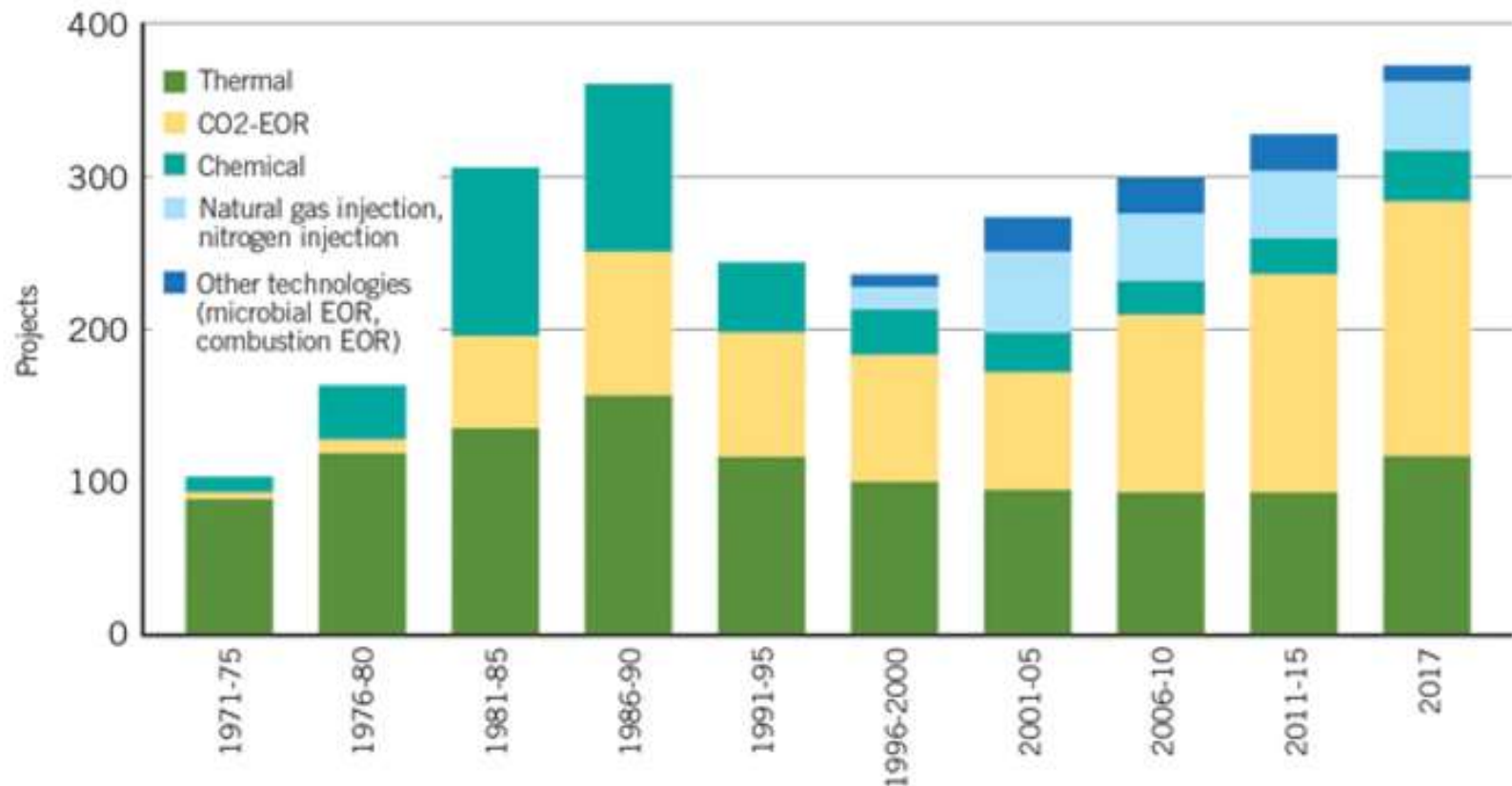


- EOR/IOR in terms of oil recovery, as defined by the Society of Petroleum Engineers (SPE).

EOR Projects Worldwide (2018)

EOR PROJECTS OPERATING WORLDWIDE

FIG. 1



Source: IEA World Energy Outlook

- The International Energy Agency (IEA) estimated about 375 enhanced oil recovery projects worldwide produced slightly more than 2 million b/d in 2018. EOR's share of global crude production seems to have remained broadly stable over time: around 2% of global oil production.

EOR Drivers & Challenges

Drivers	Challenges
• Daily oil production comes from mature or maturing oil fields thus reserves replacement is required as access to new exploration acreage is difficult/expensive. • The world average recovery factor from HC reservoirs is stuck in the mid-30% range. There is significant oil in place in discovered reservoirs that will otherwise not be recovered (roughly 65%). • In the long term EOR can make important contribution to world oil supply • Maturing (EOR) technologies provide new opportunities. EOR economics can be attractive.	• The implementation of EOR is intimately tied to the price of oil and overall economics. • EOR is capital and resource intensive, and expensive with long lead time, primarily due to high injectant costs. The risks are aggravated with the fluctuations in the price of oil. • The unit costs of EOR oil are substantially higher than those of secondary / conventional oil. • Environmental footprint

EOR TECHNICAL SCREENING GUIDES

EOR screening concept and methodology

- The objective of the screening process is to decide which EOR methods may be applicable in a reservoir and which definitely are not. The starting point is a set of basic properties that should be available for any reservoir.
- Screening criteria are based on physical principles associated with each method and on economic assumptions. There is some **subjectivity** and there will be minor variations between authors on specific requirements.

EOR screening concept and methodology

- Three complementary screening styles are highlighted in literature:
 - Conventional Screening (also known as parametric screening) which involves comparing average reservoir data with look up tables
 - Geological Screening involves using Logging RFT, MDT sampling, SCAL, Mineral study, PVT analysis
 - Advanced EOR screening which involves the use of Artificial Intelligence Techniques, analytical models to forecast EOR response and economic / cost modelling

EOR screening concept and methodology

- In some instances, only one type of enhanced recovery technique is applicable for a specific field condition but, in many instances, more than one technique is possible. The selection of the most appropriate process is facilitated by matching reservoir and fluid properties to the requirements necessary for the individual EOR techniques.
- A distinction is made between the oil properties and reservoir characteristics that are required for each process. Conventional Screening guides or criteria are among the first items considered when a petroleum engineer evaluates a candidate reservoir for enhanced oil recovery. There are several screening guides (1976 National Petroleum Council (NPC) report on Enhanced Recovery, revision by the NPC in 1984, Taber et. al (1997) etc.

EOR screening concept and methodology

– US DOE 1984 Screening Criteria

Screening Parameters *	Units	Chemical Flooding			Miscible Flooding (Carbon Dioxide)	Thermal Recovery	
		Surfactant	Polymer	Alkaline		Steam	In Situ Combustion
Oil Gravity	°API	-	-	<30	≥ 25	10 to 34	10 to 35
In Situ Oil Viscosity (μ)	cp	<40	<100	<90	-	$\leq 15,000$	$\leq 5,000$
Depth (D)	Feet	-	-	-	-	$\leq 3,000$	$\leq 11,500$
Pay Zone Thickness (h)	Feet	-	-	-	-	≥ 20	≥ 20
Reservoir Temperature (T_R)	°F	<200	<200	<200	-	-	-
Porosity (ϕ)	Fraction	-	-	-	-	$\geq 0.20^\dagger$	$\geq 0.20^\dagger$
Permeability, Average (k)	md	>40	>20	>20	-	250	35
Transmissibility (kh/ μ)	md-ft/cp	-	-	-	-	≥ 5	≥ 5
Reservoir Pressure (P_R)	psi	-	-	-	$\geq \text{MMP}^\ddagger$	$\leq 1,500$	$\leq 2,000$
Minimum Oil Content at Start of Process ($S_o \times \phi$)	Fraction	-	-	-	-	≥ 0.10	≥ 0.08
Salinity of Formation Brine (TDS)	ppm	<100,000	<100,000	<100,000	-	-	-
Rock Type	-	Sandstone	Sandstone or Carbonate	Sandstone	Sandstone or Carbonate	Sandstone or Carbonate	Sandstone or Carbonate

*Other criteria of a geological and depositional nature were also considered. Generally, reservoirs with extensive faulting, lateral discontinuities, fractures, or overlying gas caps are not prime candidates for field-wide EOR application. These factors were considered during the manual screening step when they could be identified.

† MMP denotes minimum miscibility pressure, which depends on temperature and crude oil composition.

‡ Ignored if oil saturation (S_o) $\times$ porosity (ϕ) criteria are satisfied.

EOR screening concept and methodology

– Taber 1997 Screening Criteria

		Oil Properties			Reservoir Characteristics					
Detail Table in Ref. 16	EOR Method	Gravity (°API)	Viscosity (cp)	Composition	Oil Saturation (% PV)	Formation Type	Net Thickness (ft)	Average Permeability (md)	Depth (ft)	Temperature (°F)
Gas Injection Methods (Miscible)										
1	Nitrogen and flue gas	>35 ↗ <u>48</u> ↗	<0.4 ↘ <u>0.2</u> ↘	High percent of C ₁ to C ₇	>40 ↗ <u>75</u> ↗	Sandstone or carbonate	Thin unless dipping	NC	>6,000	NC
2	Hydrocarbon	>23 ↗ <u>41</u> ↗	<3 ↘ <u>0.5</u> ↘	High percent of C ₂ to C ₇	>30 ↗ <u>80</u> ↗	Sandstone or carbonate	Thin unless dipping	NC	>4,000	NC
3	CO ₂	>22 ↗ <u>36</u> ↗ ^a	<10 ↘ <u>1.5</u> ↘	High percent of C ₅ to C ₁₂	>20 ↗ <u>55</u> ↗	Sandstone or carbonate	Wide range	NC	>2,500 ^a	NC
1-3	Immiscible gases	>12	<600	NC	>35 ↗ <u>70</u> ↗	NC	NC if dipping and/or good vertical permeability	NC	>1,800	NC
(Enhanced) Waterflooding										
4	Micellar/ Polymer, ASP, and Alkaline Flooding	>20 ↗ <u>35</u> ↗	<35 ↘ <u>13</u> ↘	Light, intermediate, some organic acids for alkaline floods	>35 ↗ <u>53</u> ↗	Sandstone preferred	NC	>10 ↗ <u>450</u> ↗	>9,000 ↘ <u>3,250</u>	>200 ↘ <u>80</u>
5	Polymer Flooding	>15	<150, >10	NC	>50 ↗ <u>80</u> ↗	Sandstone preferred	NC	>10 ↗ <u>800</u> ↗ ^b	<9,000	>200 ↘ <u>140</u>
Thermal/Mechanical										
6	Combustion	>10 ↗ <u>16</u> →?	<5,000 ↓ <u>1,200</u>	Some asphaltic components	>50 ↗ <u>72</u> ↗	High-porosity sand/ sandstone	>10	>50 ^c	<11,500 ↘ <u>3,500</u>	>100 ↗ <u>135</u>
7	Steam	>8 to <u>13.5</u> →?	<200,000 ↓ <u>4,700</u>	NC	>40 ↗ <u>66</u> ↗	High-porosity sand/ sandstone	>20	>200 ↗ <u>2,540</u> ↗ ^d	<4,500 ↘ <u>1,500</u>	NC
—	Surface mining	7 to 11	Zero cold flow	NC	>8 wt% sand	Mineable tar sand	>10 ^e	NC	>3:1 overburden to sand ratio	NC

NC = not critical.

Underlined values represent the approximate mean or average for current field projects.

^aSee Table 3 of Ref. 16.

^b> 3md from some carbonate reservoirs if the intent is to sweep only the fracture system.

^cTransmissibility > 20 md-ft/cp

^dTransmissibility > 50 md-ft/cp

^eSee depth.

EOR screening concept and methodology – Screening EOR (Dickson et. al SPE 129768, 2010)

Property	HC gas	CO2	N2/Flue	CSS	Steam	SAGD	Hot water	Polymer	ASP
Oil API	>30-40	>22	>40	8-35	8-20	7-12	10-35	>15	>20
Oil viscosity, cp	<3	<10	<0.4	$10^3 - 10^8$	$10^3 - 10^4$	4000 - 10^6	$10^3 - 10^4$	10-1000	<35
Reservoir Depth, ft	4000-16000	>2500	>10,000	400-3000	400-4500	250-3000	<3000	800-9000	500-9000
Permeability, md	--	--	--	>250	>250	>5000	>35	>100	>100
Pressure, psia	>MMP	>MMP	<MMP	400-1500	<1500	High	>2000	--	--
Oil saturation, %	>30	>20	>40	>50	>40	>50	>50	>30	>45
Thickness, ft	Thin	Thin	Thin	>20-150	15-150	50-100	>20	--	--
Salinity, ppm	--	--	--	--	--	--	--	<3000	<200000
Temperature, F	Affect MMP	Affect MMP	--	--	--	--	--	< 170	<200

EOR Screening Criteria for Miscible Gas Injection Processes

Gas Injection Process (miscible)	Fluid Properties			Reservoir Properties				
	Gravity (°API)	Viscosity (cp)	Temp. (°F)	Porosity (%)	Perm. (md)	Oil Sat. at start (% PV)	Lithology	Depth (ft)
Nitrogen/ flue gas	>30	<0.5	>250	>10	>30	>50	Carbonate or Sandstone	>7000
Hydrocarbon Gas (HC)	21–57	0.1–1.3	136–290	4–26	10–5000	30–98	Carbonate or Sandstone	4000–14500
Carbon Dioxide (CO ₂)	28–44	0.4–3.0	100–250	4–26	2–500	25–90	Carbonate or Sandstone	2000–12000

EOR Screening Criteria for Chemical Flooding Processes

Chemical Flooding Process	Fluid Properties			Reservoir Properties				
	Gravity (°API)	Viscosity (cp)	Temp. (°F)	Porosity (%)	Perm. (md)	Oil Sat. at start (% PV)	Lithology	Depth (ft)
Polymer/Surfactant Flooding	14–34	5–80	80–160	20–30	170–900	60–75	Sandstone	1300–4600
Alkali/Surfactant/Polymer (ASP) Flooding	14–34	5–80	80–160	20–30	170–900	60–75	Sandstone	1300–4600
Polymer Flooding	14–34	5–80	80–160	20–30	170–900	60–75	Sandstone	1300–4600
Microbial (MEOR)*	—	5–50	<176	≥20	>50	—	Sandstone or Carbonate	<7700

EOR Screening Criteria for Thermal Flooding Processes

Thermal Process	Fluid Properties			Reservoir Properties				
	Gravity (°API)	Viscosity (cp)	Temp. (°F)	Porosity (%)	Perm. (md)	Oil Sat. at start (% PV)	Lithology	Depth (ft)
ISC or HPAI	19–33	2–660	110–230	17–32	10–1265	50–94	Sandstone or Carbonate	400–8300
Steam	8–30	50–500,000	45–290	15–65	100–10000	44–90	Sandstone* or Tripolite	200–3600

EOR TECHNOLOGY / PROCESSES

EOR Technology Matrix

- EOR methods are classified by the main mechanism of oil displacement. There are really just three basic mechanisms for recovering oil from rock other than by water alone. The methods are grouped according to those which rely on:
 - a) A reduction of oil viscosity,
 - b) The extraction of the oil with a solvent, and
 - c) The alteration of capillary and viscous forces between the oil, injected fluid, and the rock surface.

MISCIBLE GAS INJECTION EOR

CO₂ EOR Projects

- A miscible displacement process is defined as a fluid displacement where there is **no** interface (or boundary) between the displacing fluid and the displaced fluid.
- Gas injection, especially CO₂, is another popular EOR method, and is applicable to light oil reservoirs, in both carbonates and sandstones.
- Its popularity is expected to increase for two reasons:
 - increased oil recovery through miscibility and
 - disposal of a greenhouse gas.
- There are over 100 commercial CO₂-EOR projects, the bulk of them concentrated in the west Texas carbonates of the Permian Basin in the US.

Basic Concepts of Miscibility for Gas Displacement Processes

- The concept of miscibility in a gas displacement process can be demonstrated with a ternary phase diagram. A ternary-phase diagram for the methane/n-butane/decane system at 2500 psia and 160 °F is shown in the Figure 6.1
- This system has a two-phase region which is enclosed by a bubble point line and a dew point line that meet at the plait (or critical) point. At 2500 psia and 160 °F, mixtures of this system with compositions within the two-phase envelope are immiscible, while mixtures of the system with composition outside the two-phase envelope are miscible.

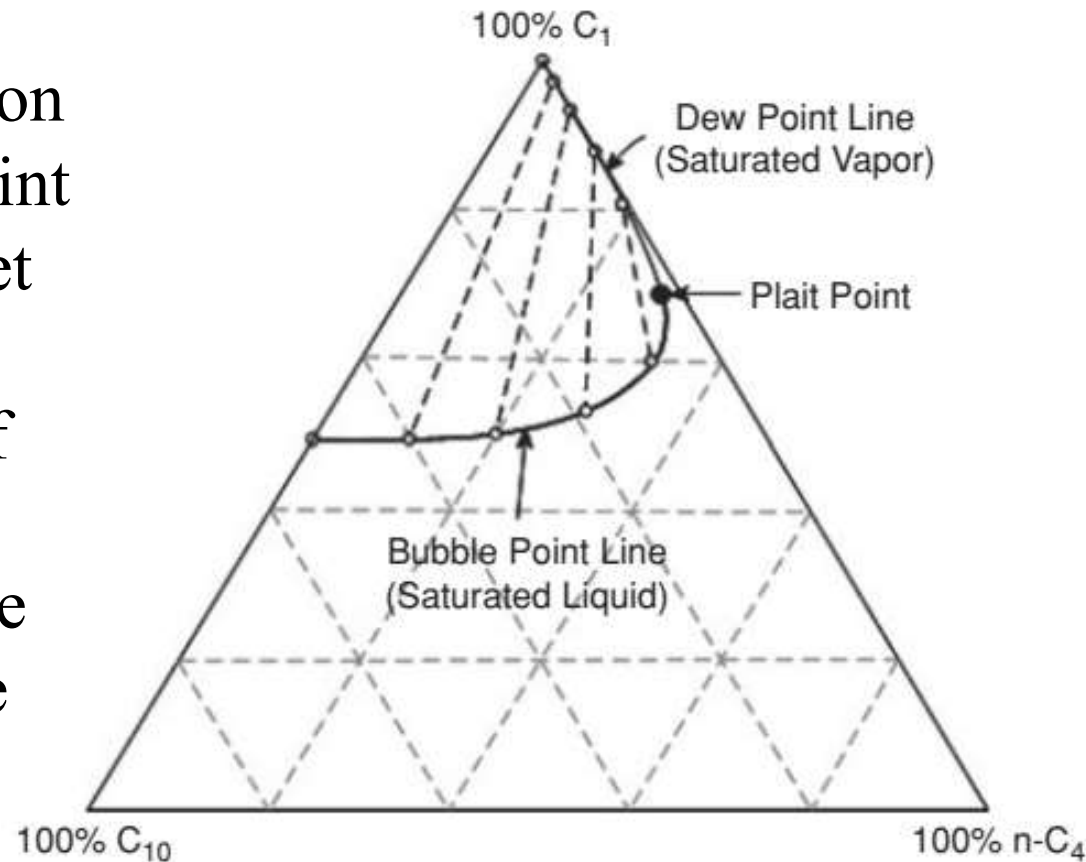


Fig. 6.1 Ternary-phase diagram of methane/n-butane/decane system at 2500 psia and 160 F

Basic Concepts of Miscibility for Gas Displacement Processes

- The relative size of the two-phase envelope in Figure 6.1 depends on temperature and pressure of the system.
- Increasing the pressure of the system at a constant temperature will shrink the two-phase envelope. It also means that at higher pressures more compositions of the mixture are miscible at 160°F.

For simplicity, complex hydrocarbon systems can be represented with a pseudo-ternary diagram, as shown in Figure 6.2. In a pseudo-ternary diagram, pseudo-components are used to represent the multicomponent hydrocarbon system

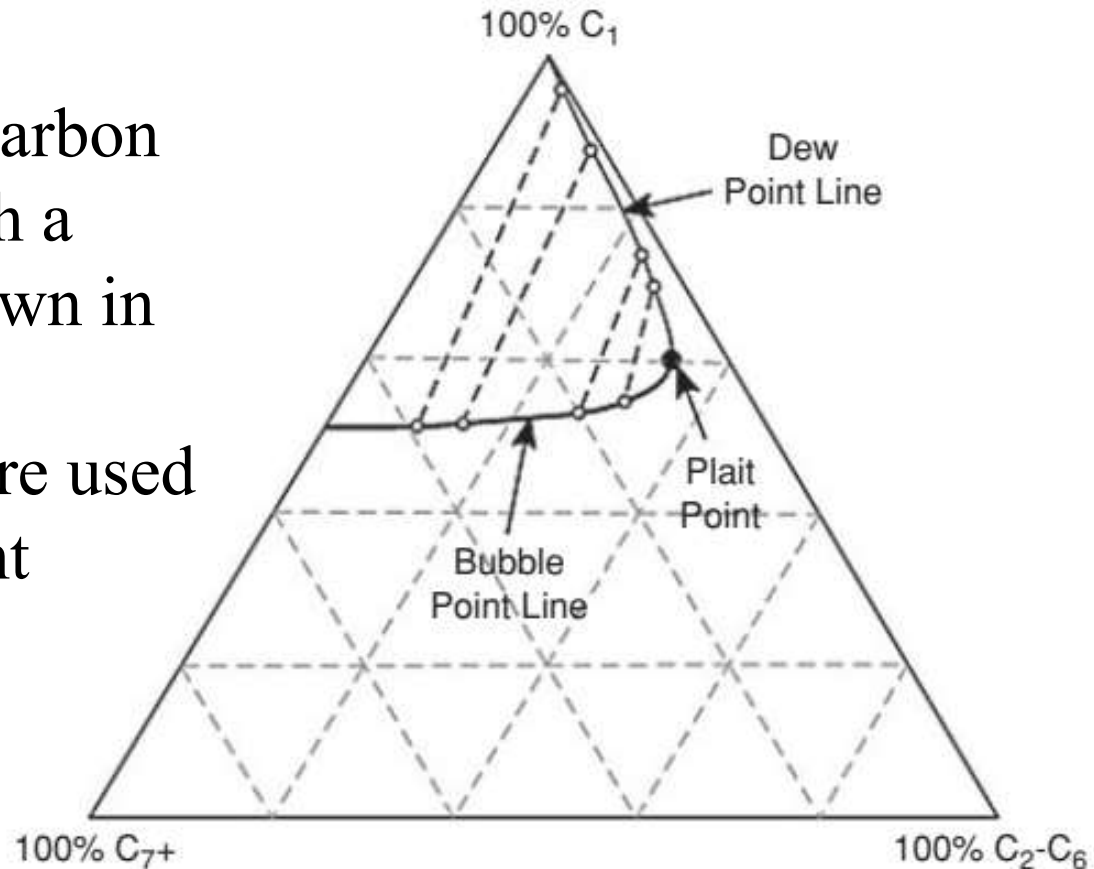


Figure 6.2 Pseudoternary phase diagram for a complex hydrocarbon system

Basic Concepts of Miscibility for Gas Displacement Processes

- The phenomenon of **first-contact miscibility** (FCM) occurs when the injected fluid is completely miscible with the reservoir fluid under the existing reservoir pressure and temperature.

Under FCM, the injected fluid completely mixes with the reservoir fluid to create a transition fluid that miscibly displaces the reservoir fluid. FCM can be considered as an instantaneous process of complete mixing of the injected fluid with the reservoir fluid. The factors that determine FCM are reservoir temperature and pressure, composition of the injected fluid, and composition of the reservoir fluid.

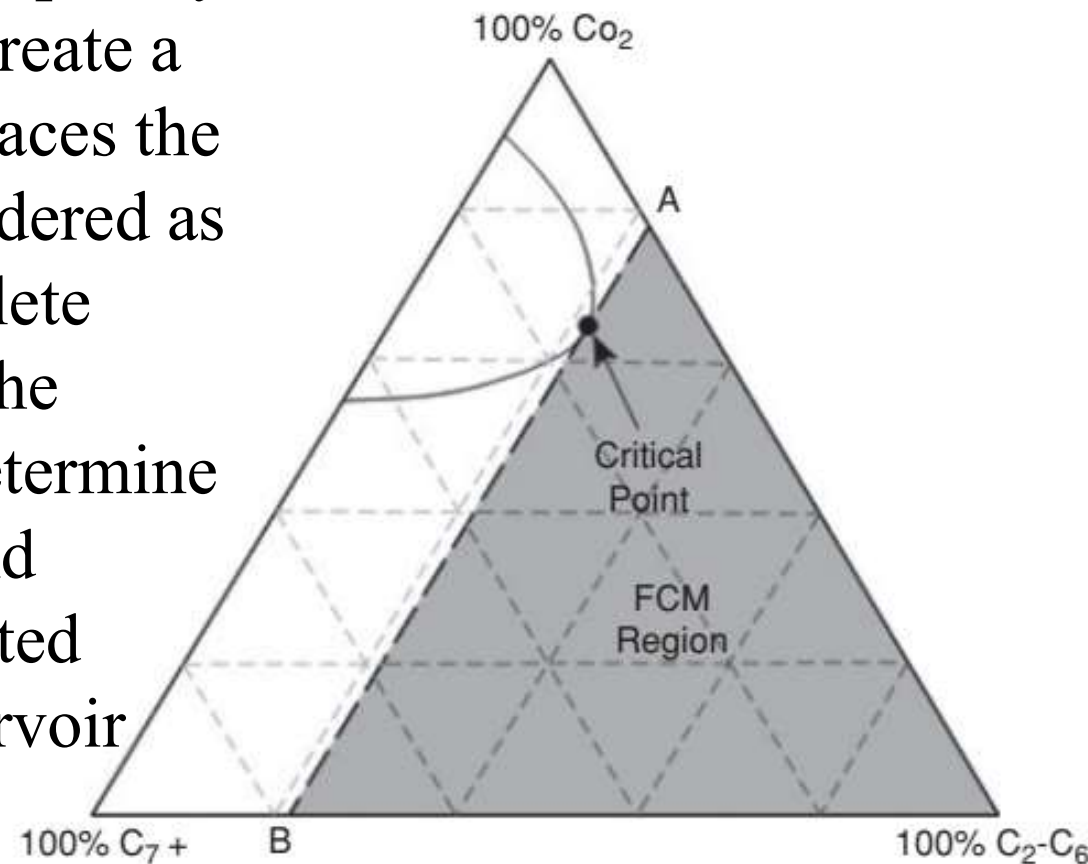


Fig. 6.3 Pseudoternary diagram for CO₂-hydrocarbon system 255

Basic Concepts of Miscibility for Gas Displacement Processes

- **Multiple-contact miscibility (MCM)** is achieved as the result of repeated contacts in the reservoir between the reservoir oil and the injected fluid, and also fluids generated in-situ by interactions between the injected fluid and the reservoir oil.
- There are two main processes by which MCM is achieved in the reservoir: the vaporizing gas drive and the condensing gas drive.
 - In the **vaporizing gas drive**, the composition of the injected fluid is progressively modified by vaporization of the intermediate hydrocarbon components (C_2 to C_6) in the reservoir oil until the modified fluid becomes ultimately miscible with the original reservoir oil.
 - In the **condensing gas drive**, intermediate hydrocarbon components in the injection fluid condense in the reservoir oil to generate a modified fluid that becomes miscible with the injection fluid.
- This third MCM process is termed the combined condensing and vaporizing (CV) gas drives. The type of MCM process that is dominant in a displacement drive depends on the composition of the reservoir oil, the composition of the injection fluid, and the temperature and pressure of the reservoir.

Basic Concepts of Miscibility for Gas Displacement Processes

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Methods for Determination of Minimum Miscibility Pressure

- **Minimum Miscibility Pressure (MMP)** The MMP represents the minimum pressure at which MCM can be achieved in the reservoir for the fluid system. At pressures less than MMP, the displacement process will be immiscible. The MMP is defined as the minimum pressure at which the reservoir oil composition lies on a critical tie-line extension for a specified fluid system.
- **Minimum Miscibility Enrichment (MME)**. This describes the process of enriching the composition of the injection gas with intermediate hydrocarbons (C_2 to C_6) to achieve miscibility with the reservoir oil. The MME at a given pressure is defined as the minimum enrichment of the injection gas composition, such that the gas composition lies on an extension of a critical tie-line.

Methods for Determination of Minimum Miscibility Pressure

- The two main techniques used in the determination of MMP or MME can be broadly classified as analytical techniques and experimental methods.
- Analytical techniques, which are based on numerical computations, can be subdivided further into empirical correlations and compositional simulations. Empirical correlations are based on fitting equations to MMP data obtained from experimental methods, and/or computed from compositional simulations. Compositional simulations methods generally use equations of state to predict MMP or MME from the compositions of the injection gas and the reservoir oil.
- The two most common experimental methods are the slimtube method and the rising bubble apparatus method.

Types of Miscible Gas Flooding

- The types of gases used for miscible flooding include N_2 , flue gas, light HC (C_1 to C_6), & CO_2 . Flue gas is generated from power plants, internal combustion engines, etc. & contains a mixture of gases including a high percentage of N_2 (66%), CO_2 (9%–12%), & other gases, such as trace amounts of SO_2 and O_2 .
- The MMP required for flue gas injection is close to the requirements for nitrogen injection. For this reason, flue gas is grouped with nitrogen in the discussion of miscible gasflooding. Note however that flue gas is very corrosive because it contains carbon dioxide and sulfur dioxide.
- The manner by which miscibility is achieved depends on the type of injection gas. Generally, nitrogen and flue-gas achieve MCM by the vaporizing gas drive mechanism. HC gas and CO_2 can achieve MCM by vaporizing gas drive, condensing gas drive or combined condensing-vaporizing gas drive mechanisms, depending on the compositions of the reservoir fluid and injection gas, and injection pressure.

Nitrogen and Flue Gas Flooding

- Nitrogen and flue gas (about 87% N_2 and 12% CO_2) are sometimes used in place of HC gases or CO_2 because of economics. The economic appeal of nitrogen stems not only from its lower cost on a standard Mcf basis, but also because its compressibility is much lower.
- However, both nitrogen or flue gas are inferior to HC gases (and much inferior to CO_2) from an oil recovery point of view. N_2 has a lower viscosity and poor solubility in oil & requires a much higher pressure to generate or develop miscibility.
- The mechanism by which N_2 /flue-gas achieves miscibility with oil is through vaporization of the light intermediate components present in the oil (the mechanism of vaporizing gas drive). For this reason, N_2 /flue-gas is suitable for light oil reservoirs as candidates for miscible flooding in deep reservoirs.

Hydrocarbon Gas Injection

- Many pressure maintenance projects with HC gas injection involve some aspects of miscible gasflooding, if reservoir pressures are maintained above MMP. HC gas injection can be classified broadly into three categories.
 - The first category involves the injection of liquefied petroleum gas (LPG), such as propane at pressures sufficiently high to achieve FCM.
 - The second category involves the injection of enriched gas (containing C_2 to C_6 at enriched levels) to achieve MCM by predominantly condensing gas drive mechanism.
 - The third category entails the injection of lean gas (containing mostly methane) at high pressures to achieve MCM by vaporizing the light components (C_2 to C_6) in the oil using predominantly the vaporizing drive mechanism.
- In addition to enhanced oil recovery due to miscible conditions, HC gasflood improves oil recovery by swelling the oil, reducing oil viscosity, and physical displacement of the oil from the pore spaces.
- The application of HC gasflooding is limited by poor sweep efficiencies due to potential for viscous fingering, and the high economic value of hydrocarbon gases.

CO₂ EOR Projects

- Carbon dioxide (CO₂) gas is a very effective agent for miscible displacement of oil. In most cases, CO₂ achieves miscibility with reservoir oil by extracting light to intermediate (C₁ to C₆) components of the oil in a MCM, vaporizing gas drive mechanism. Also, CO₂ improves oil recovery by swelling the oil, reducing its viscosity, and lowering interfacial tension between the oil and the CO₂-enriched fluid.
- CO₂ can cause extensive corrosion and should be used in a field project after extensive corrosion preventative measures have been implemented. Carbon dioxide can be obtained from naturally occurring CO₂ reservoirs or separated from flue gases for injection. CO₂-EOR projects are attractive as a result of environmental reasons (sequestration).

CO₂ EOR Projects

- Oil and gas reservoirs are good candidates for storage or sequestration of carbon dioxide because these reservoirs had geologic seals that originally trapped the hydrocarbons provided the seals had not been breached by field operations. It is anticipated that large-scale operations in the future will combine the use of carbon dioxide for oil recovery and sequestration of large volumes of carbon dioxide in the same reservoir and its associated aquifers.

CO₂ Flooding

- CO₂ is effective for recovery of oil for a number of reasons.
 - In general, carbon dioxide is very soluble in crude oils at reservoir pressures; therefore, it swells the net volume of oil and reduces its viscosity even before miscibility is achieved by the vaporizing gas drive mechanism.
 - As miscibility is approached, both the oil phase and the CO₂ phase (which contains many of the oil's intermediate components) can flow together because of the low interfacial tension and the relative increase in the total volumes of the combined CO₂ and oil phases compared to the water phase.

CO₂ Flooding

- Therefore, at higher temperatures, the higher pressures are needed only to increase the CO₂ density to the same value as observed for the MMP at the lower temperature.
- Although the mechanism for CO₂ flooding appears to be the same as that for hydrocarbon miscible floods, CO₂ floods may give better recoveries even if both systems are above their required miscibility pressures, especially in tertiary floods.

Types of Miscible Gas Injection Strategies

- Mobility control is a major concern in miscible gas injection processes because of the relatively low viscosity of the injected gas. The mobility ratio between injected gas and displaced oil is unfavorable in most applications. This causes viscous fingering which creates low volumetric sweep efficiency. To remedy this condition, most miscible gas injection processes, inject gas and water to improve volumetric sweep of the reservoir. If water and gas are injected in alternating cycles, the process is called water-alternating-gas (WAG) injection strategy.
- In some application, water and gas are injected simultaneously. This injection strategy is called simultaneous-water and-gas (SWAG). SWAG injection strategy is considered to be a variant of WAG. WAG injection strategy is used in most miscible flood projects around the world. A WAG ratio of 1:1 (pore volume of water injected versus pore volume of gas injected) is used in most field applications.

Types of Miscible Gas Injection Strategies

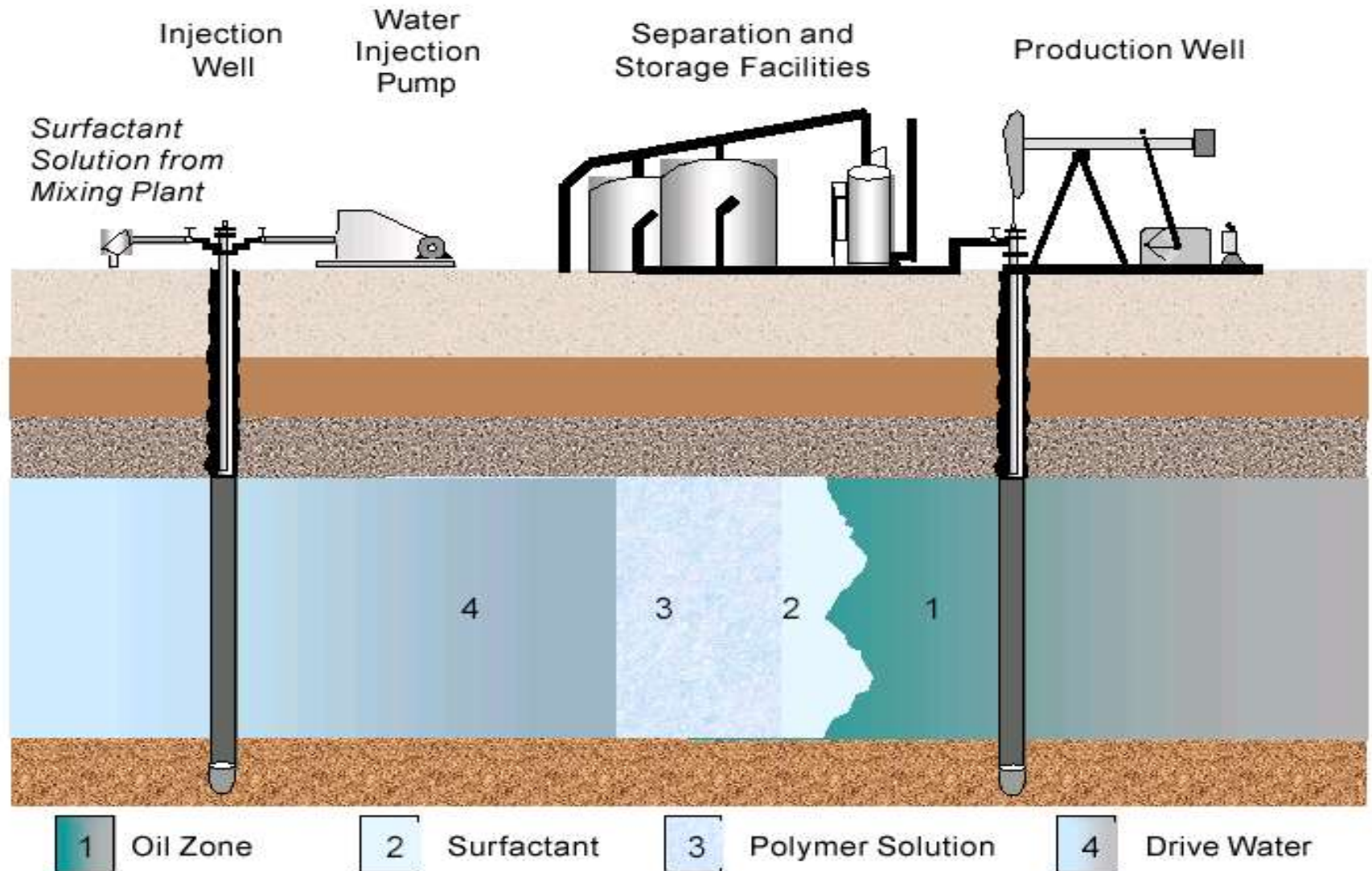
- Note that application of WAG results in loss of injectivity of water and gas over time. Some miscible gasfloods are initiated with the strategy of continuous gas injection (CGI) which can be switched later to WAG or SWAG to achieve better mobility control. The volume of gas injected can be reduced as the flood progresses to reduce cost and improve mobility control. This injection strategy is described as tapering. Tapering is used in most carbon dioxide and hydrocarbon injection projects to reduce the cost of injecting these gases into reservoirs for hydrocarbon recovery.

CHEMICAL EOR

Chemical EOR

- In chemical flooding, the primary goal is to recover more oil by either one or a combination of the following processes:
 - Mobility control by adding polymers to reduce the mobility of the injected water, and
 - Interfacial tension (IFT) reduction by using surfactants, and/or alkalis.
- Therefore, these methods require conditions that are very favourable for water injection: low-to-moderate oil viscosities, and moderate-to-high permeabilities. Hence, chemical flooding is used for oils that are more viscous than those oils recovered by gas injection methods but less viscous than oils that can be economically recovered by thermal methods.
- Chemical flooding processes are displacement processes that improve oil recovery from reservoirs with the aid of chemicals that are generally called surfactants, polymers, and alkalis. These chemicals are combined in various proportions to create injection systems which are termed polymer/surfactant (micellar) process, and alkali/surfactant/polymer (ASP) process. All of these methods involve mixing chemicals (and sometimes other substances) in water prior to injection.

Chemical EOR



- Figure CE: polymer/surfactant (micellar) process

Chemical EOR

- Reservoir permeabilities for chemical flood conditions need to be higher than for the gas injection methods, but not as high as for thermal methods. Since lower mobility fluids are usually injected in chemical floods, adequate injectivity is required. If previously waterflooded, the chemical flood candidate should have responded favourably by developing an oil bank.
- Generally, active water-drive reservoirs should be avoided because of the potential for low remaining oil saturations. Reservoirs with gas caps are ordinarily avoided since mobilized oil might re-saturate the gas cap.
- Formations with high clay contents are undesirable since the clays increase adsorption of the injected chemicals. In most cases, reservoir brines of moderate salinity with low amounts of divalent ions are preferred since high concentrations interact unfavourably with the chemicals that are injected.

Chemical EOR – Polymer / Surfactant Waterflooding

- In a typical polymer/surfactant flood (also called micellar/polymer flood), a slug of micellar solution (consisting of water, surfactant, electrolytes, etc.) is injected into the reservoir. The size of the micellar slug can vary from 3% to 30% of the flood pattern pore volume (PV) depending on the flood design. The micellar slug is followed with a mobility buffer slug made up of polymer & water for mobility control. The entire micellar/polymer slug is then chased with drive water (Figure CE).
- The polymer solution sometimes contains a biocide to control micro organisms that can degrade the polymer & reduce the viscosity of the solution. The micellar slug is formulated such that a favorable mobility ratio exists between the slug and the oil bank that is formed by the flooding process. Adsorption of surfactants on rock surfaces can be reduced by preflushing the flood zone with fresh water. This also reduces incompatibilities between the surfactant and formation brine.
- The application of micellar/polymer floods have declined because of the high cost of micellar solution & limited economic success of field tests 273

Chemical EOR – Polymer / Surfactant Waterflooding

- Partially hydrolyzed polyacrylamides (HPAM) and xanthan gum (XG) polymers both reduce the mobility of water by increasing viscosity. In addition, HPAM can alter the flow path by reducing the permeability of the formation to water. The reduction in permeability to water that is achieved with HPAM solution can be fairly permanent while the permeability to oil can remain relatively unchanged.
- The resistance factor is a term that is commonly used to indicate the resistance to flow that is encountered by a polymer solution as compared to the flow of plain water. For example, if a resistance factor of 10 is observed, it is 10 times more difficult for the polymer solution to flow through the system, or the mobility of water is reduced 10-fold.
- A properly sized polymer treatment may require the injection of 15%–25% of a reservoir pore volume; polymer concentrations may normally range from 250 to 2,000 mg/L. For very large field projects, millions of pounds of polymer may be injected over a 1–2 year period of time; the project then reverts to a normal waterflood.

Chemical EOR – Alkali/Surfactant/Polymer (ASP) Flooding

- Alkali/surfactant/polymer (ASP) flooding is more widely used than micellar/polymer flooding in field applications. ASP formulation utilizes the favorable mechanisms of its key components (alkali, surfactant and polymer) to improve oil recovery.
- The main functions of the alkaline component are to promote emulsification of the crude oil, reduce IFT, reduce adsorption of the surfactants, and regulate phase behavior of the mixtures.
- The main function of the surfactants is to reduce IFT between the oil and the injected slugs.
- The polymers are used for mobility control and improvement of sweep efficiencies. The favorable attributes of ASP formulations have been tested and proven in many laboratory and field tests. ASP flooding has been applied in most cases to sandstone reservoirs but laboratory and pilot tests suggest that it can also be applied to carbonate reservoirs.

Chemical EOR – Polymer Flooding

- Polymer flooding is the most widely used chemical flooding process for both sandstone and carbonate reservoirs. Polymer flooding can be considered as a waterflooding process in which polymers have been added to the solution to improve mobility control.
- Thus, polymer flooding is able to achieve higher oil recovery than ordinary waterflooding due to improved volumetric sweep efficiency resulting from better mobility control.
- Factors affecting the application of polymer flooding are the cost of the polymers, the permeability of the reservoir, and the temperature of the reservoir. The cost of polymers limits the concentration & slug size of the polymers that can be used to achieve higher oil recovery.
- Low rock permeability can limit the use of more viscous polymers to improve mobility control, and can cause low injectivity. Polymers are not stable at reservoir temperatures higher than 175 F.

Chemical EOR – Microbial EOR

- Microbial activities in the reservoir are believed to increase oil recovery by altering the wettability of the rock, reducing interfacial tension, reducing oil viscosity, generation of gases such as CO₂, production of surfactants, etc. These mechanisms, which combine to improve oil recovery by the MEOR process, are complex and not yet fully understood. Consequently, after many years of laboratory and pilot tests, MEOR processes have not been applied on a large scale in the field in comparison with other EOR processes.
- Generally, MEOR processes are applicable to reservoirs with injected and connate water salinities 150,000 ppm; oil viscosity in the 5–50 cp range; reservoir temperature 176 F; rock porosity 20%; rock permeability 50 md.; and reservoir depth 7700 ft.
- Most MEOR tests have been on sandstone reservoirs, although there have been tests in carbonate reservoirs.

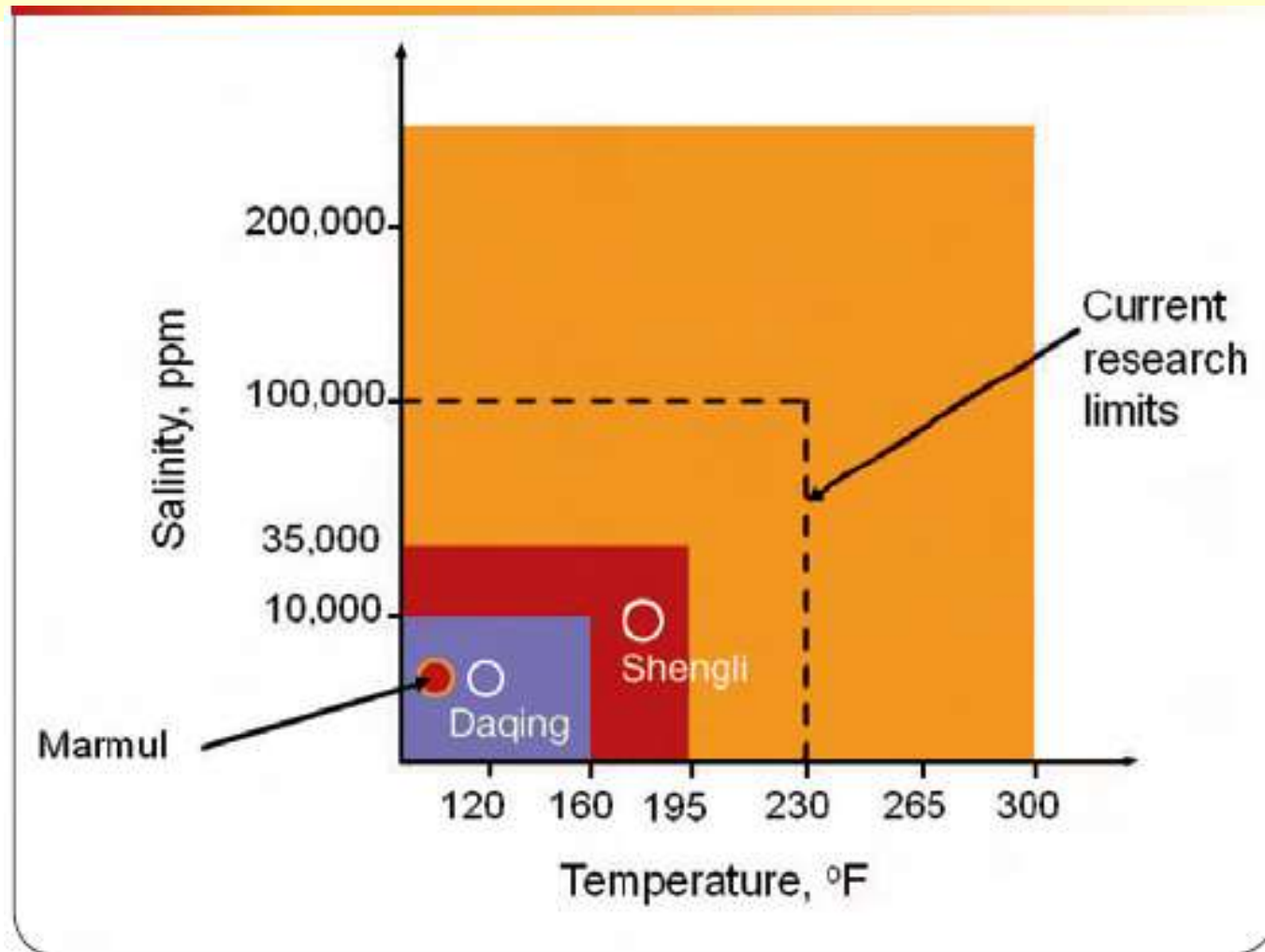
Chemical EOR – Microbial EOR

- The bacteria source in MEOR processes can be indigenous or exogenous. Indigenous bacteria are already present in the reservoir while exogenous bacteria are cultivated & injected into the reservoir.
- Microbial processes in the reservoir can be aerobic or anaerobic. For aerobic processes, oxygen must be injected to maintain the required concentration of bacteria.
- Injection of oxygen into the wellbore may cause severe corrosion problems. Anaerobic processes require injection of large amounts of nutrients in the form of sugars to sustain the growth of bacteria. This will increase the cost of the MEOR process due to the additional costs of purchasing, processing, and injecting the nutrients.
- 47 Microbial Flooding Recovery (MFR) field tests have been carried out in China, and 12 field tests are conducted in Daqing Oilfield. MFR field test's incremental oil recovery is as high as 4.95% OOIP, with a typical slug size less than 0.1 PV.

Challenges with Chemical EOR

- Surfactant injection has produced limited successes and remains challenging, especially in a high salinity, high temperature environment.
- Alkalis, although cheap, bring along a string of operational headaches (scaling, emulsions, plugging, etc.).
- Nearly all of the polymer floods have been implemented in sandstones, and carbonates remain a major challenge.
- Chemical EOR faces significant challenges, especially in light oil reservoirs. One of the reasons is the availability, or lack of, compatible chemicals in high temperature and high salinity environments.

Challenges with Chemical EOR



- This Figure shows the current limitation on a salinity-temperature plot. R&D will play a critical role in the future of chemical EOR.

THERMAL EOR

Thermal EOR Methods

- Thermal EOR processes are defined to include all processes that supply heat energy to the rocks and fluids contained in a reservoir thereby enhancing the ability of heavy, viscous crude oil (including other fluids) to flow by primarily reducing its viscosity. The oil caused to flow by the supply of thermal energy is produced through nearby wells.
- **Steam (or hot water) injection** and **in-situ combustion** are the popular thermal recovery methods.
- Steam flooding has achieved substantial commercial successes around the world and is ranked as one of the top EOR processes. In-situ combustion process, by comparison, has attained limited commercial success, and is actively applied in few reservoirs around the world. The main obstacle to the application of in-situ combustion technology is ability to manage and control the advancement and progression of the combustion process.

Thermal EOR Methods

- The three main methods of steam flooding are
 - cyclic steam stimulation (CSS),
 - steam drive (SD), and
 - steam assisted gravity drainage (SAGD). SAGD can be considered as a modification of SD for heavy oil reservoirs including tar sands.
- In-situ combustion involves the injection of air, where the oil is ignited, generates heat internally and also produces combustion gases, which enhance recovery.

Thermal EOR – Cyclic Steam Stimulation (CSS)

- Cyclic steam stimulation (CSS) is a simple and cheap method of applying thermal recovery process on a reservoir. It involves injecting steam into a well for several weeks, shutting the well in as long as necessary to allow the steam to heat the oil in the areas around the well, and putting the well back on production to recover the heated oil. This process is repeated when the production from the well declines to a low level. The cycle is repeated many times until the ratio of oil produced to steam injected termed oil-steam ratio (OSR) drops to a level that is considered uneconomic (see Figure TE1).
- CSS was the first steam flooding technique used in heavy oil reservoirs. It can still be used in a limited pilot test on a well to investigate whether steam injection can enhance oil recovery in a particular field before undertaking more expensive pattern pilot tests that will require more wells and more equipment.

Thermal EOR – CSS & SD

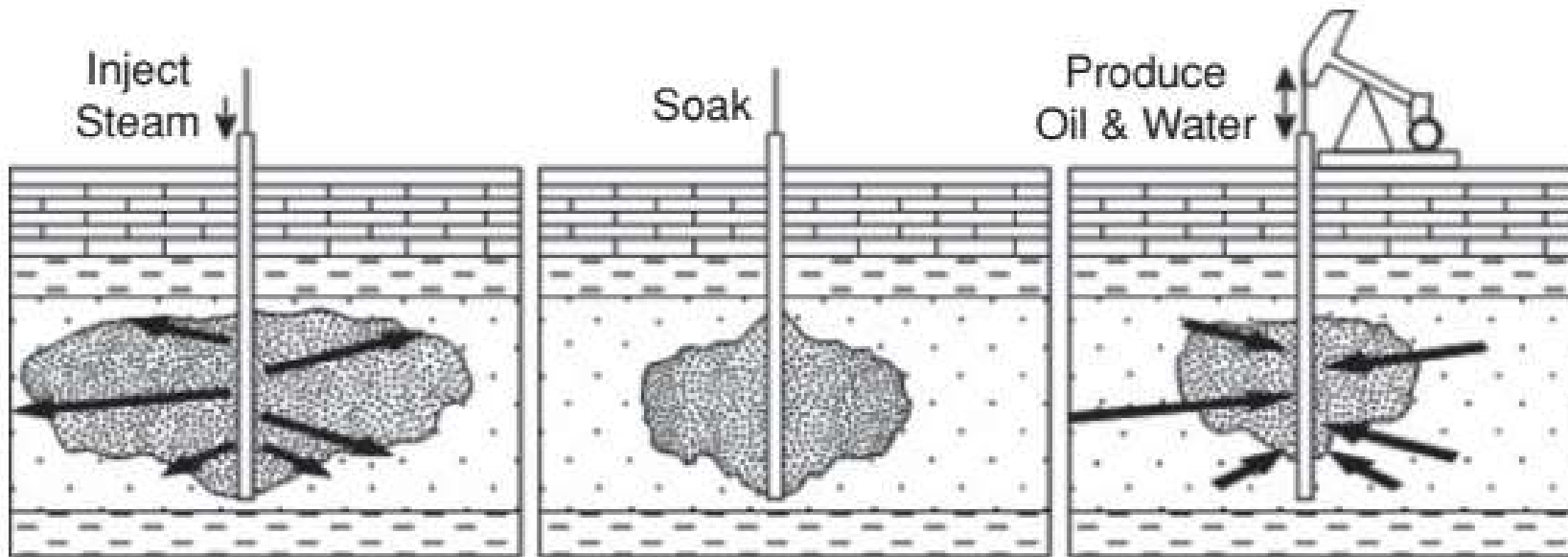


Figure TE1:
Cyclic steam stimulation
(CSS) process
[huff and puff]

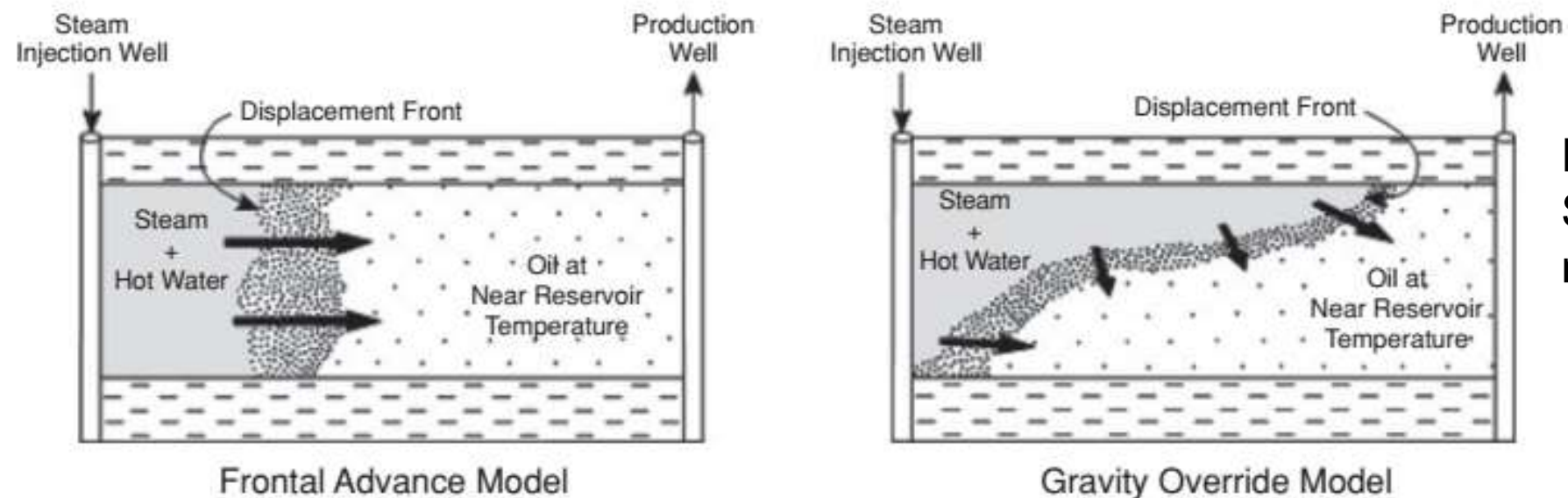


Figure TE2:
Steam Drive
models

Thermal EOR – Steam Drive (SD)

- In steam drives, steam is injected continuously at injectors with the aim of driving oil towards producers. Typically in most projects, steam injection is organized in patterns. For instance, in a normal five-spot pattern, the producer is located in the middle of the pattern and surrounded by four steam injectors. Steam drive was initially described as essentially a frontal displacement process similar to waterflooding, whereby steam displaces oil towards a producer. Later, experimental and field evidence showed that the mechanism of oil recovery by steam is dominated by gravity drainage caused by steam rising towards the top of the formation due to gravity override (Fig. TE2). The accumulation of steam at the top of the formation creates a steam zone which continues to expand with more steam injection. This steam zone can propagate and breakthrough at the producers to cause inefficient utilization of the energy from the steam due to cycling. The main challenges of managing a steam drive project are monitoring the size and growth of the steam zone, monitoring producers for early steam breakthrough, adjusting steam inject rates, and measuring the trend of oil-steam ratio for each pattern.

Thermal EOR – Steam-Assisted Gravity Drainage (SAGD)

- The SAGD process is applicable to other heavy oil and oil sands deposits in other countries such as China, Venezuela, and Russia. The SAGD process consists of two horizontal wells about 15 feet apart located close to the bottom of the formation (Fig. TE3). Steam is injected into the top horizontal well, while the horizontal well below it functions as the producer. The steam creates an expanding steam chamber around the injector as more steam is injected. Within the steam chamber and at its boundaries, as the viscosity of the oil is reduced, its mobility increases causing it to drain under gravity towards the production well. The SAGD process should be applied to reservoirs with formation thickness greater than 50 feet, good vertical permeability, and absence of thief zones. In some SAGD projects, a hydrocarbon solvent (such as methane) is added to the steam to improve oil recovery. This variation of the SAGD process has been described as the expanding solvent-SAGD (ES-SAGD).

Thermal EOR – Steam-Assisted Gravity Drainage (SAGD)

- In another variation, vertical wells are used for steam injection, while the producer remains a horizontal well as shown in Fig. TE4. This configuration of injectors and producer is termed Cross SAGD (XSAGD). Oil recoveries with SAGD in pilot tests can reach 65% of the oil-in-place in the project area in comparison with CSS that achieves oil recovery in the range of 25% to 30%.

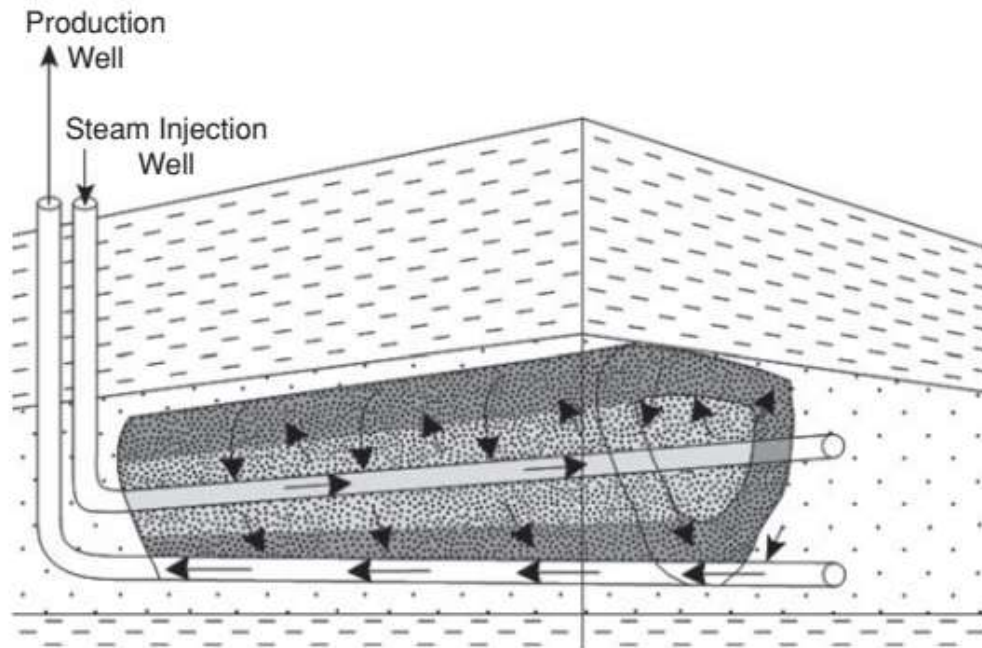


Figure TE3: SAGD process diagram

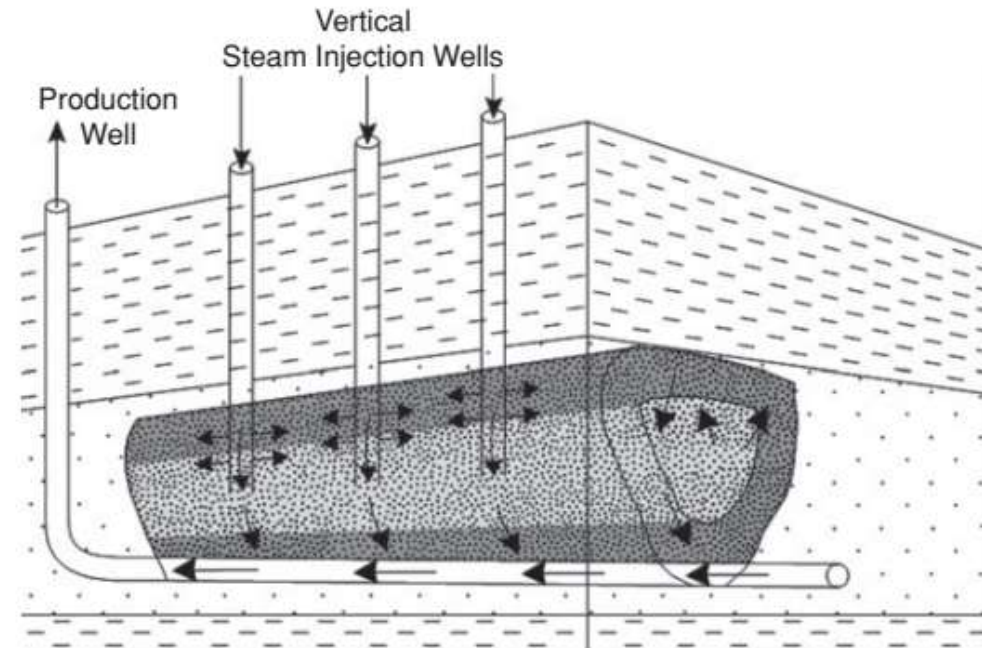
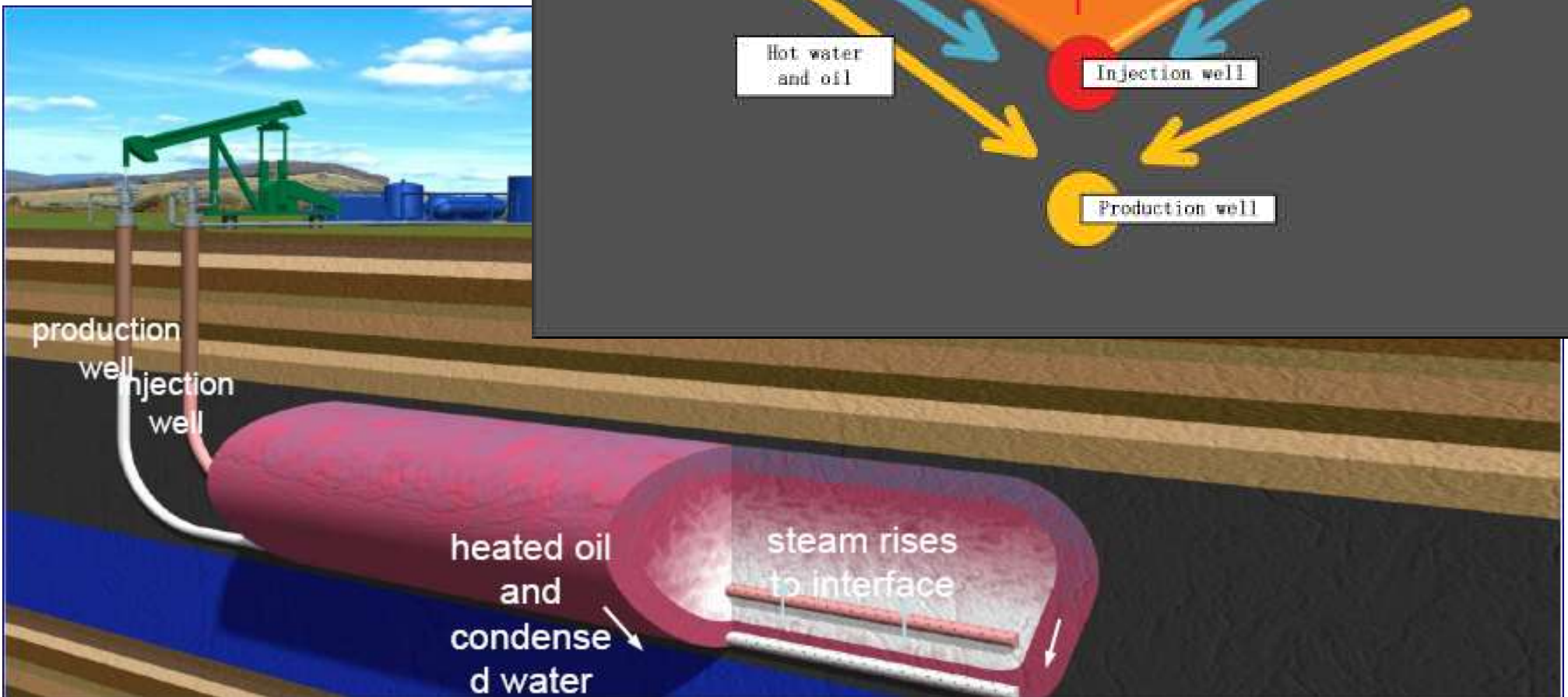
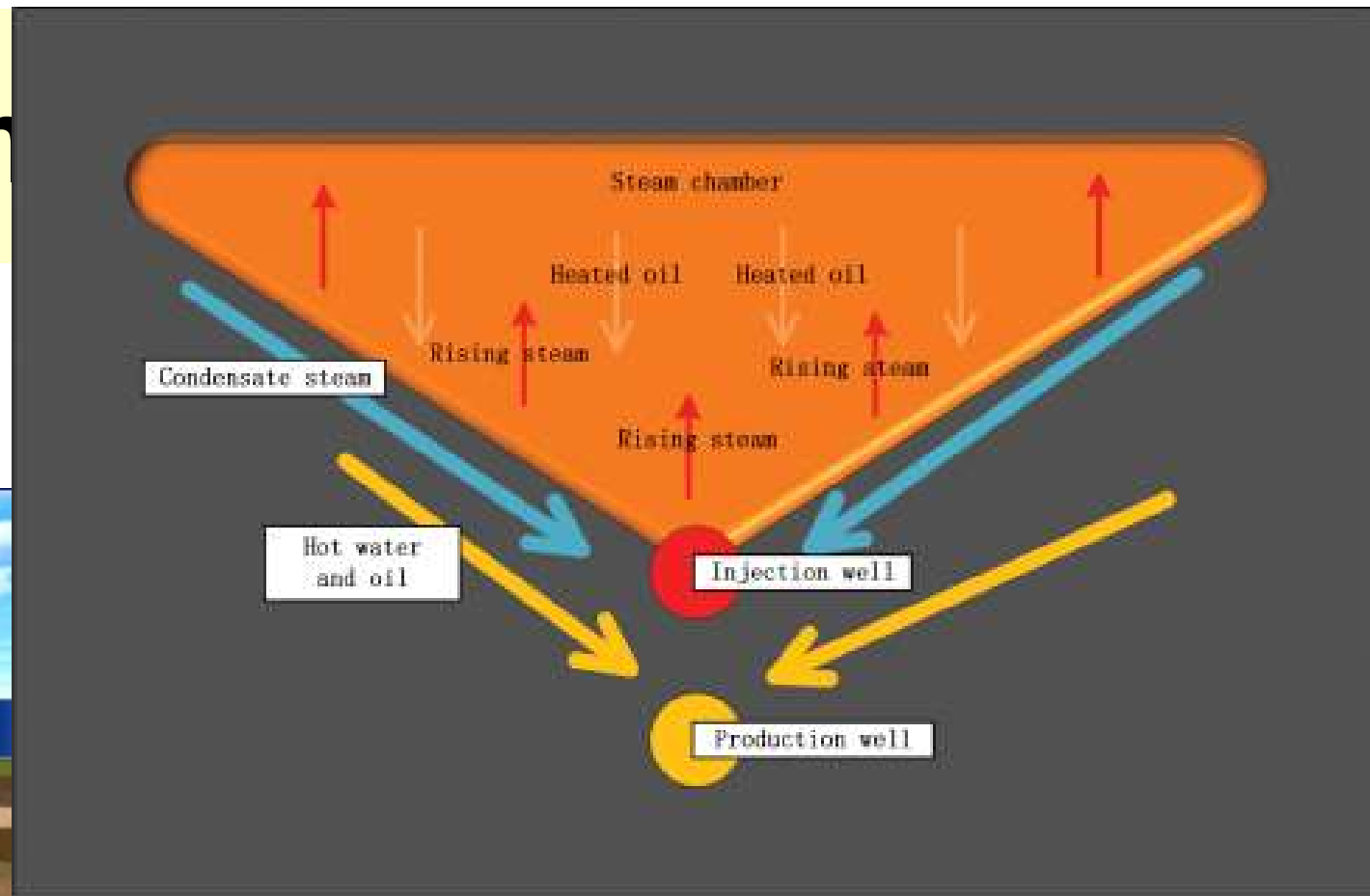


Figure TE4: XSAGD process diagram₂₈₈

Steam



- Steam Assisted Gravity Drainage further illustrated

Thermal EOR – Water-Alternating-Steam Process (WASP)

- Water-alternating-gas (WAG) was described earlier as an injection strategy used in gasfloods to reduce viscous fingering, improve vertical sweep, and thereby increase oil recovery.
- This concept is also used in steam flooding by injecting water alternating with steam to reduce fingering of steam, and improve its vertical sweep efficiency. This strategy is called water-alternating-steam process or WASP. The main benefit of WASP over continuous steam injection is reduction or elimination of premature steam breakthrough at producing wells. Early steam breakthrough at producers wastes heat energy, and reduces productivity of the wells.

Management of Steamflood Projects

- Management of a steamflood project offers unique challenges because the heat energy supplied to the reservoir must be utilized efficiently and effectively to maximize oil recovery. The management of steamflood projects can be subdivided into the management of three interrelated areas: surface equipment management, well management, and reservoir management. These areas are presented briefly:
- The key objective of **surface equipment management** in steamfloods is to minimize heat losses and optimize the process of generating steam. To minimize heat losses, most projects use insulated tubing and coupling joints coated with Teflon. Many projects have also incorporated cogeneration of steam and power using cogeneration plants as a cost effective means of improving the economics of steamflooding

Management of Steamflood Projects

- **Well Management:** The well system in a typical steamflood project consists of steam injectors, oil producers, and observation wells. Management of steam injectors usually requires profile control so that the steam can reach the targeted intervals. This is achieved by using surfactants or high temperature polymers to reduce mobility of steam and re-direct steam to unswept intervals, thereby improving sweep efficiency.
- Another practice is the use of multizone completion techniques to isolate and inject steam into separate target zones. Steam injectors can be remediated by cementing existing perforated intervals, and recompleting the well in other intervals that has not been flooded with steam. Similar remedial practice is conducted at the producers by either shutting-in updip producers as steam breaks through their perforated intervals or recompleting them to produce from lower intervals.

Management of Steamflood Projects

- **Steamflood Reservoir Management Strategy:** Producers should be monitored regularly to measure fluid production rates and detect steam breakthrough. One of the important activities in managing steamfloods is the use of observation wells to monitor the temperature and growth of the steam zone. This can be accomplished by running frequent temperature surveys or equipping the observation wells with fiberoptic distributed temperature sensing (DTS) technology.
- Steam-flooded reservoirs require higher levels of monitoring and surveillance because of the close spacing of injectors and producers, the potential for early steam breakthrough from gravity override, and the need to ensure that the energy supplied by the steam is effectively used in heating the reservoir to produce oil. The recommended practice is to manage the steamflooding process with a numerical thermal simulator with the goal of optimal utilization of the heat energy from steam.

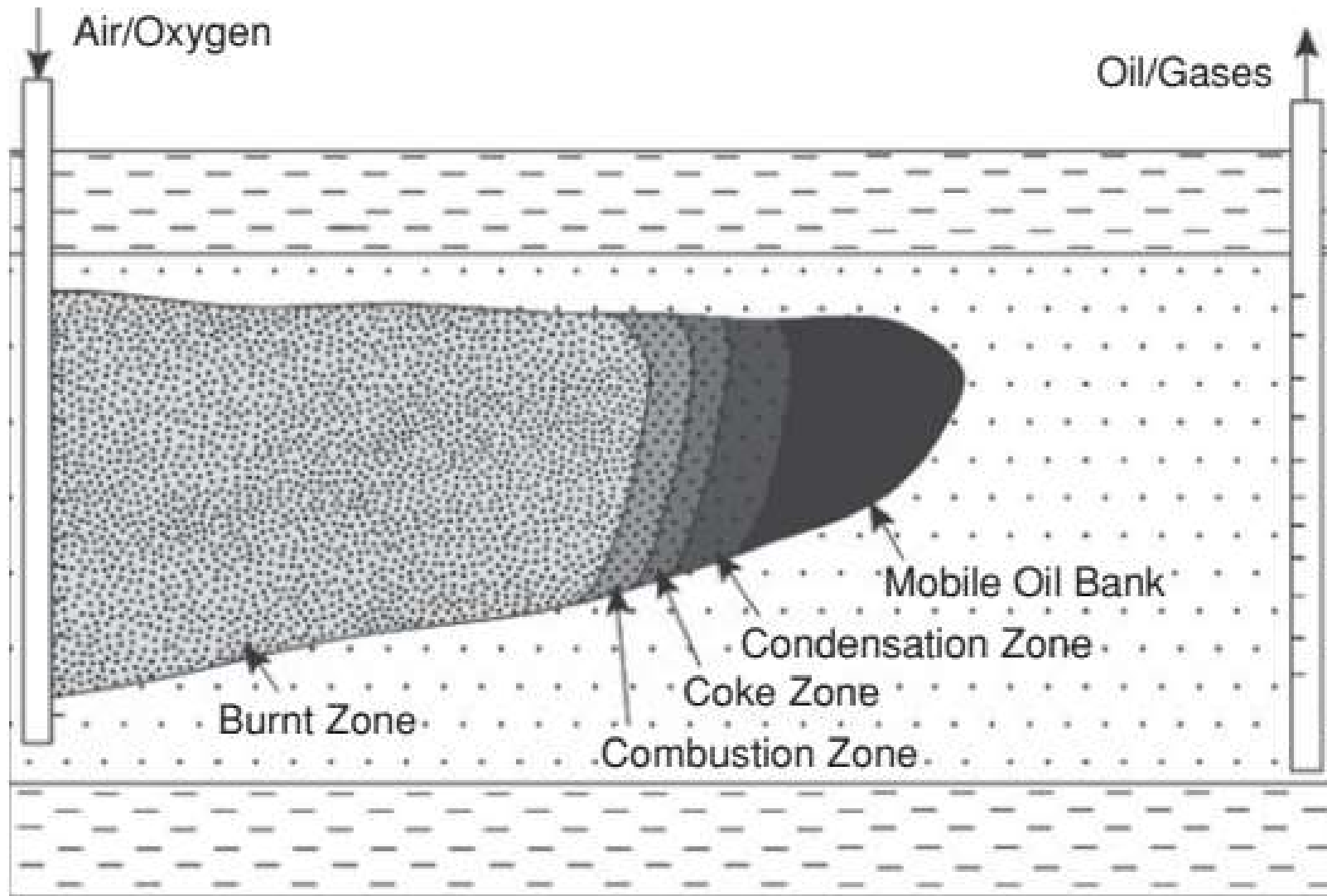
In-Situ Combustion (ISC) / High Pressure Air Injection (HPAI)

- In-situ combustion (ISC)/High pressure air injection (HPAI) are EOR processes in which compressed air is injected into heavy and light oil reservoirs under conditions, such that the oxygen in the injected air reacts with a small fraction of the crude oil at high temperatures to create a combustion front.
- The distinction between ISC and HPAI processes is blurred, although both processes are definitely related. Turta et al. state that an ISC process is associated with high peak temperatures (350°C to 600°C), and may require ignition to initiate it, whereas an HPAI process may not have high peak temperature and does not require ignition.
- The ISC process appears to be more applicable to heavy oil reservoirs, whereas HPAI is better suited for light oil reservoirs. Both processes are considered to be air injection processes (AIP). A schematic diagram of an air injection displacement process is shown in Figure TE4.

In-Situ Combustion (ISC) / High Pressure Air Injection (HPAI)

- The displacement path consists of an air zone behind the combustion/reaction zone. The combustion/reaction zone is preceded by a coke zone, a condensation zone, and mobilized oil bank.
- High oil recoveries have been reported from laboratory and field tests of AIP. However, there are few air injection projects around the world. In the United States, the remaining few air injection projects are based on relatively small reservoirs located in North and South Dakota.
- Steamflooding have surpassed air injection as the preferred EOR process even though both processes can satisfy the screening criteria for the same light or heavy oil reservoirs. This is the result of air injection being considered as more risky than steamflooding because the propagation of its combustion/reaction front is more difficult to predict and control.

In-situ Combustion



- Fig. TE4: Air Injection process diagram

Implementation of EOR Projects

- Taking an EOR project from initial screening to field implementation is a long, labor-intensive process. It is one of the most elaborate processes in project management and execution that requires the involvement of all the major areas of petroleum engineering.
- EOR projects require input from reservoir engineers, geologists, petrophysicists, geophysicists, drilling engineers, production engineers, facilities engineers, field personnel, economists, etc. for proper evaluation and implementation of the process. Assembling these professionals in a functional team is more efficient and effective for execution of the project.
- The procedure of screening, testing, and implementing an EOR process is divided into stages. The stages are organized sequentially as follows:

Implementation of EOR Projects

- 1) Process screening and selection.
- 2) Quick economic evaluation of selected processes.
- 3) Geologic and reservoir modeling of selected processes.
- 4) Expanded economic evaluation of processes.
- 5) Pilot testing, if necessary.
- 6) Update/upgrade simulation models with pilot test data/results.
- 7) Detailed economic evaluation of selected process.
- 8) Field implementation of process.
- 9) Project management.

Adopting EOR Methods

- Total costs and their breakdown between investment and operation for the various EOR methods depend on several factors.
 - are ships or new/existing installations used?
 - are the field(s) concerned on stream or in the planning phase?
 - are new wells required?
 - is the time frame for the project long or short?
- It can generally be assumed that EOR methods such as CO₂ and low-salinity injection are the most capital-intensive, while operating costs form a larger part of the overall bill for injecting polymers, surfactants, gels and alkalines.

Adopting EOR Methods

- An example of a financially sound EOR project is the low-salinity scheme on the BP-operated Clair Ridge field on the UK continental shelf which came on stream in 2018, it is estimated to have a breakeven oil price of USD 3 per barrel.

Conclusion

- Realisation of EOR potential can only be achieved through long-term commitments, both in capital and human resources, a vision to strive towards ultimate oil recovery instead of immediate oil recovery, research and development, and a willingness to take risks.
- Environmental considerations are also likely to shape EOR deployment. While shale has taken much of the limelight in recent years, it is much too soon to write EOR technologies out of the future of global energy.

THE END OF THE COURSE